Spring 2026: Noncommutative Algebra

Yutong Chen *

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1 Basics of Noncommutative Rings

1.1 First properties and Examples

Definition 1.1. A skew field is a nonzero ring R with all nonzero elements invertible. In this cases, modules are vector spaces over R.

For a ring R, I can attach $\mathrm{Mat}_n(R)$ the matrix ring.

Definition 1.2. A R-bimodule is an abelian group M with a left and right R-module structure which commutes. This is the same data as a left $R \times R^{\mathrm{op}}$-module. In general a (R, S)-bimodule is an abelian group with two commuting left R-module structure and right S-module structure.

Remark. *Kernel of a ring map is a two-sided ideal. Kernel, image, cokernel of a morphism between modules are modules.*

Definition 1.3. An R-module M is free if $M \simeq R^I$ for some set I, where R^I mean the set of functions $I \to R$ with almost all values 0. In other words, a free module is a direct sum of copies of R.

Remark. *Rank of a free module is not well-defined in general for noncommutative algebras.*

Definition 1.4. A ring R has an IBN property (invariance basis number) if $R^I \simeq R^J \implies I \simeq J$.

Remark. *Any skew field has IBN property.*

Example. *Let $V = \mathbb{C}^\infty$ be the space of sequences of $\mathbb{C}$-numbers with almost all terms zero. So that $e_i = (0, 0, \cdots, 1, 0, \cdots), i \in \mathbb{N}$ form a basis for V. Now consider $R = \mathrm{End}_{\mathbb{C}}(V)$, it consists of $A = (a_{ij}), i, j \in \mathbb{N}$, every column has only finitely many nonzero entries. We claim that this R does not have IBN property. Consider $V_+ = \mathbb{C}^\infty_{\mathrm{even}}, V_- = \mathbb{C}^\infty_{\mathrm{odd}} \subset \mathbb{C}^\infty$, so that $V = V_+ \oplus V_-$. But on the other hand $V_+ \simeq V_- \simeq V$. So that:*

$$\mathrm{End}(V) \simeq \mathrm{Hom}(V \oplus V, V) \simeq \mathrm{Hom}(V, V) \oplus \mathrm{Hom}(V, V) \simeq \mathrm{End}(V) \oplus \mathrm{End}(V) = R \oplus R$$

as R-modules.

Proposition 1.5. *If $\varphi : R \to S$, and S has IBN, then so does R.*

Corollary 1.6. *Each commutative ring has IBN.*

Proof. Pick a maximal ideal $\mathfrak{m}$ and consider $R \to R/\mathfrak{m}$. □

Proposition 1.7. *If R has IBN, then $S = \mathrm{Mat}_n(R)$ has IBN for all $n \in \mathbb{Z}_{>0}$.*

Proof. Note that: $S^I|_R = R^{n^2 \cdot I}$. So if $S^I \simeq S^J$ as S-module, we know $R^{n^2 \cdot I} \simeq R^{n^2 \cdot J}$ as R-modules. So that $n^2 \cdot I \simeq n^2 \cdot J \implies I \simeq J$. □

Corollary 1.8. *If D is a skew field, then $\mathrm{Mat}_n(D)$ has IBN. There are no homomorphism from $\mathrm{End}(\mathbb{C}^\infty) \to \mathrm{Mat}_n(D)$.*

Lemma 1.9 (Schur's lemma)**.** *If M is simple R-module, then $\mathrm{End}_R(M)$ is a skew field.*

Lemma 1.10. 1. *Every nonzero ring has a simple module.*

2. *Every proper left ideal in a nonzero ring is contained in a maximal left ideal. More generally, for every finitely generated R-module M, every proper submodule of M is contained in a maximal one.*

3. *A proper submodule in a R-module in an R-module M is maximal iff M/N is simple.*

Corollary 1.11. *Every finitely generated module has a simple quotient.*

Remark. *This is not true in general. Consider $\mathbb{Q}$ as a $\mathbb{Z}$-module.*

Theorem 1.12. *Let R be a ring such that every R-module is free. Then R is a skew-field.*

Proof. Let L be a simple R-module. If L is free, then necessarilty $L \simeq R$. So that:

$$\mathrm{End}_R(L) = \mathrm{End}_R(R) = R^{\mathrm{op}}$$

is a skew-field. We're done. □

1.2 Semisimple Modules

Definition 1.13. We call M semisimple, if I can present it as a direct sum of simple modules.

Example. *$\mathbb{Z}/4$ is not semisimple $\mathbb{Z}$-module.*

Lemma 1.14. *For $R \neq 0$ any ring, and assume $R^I \simeq R^J$, then $I \simeq J$ if either I or J is infinite.*

Lemma 1.15. *Let $M = \bigoplus_{i \in I} M_i$ be direct sum of simple modules. Let $N \subset M$ be a submodule. Then:*

$$M = N \oplus (\bigoplus_{j \in J} M_j)$$

for some $J \subset I$.

Proof. For $J \subset I$, let $M_J = \bigoplus_{i\in J} M_i$. Let T be the net of J with $M_J \cap N = 0$, orderd by inclusion. If J_α is a chain in T and $J = \cup_\alpha J_\alpha$, then $J \in T$. So we can apply Zorn's lemma to T to get a maximal J. Now I claim that $M_J + N = M$. Suppose not, then there exists i such that $M_i \nsubseteq M_J + N$. I necessarily have $i \notin J$. Then if

$$0 \neq n \in M_{J\cup\{i\}} \cap N, \quad n = \sum_{j\in J} m_j + m_i$$

$$\implies m_i = n - \sum_{j\in J} m_j \in M_J + N \implies M_i \subset M_J + N$$

this is a contradiction. □

Corollary 1.16. *Subquotients and sums of semisimple R-modules are semisimple.*

Proof. Let N be a submodule. Then there exists decomposition:

$$M = N \oplus (\bigoplus_{i\in J} M_i)$$

So that the composition:

$$N \hookrightarrow M \to \bigoplus_{i\in I\setminus J} M_i$$

is an isomorphism. Therefore N is semisimple. Also clearly M/N is semisimple. So every subquotients are semisimple. Finally given $M_\alpha \alpha \in A$ with each M_α semisimple, we have $\sum M_\alpha$ is a quoteint of $\bigoplus_{\alpha\in A} M_\alpha$, hence it is semisimple. □

Example. *Let $R = \mathrm{Mat}_n(D)$ for D skew-field. So that $R \simeq \oplus_{i=1}^n D^n$, with each D^n simple module. So that R-module R is semisimple.*

Definition 1.17 (socle)**.** The socle of an R-module M is the sum of all its simple (or semisimple) submodules. Denote by $\mathrm{soc}(M)$

Example. *Let $R = \mathbb{C}[t] = M$, then $\mathrm{soc}(M) = 0$.*

Example. *Let $R = \mathbb{C}[t]/(t^n) = M$, then $\mathrm{soc}(M) = \mathbb{C} \cdot t^{n-1}$.*

Example. *Let G be a p-group, and let k be a field of characteristic p. Consider $R = kG$, recall in this case irreducible representation of G is just trivial representation. Note that*

$$\mathrm{Hom}(k, kG) = k \implies \mathrm{soc}(kG) = k$$

Definition 1.18. If L is a simple R-module, then the L-isotypic component of a semisimple module M is the sum of the images of the embeddings $L \to M$. Equivalently, if $M = \oplus_{i\in I} M_i$, we have:

$$M_L = \bigoplus_{i\in I : L\simeq M_i} L$$

Proposition 1.19. *M is semisimple iff every short exact sequence:*

$$0 \to M_1 \to M \to M_2 \to 0$$

splits.

Proof. $\implies$ follows from lemma before. Conversely, consider:

$$0 \to \operatorname{soc}(M) \to M \to N \to 0$$

so that $M \simeq \operatorname{soc}(M) \oplus N$. So that N cannot have a simple submodule.
Claim: If $Q \subset N$ is a submodule, then there exists $P \subset N$ such that $N = P \oplus Q$.
Indeed, consider:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \operatorname{soc}(M) \oplus Q & \longrightarrow & M & \underset{\tilde{s}}{\rightleftarrows} & T & \longrightarrow & 0 \\ & & & & \downarrow & \swarrow & & & \\ & & & & N & & & & \end{array}$$

we just take $P = \tilde{s}(T)$ and prove the claim. Now we must prove $N = 0$. Suppose not, take $a \in N$ and consider $R \cdot a \subset N$. We know $R \cdot a$ has a simple quotient L:

$$0 \to Q \to N \to L \to 0 \implies N \simeq Q \oplus L$$

In particular $L \subset N$, contradiction. □

Definition 1.20. R is semisimple if it is a semisimple R-module.

Theorem 1.21. *The following are equivalent:*

1. *Every R-module is semisimple.*
2. *R is semisimple.*
3. *$R \simeq \bigoplus_{j=1}^{m} \operatorname{Mat}_{n_j}(D_j)$ (Exercise: this decomposition is unique.)*

Proof. (1) $\iff$ (2), (3) $\implies$ (2) are clear. It suffices to show (2) $\implies$ (3). Consider:

$$R = \bigoplus_{i \in I} M_i$$

be a decomposition into simple modules. It's clear that I is finite. So we can write:

$$R = \bigoplus_{j=1}^{m} L_j^{\oplus n_j}$$

with each L_j pairwise non-isomorphic. We can thus compute:

$$R^{\mathrm{op}} = \operatorname{End}_R(R) = \bigoplus_{j=1}^{m} \operatorname{Mat}_{n_j}(D_j), \quad D_j = \operatorname{End}_R(L_j)$$

□

Definition 1.22. We call R a simple ring if it has no 2-sided ideals other than $0, R$.

Example. *1. Skew fields are simple.*

2. If R is simple, then $\mathrm{Mat}_n(R)$ is simple. To see this, consider $e_i = E_{ii}$ idempotents. Let $a \in \mathrm{Mat}_n(R)$, we see that there exists:

$$e_i a e_j \neq 0$$

We "rescale" the entry to be 1. We can thus obtain other matrices by row and column operations to generate the whole matrix.

3. $\mathrm{End}(\mathbb{C}^\infty)$ is not simple, since it has an ideal of finite rank opeartors.

4. The Weyl algebra $\mathcal{D}(\mathbb{A}^1)$ is simple.

Remark. *A simple algebra need not be semisimple, for example look at $\mathcal{D}(\mathbb{A}^1)$:*

$$0 \to \mathbb{C}[x] \to \mathbb{C}[x, x^{-1}] \to \mathbb{C}[x, x^{-1}]/\mathbb{C}[x] \to 0$$

doesn't split.

Theorem 1.23 (Maschke's theorem)**.** *Let G be a finite group, $\mathrm{char}(k) \nmid |G|$, then kG is semisimple.*

Proof. Let $0 \to N \to M \to L \to 0$ be exact. We need to present a splitting $s \in L^* \otimes M$. This is equivalent to show that the map:

$$(L^* \otimes M)^G \to (L^* \otimes L)^G$$

is surjective. More generally we wish to show if I have $V \to W \to 0$, we will have

$$V^G \to W^G \to 0$$

This is again done by averaging. □

Given M semisimple module, the isotypic decomposition is canonical:

$$M = \bigoplus M_i, \quad M_i \simeq L_i^{m_i}$$

Note that fixing an isomorphism $M_i \simeq L_i^{m_i}$ amounts to fixing a basis in the vector space $V_i \coloneqq \mathrm{Hom}_R(L_i, M_i)$ over D_i^{op}, $D_i = \mathrm{End}_R(L_i)$. A submodule of M has the following form:

$$N = \bigoplus N_i, \quad N_i \subset M_i$$

and $N_i \subset M_i$ is determined by a vector subspace on $\mathrm{Hom}(L_i, N_i) \subset V_i = \mathrm{Hom}(L_i, M_i)$. It's given by linear equations:

$$\sum m_j d_j^k = 0, \quad m = (m_1, \cdots, m_{m_i}) \in M^{\oplus m_i}$$

Theorem 1.24 (Density Theorem)**.** *1. Let L be a simple module over R. Given*

$$x_1, \cdots, x_n \in L$$

linearly independent over $D = \mathrm{End}_R(L)$. For any $y_1, \cdots, y_n \in L$, there exists $r \in R$ such that $rx_i = y_i$.

2. More generally, given $L_1, \cdots, L_k$ a finite set of pairwise non isomorphic irreducible modules. Suppose for each $i = 1, 2, \cdots, k$ we have:

$$x_1^i, \cdots, x_{n_i}^i \in L_i$$

be linearly independent over $D_i = \mathrm{End}(L_i)$, then for all $y_1^i, \cdots y_{n_i}^i \in L_i$, there exists $r \in R$ with $rx_j^i = y_j^i$.

Proof. We only prove the first statement. The proof for the second statement is similar. Consider $M = L^{\oplus n}$ has submodule N generated by $\overline{x} = (x_1, \cdots x_n)$. Now if $N \nsubseteq M$, then it is annhilated by $d_1, \cdots, d_n \in D^n$, namely

$$\sum x_i d_i = 0$$

contradicting the linear independence over D. So $N = M$, this is precisely the statement of the theorem. □

Corollary 1.25. *If L is finite dimensional over D, then:*

$$R \twoheadrightarrow \mathrm{End}_D(L) = \mathrm{Mat}_n(D^{\mathrm{op}})$$

Remark. *The corollary is clearly false for infinite dimensional case. Consider R, L be countable dimensional over $\mathbb{C}$, then $\mathrm{End}_R(L) = \mathbb{C}$. But $\mathrm{End}_{\mathbb{C}}(L)$ is not countable dimensional.*

1.3 Finiteness Conditions

Definition 1.26. We call R is left (resp. right) noetherian, if every ascending chain of left (resp. right) ideals stablizes. By Zorn's lemma this is equivalent to saying every collection of ideals has a max element.

Remark. *Left noetherian is not equivalent to right noetherian.*

Definition 1.27. We call R is left (resp. right) Artinian, if every descending chain of left (resp. right) ideals stablizes. By Zorn's lemma this is equivalent to saying every collection of ideals has a min element.

Remark. *Artinian implies noetherian.*

Theorem 1.28 (Wedderburn)**.** *The following are equivalent:*

1. R is simple and (left or right) Artinian.

2. Every R-module is semisimple and R has a unique simple module up to isomoprhism.

3. $R \simeq \mathrm{Mat}_n(D)$ *for* D *some skew-field.*

Proof. $(2) \iff (3), (3) \implies (1)$ are clear. We prove $(1)implies(3)$. Let's assume R to be left Artinian. Choose a minimal left ideal $L \subset R$, this is a simple R-module. By simplicity, we see $LR = R$. Therefore

$$R = LR = \sum_{a \in R} La = \bigoplus_{i \in I} L$$

and I must be finite since R is Artinian. Hence we see:

$$R^{\mathrm{op}} \simeq \mathrm{End}(\oplus L) = \mathrm{Mat}_n(D).$$

□

Definition 1.29. A module M is called Noetherian (resp. Artinian) if it satisifes ascending (resp. descending) chain termination condition.

Example. *Consider* $R = \mathbb{Z}, M = \mathbb{Z}$*, this is noetherian but not artinian. Dually, consider* $R = \mathbb{Z}, M = \mathbb{Z}[\frac{1}{2}]/\mathbb{Z}$ *is artinian but not noetherian:*

$$\frac{1}{2}\mathbb{Z}/\mathbb{Z} \subset \frac{1}{4}\mathbb{Z}/\mathbb{Z} \subset \cdots$$

Another example of artinian but not noetherian module is:

$$R = \mathbb{C}[t], M = \mathbb{C}[t, t^{-1}]/\mathbb{C}[t]$$

Lemma 1.30. *A module* M *is Noetherian iff every submodule* M *is finitely generated.*

Proof. The proof in the commutative case applies. □

Given a finite filtration of a module M, we can form $\mathrm{gr}M$ its associated graded.

Definition 1.31. We say M has finite length if it has a finite filtration with $\mathrm{gr}_i(M)$ is simple for $i = 1, 2, \cdots, n$.

Proposition 1.32. M *has finite length iff* M *is both Artinian and Noetherian.*

Lemma 1.33. *Given* $0 \to M_1 \to M_2 \to M_3 \to 0$ *with* M_1, M_3 *noetherian (resp. artinian), then so is* M_2*. In fact this is an if and only if.*

Proof. (of the proposition). Note that simple module is both artinian and noetherian, so that the direction $\impliedby$ follows by induction and applying the lemma. Conversely, suppose M is artinian and noetherian. The artinian condition implies that for every collection of submodules, it has a minimal element. So that we can pick a minimal element in the collection of nonzero submodules $L = M_1$, it is simple. So that we can build M_2 by considering M/M_1 and continue this way. We get a strictly increasing chain with simple quotients:

$$0 \subset M_1 \subset M_2 \subset \cdots$$

By noetherian hypothesis this will stablize. □

Theorem 1.34 (Jordan-Holder)**.** *Any two simple composition series of a finite length module have the same length, moreover with respect to the two filtrations:*

$$\mathrm{gr}M \simeq \mathrm{gr}'M$$

So from each module M, we get a multi-set of irreducibles, this is called Jordan Holder content of M.

Proof. This proof works when R is a k-algebra for algebraically closed field of characteristic 0, and M finitely dimensional over k. Consider two composition series. Let $L_1, \cdots, L_t$ be the list of irreducibles appearing as gr_i in either filtrations. By density theorem, there exists $r \in R$ which acts as E_{11} for some basis in L_i, and 0 in L_j for $j \neq i$. We see:

$$\mathrm{Tr}(r, L_i) = 1, \quad \mathrm{Tr}(r, L_j) = 0 \text{ for } i \neq j$$

Then we can just "spot" the multiplicity by $\mathrm{Tr}(r, M)$. □

Proof. Consider two filtrations:

$$0 \subset M_1 \subset \cdots \subset M_n = M$$

$$0 \subset M_1' \subset \cdots \subset M_{n'}' = M$$

Let i be minimal such that $M_1 \subset M_i'$, we see that:

$$M_1 \simeq \mathrm{gr}_i'(M) \implies M_i' = M_1 \oplus M_{i-1}'$$

Now M_i/M_1 gives a filtration of M/M_1. But the image of M_i' in M/M_1 also form a filtration of M/M_1. We conclude by comparing these two filtrations and induction. □

Remark. *This procedure produces the relative position of two flags when we consider $R = k$.*

1.4 Grothendieck Group

Let $\mathcal{C}$ be a class of modules closed under extensions. For example we can take:

1. finite length modules.
2. For R noetherian, finitely generated modules.
3. For R noetherian, finitely generated projective modules.

We can define $K(\mathcal{C})$ be the free abelian group generated by isomorphism classes of M modulo the relation:

$$[M] = [M_1] + [M_2], \text{ when } 0 \to M_1 \to M \to M_2 \to 0$$

We can look for invariant of a module $r(M) \in A$ for some abelian group A, additive on short exact sequence. We see that:

$$M \mapsto [M]$$

is universal, meaning every r like that comes from a map $K(\mathcal{C}) \to A$.

Remark. *If we take $\mathcal{C}$ be all modules, then $K(\mathcal{C}) = 0$:*

$$0 \to \oplus_{i=1}^{\infty} M \to \oplus_{i=0}^{\infty} M \to M \to 0$$

Jordan Holder theorem implies:

$$K(R\text{-mod}) = \oplus_{\text{Irr}(R)} \mathbb{Z}$$

Example. *Let G be a finite group. We have:*

$$K(k[G]\text{-mod}_{\text{fin}}) \simeq k[G]^G$$

for char$(k) = 0$. For char$(k) = p$, we know:

$$K(k[G]\text{-mod}_{\text{fin}}) \simeq k[G]^G_{(p)}$$

from Brauer character theory.

Example. *If $R = \mathbb{C}[x_1, \cdots, x_n]$, then:*

$$K(R\text{-mod}_{\text{fin}}) \simeq \mathbb{Z}$$

If $R = \mathbb{C}[x, y]/(y^2 - x^3 - x)$, we see:

$$K(R\text{-mod}_{\text{fin}}) \simeq \mathbb{Z} \oplus \mathbb{C}/\mathbb{Z}^2$$

1.5 Socle and Cosocle Filtration

Note that it is possible that $M \neq 0$ but $\text{soc}(M) = 0$. But if $M \neq 0$ is Artinian, we always have $\text{soc}(M) \neq 0$. Therefore we can define **socle filtration**:

$$0 \subset M_1 \subset M_2 \subset \cdots$$

with $M_1 = \text{soc}(M)$, $M_2/M_1 = \text{soc}(M/M_1)$,etc. If M is finite length, then socle filtration is an exhaustive filtration of M, with semisimple quotients.

There is a dual procedure called cosocle filtration, using semisimple quotients. We define:

$$M^1 = \{m \in M \mid \forall M \to L \text{ simple }, m \mapsto 0\}.$$

In general $M/M^\perp$ may not be simple. For example consider: $R = M = \mathbb{Z}$, then:

$$\mathbb{Z} \hookrightarrow \prod_p \mathbb{Z}/p$$

so that $M^\perp = 0$. But if M is Artinian, then:

$$M^1 = \ker(M \to \oplus L_i)$$

for some finite collection of L_i simples. We can therefore build $M^2, M^3, \cdots$ and get the cosocle filtration.

Example. *Let $R = \mathbb{C}[t]$, M finite dimensional with t acting nilpotently. The socle filtration is:*

$$\ker t \subset \ker t^2 \subset \ker t^2 \subset \cdots$$

The cosocle filtration is:

$$\text{im } (t) \supset \text{im } (t^2) \supset \text{im } (t^3)$$

Remark. *Some people say "Loewy series" for socle / cosocle filtrations.*

1.6 Jacobson Radical

Definition 1.35. The Jacobson radical of R is defined to be:

$$J(R) = \{x \in R \mid xL = 0, \forall L \text{ simple}\}$$

Remark. *Jacobson radical is a two-sided ideal.*

Example. *Let $R = \mathfrak{b}_n(k)$ upper triangular matrices, its Jacobson radical is $\mathfrak{u}_n(k)$ strictly upper triangular matrices.*

Theorem 1.36. *Given $a \in R$. TFAE:*

1. *$a \in J$*
2. *a lies in all maximal left ideals.*
3. *$1 - xa$ has a left inverse for all x*
4. *a lies in all maximal right ideals.*
5. *$1 - ax$ has a right inverse for all x.*
6. *$1 - xay$ is invertible for all $x, y \in R$.*

Corollary 1.37. $J(R) = J(R^{\text{op}})$.

Proof. (of the theorem). (1) $\iff$ (2) is clear. Now that:

$$(3) \iff R(1 - xa) = R \iff 1 - xa \notin \text{any max ideal} \iff (2)$$

It remains to show (3) $\implies$ (6). Given a such that $1 - xa$ has as a left inverse for all x. Note $ay \in J$, so we can find b such that:

$$(1 - b)(1 - xay) = 1 \implies b = -xay + bxay \in J \implies (1 - b) \text{ has a left inverse}$$

Therefore the left inverse must again bu $1 - xay \implies 1 - b$ is again the right inverse of $1 - xay$. □

Example. *Let $R = k[G]$ for $char(k) = p$ and G a finite p-group. Then:*

$$J(R) = \{\sum a_i g_i \mid \sum a_i = 0\}$$

Definition 1.38. We call R local if all noninvertible elements form a two-sided ideal. Then $J \coloneqq \{\text{noninvertible ideals}\}$. It follows any two sided-ideal must lie in J, and $J = J(R)$ is its Jacobson radical.

1.7 Indecomposable Modules

Theorem 1.39 (Krull-Schmidt)**.** *Every finite length module can be written as direct sum of indecomposable modules. Moreover, if*

$$M \simeq \bigoplus_{i=1}^{n} P_i \simeq \bigoplus_{j=1}^{m} Q_j$$

then $n = m$ and $P_i \simeq Q_i$ after reindexing.

Remark. *We have existence if M is either noetherian or artinian. Indeed, if not, then $M = M_1 \oplus M_2$ where at least one summand is not a sum of indecomposable. Then we just continue and M can be written as direct sum of arbitrarily many pieces.*

To prove this, we need:

Proposition 1.40 (Fitting Lemma)**.** *If M is indecomposable finite length, then* $\mathrm{End}_R(M)$ *is local*

Remark. *This is in fact an if and only if. Indeed, if $M = M_1 \oplus M_2$, then we get projectors e_1, e_2. Then e_1, e_2 is not invertible, but their sum is. So* $\mathrm{End}(M)$ *is not local.*

Remark. *In general, we know M is decomposable iff we can find two orthogonal idempotents $e_1, e_2 \in \mathrm{End}_R(M)$.*

Lemma 1.41. *Suppose $x \in \mathrm{End}(M)$, and* $\ker(x) = \ker(x^2), \mathrm{im}\ (x) = \mathrm{im}\ (x^2)$*. Then $M = \ker x \oplus \mathrm{im}\ x$*

Proof. (of lemma). We check:

$$m \in \ker(x) \cap \mathrm{im}\ (x) \implies m = xn, b \in M \implies xm = x^2 n = 0 \implies m = xn = 0$$

the last step is because $\ker x = \ker x^2$. On the other hand, given any $m \in M$, we can write:

$$xm = x^2 m' \implies x(m - xm') = 0$$

So I can present:

$$m = xm' + (m - xm') \in \mathrm{im}\ (x) + \ker(x)$$

□

Proof. (of proposition) Suppose $a \in \mathrm{End}(M)$ is not invertible. We will show $a \in J = J(\mathrm{End}(M))$. Consider:

$$\ker(a) \subset \ker(a^2) \subset \cdots$$

$$\mathrm{im}\ (a) \supset \mathrm{im}\ (a^2) \supset \cdots$$

Both will stablize, hence we can find n such that:

$$\mathrm{im}\ (a^{n+1}) = \mathrm{im}\ (a^n), \quad \ker(a^{n+1}) = \ker(a^n)$$

Now if we set $x = a^n$, then byt the previous lemma we see:

$$M = \ker(x) \oplus \text{im } (x)$$

Therefore one of them vanishes by indecompibility. If $\ker(x) = 0, \text{im } (x) = M$, so x is an ismorphism, which contradicts our assumption that a is noninvertible. So we must have $\ker(x) = M, \text{im } (x) = 0$, namely $x = 0 \implies a$ is nilpotent. We claim that xay is not invertible hence nilpotent, for all $x, y \in \text{End}(M)$. Indeed, since:

$$a : M \to M$$

is not an isomorphism, a is either not 1-1 or not onto. Thus this implies length(im (M)) $<$ length(M). This implies a is not 1-1 and not onto. But then:

a is not 1-1 $\implies xa$ is not 1-1 $\implies xa$ is not onto $\implies xay$ is not onto.

Now xay is nilpotent, so that $1 - xay$ is invertible. This implies $a \in J(R)$, we're done. □

Proof. (of Krull-Schmidt) The existence can be done by induction in the length. Now let's prove uniqueness. Let P, Q be two indecomposable finite length modules. Consider:

$$\text{Hom}_R(P, Q) \times \text{Hom}_R(Q, P) \to E \coloneqq \text{End}(P)$$

We've seen End(P) is local, say has maximal ideal J. We claim that $P \not\simeq Q \iff$ the pairing lands in J. The direction $\impliedby$ is clear. For the other direction, look at:

$$P \xrightarrow{\alpha} Q \xrightarrow{\beta} P$$

If $\beta \circ \alpha$ is an isomorphism, withoutloss we may assume $\beta \circ \alpha = \text{id}_P$, this provides a splitting:

$$Q = \text{im } (\alpha) \oplus \ker(\beta)$$

But Q is indecomposable, we know $\ker(\beta) = 0 \implies P \simeq Q$, contradiction. Therefore $\beta \circ \alpha \in J$ as desired.

Now given a finite length module M, P an indecomposable finite length. We need to check if

$$M = \bigoplus Q_i$$

the number of $\{i \mid Q_i \simeq P\}$ is indepedent of the decomposition. Let's use the above pairing:

$$\text{Hom}(P, M) \times \text{Hom}(M, P) \to E = \text{End}(P)$$

Now set $D = E/J$ a skew-field. We have:

$$\overline{\text{Hom}}(P, M) \times \overline{\text{Hom}}(M, P) \to D$$

by modulo everthing by J. This is a D-blinear pairing. Now if $M = Q$ indecomposable, then: for $Q \not\simeq Q$ this pairing is zero; for $Q \simeq P$ this is standard rank 1 pairing. Therefore we can write out the matrix of the pairing, and the number we desire is nothing but the rank of the matrix. □

Remark. *Given a D-vector space V and D^{op}-vector space W. A blinear pairing:*

$$V \times W \to D$$

is the same as a D-linear map $V \to W^*$. The rank of the pairing is the rank of this matrix with coefficients in D.

Example. *Let $R = \mathbb{Z}$. The simple modules are $\mathbb{Z}/p$; the indecomposable modules are $\mathbb{Z}/p^n$, Then Krull-Schmidt tells us the structure of finite length modules (i.e. finite abeliang groups).*

Example. *Let $R = \mathbb{C}[t]$, the finite length modules are finite dimensional ones. The indecomposable modules are*

$$\mathbb{C}[t]/(t-a)^n$$

These are the Jordan blocks. So Krull-Schmidt in this case is just Jordan normal form theorem.

For most rings, there will be no classification of indecomposable modules. Here is one such wild problem. Let k be algebraically closed, $R = k\langle x, y\rangle$. Then finite dimensional modules are nothing but two matrices. M up to isomorphism is the same as the two matrices up to conjugacy.

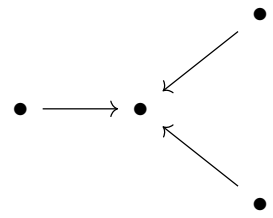
Definition 1.42. A **quiver** Q is an oriented graph. A **representation** of a quiver Q (over k) is a collection of vector spaces indexed by vertices, and for each edge we assign a linear map. The **path algebra** A_Q has a basis indexed by oriented paths, with multiplications just concatenation (include length 0 paths).

For every vertex v, we denote e_v the element corresponding to the length 0 path at v. Then $e_v^2 = e_v$, $e_v e_{v'} = 0$ for $v \neq v'$. So it follows:

$$\sum_v e_v = 1_{A_Q}$$

Now given a module M for A_Q, we obtain a quiver representation: $M_v = e_v M$. This in fact loses no information.

Example. *Consider:*

$Q =$ 

a star shape quiver with $n+1$ vertices. We wish to classify indecomposable representations. There are two types of indecomposables:

1. *Assign each vertex a one-dimensional space.*

2. *All arrows are 1-1. Such a representaiton is the same data as a vector space with n subspaces. Such object has no finite classifications for $n \geq 4$. But for $n = 3$ it does. There are 12 indecomposables.*

   When $n = 4$, assume the central vertex corresponds to a vector space of dimension 2, then the 4 outsides nodes corresponds 4 points on the projective line $\mathbb{P}^1$. They're therefore classified by cross-ratios.

This is a special instance of **Gabriel's Theorem**: A connected quiver has finitely many indecomposable representations iff the corresponding graph is a Dynkin graph.

For an algebra / $k = \overline{k}$, we have a trichotomy (Drozd):

1. Finite. There are finitely many finite dimensional indecomposables.
2. Tame. "Good classification". The family of indecomposables depends on one parameter.
3. Wild. No managable classification. The family depends on more than two parameters.

In fact, a quiver is tame iff it's an affine Dynkin diagram. The rest are wild.

Remark. *Same graphs appear in classification of root systems, simple Lie algebras, compact Lie groups etc. The Roots tell the dimension of indecomposable modules.*

Example. *Consider the Kronecker quiver:*

$$Q = \quad \bullet \;\overset{\longrightarrow}{\longrightarrow}\; \bullet$$

Given a vector space V with linear operators x, y, we can form the following representaiton for Q: Assign V to the left vertex and $V \oplus V$ to the right vertex. The three arrows are:

$$\mathrm{id} \oplus 0, 0 \oplus \mathrm{id}, (x, y)$$

One can show that isomorphism classes of $k\langle x, y\rangle$ modules embed into those for A_Q in the above construction. In other words, we define a fully faithful functor:

$$k\langle x, y\rangle\text{-mod} \to A_Q\text{-mod}$$

1.8 Primitive and Semiprimitve Rings

Definition 1.43. A ring R is called **semiprimitve** if if $J(R) = 0$. This is equivalent to for all $r \in R$, there exists L simple such that $rL \neq 0 \iff$ there exists a faithful semisimple module. A ring R is called **primitive** if there exists a faithful simple module.

Remark. *Since the definition of Jacobson radical is left / right symmetric. So the notion of semiprimitive is left / right symmetric. But there are example such that R is primitve, but R^{op} is not.*

Example. *Consider* $V = \mathbb{C}[t, t^{-1}]$, $R \subset \mathrm{End}(V)$ *is the ring generated by:*

$$P(t) \mapsto tP(t)$$

$$P(t) \mapsto P(t^2)$$

Then one can show R *is right primitive but not primitive. See PSET.*

Recall R is simple if the regular bimodule is simple. Note that every simple ring is primitive. Indeed, let R be simple, and pick any maximal ideal M inside R and set $L = R/M$. Then we look at:

$$R \to \mathrm{End}(L)$$

The kernel is a two-sided ideal and not everything, it follows $\ker = 0$.

Example. *Let's construct a primitive but not simple algebra. Recall we have the Weyl algebra* $W = \mathbb{C}[x, \partial]/[\partial, x] = 1$. *Consider the delta module:*

$$M_0 = \mathbb{C}[t, t^{-1}]/\mathbb{C}[t]$$

Now consider the subalgebra $A \subset W$ *generated by* $x, (x-1)\partial$. *Then one can check* M_0 *remains simple as an algebra over* A. *Now:*

$$A \subset W \hookrightarrow \mathrm{End}(M_0)$$

proving A *primitive. But on the other hand,* A *acting on* $\mathbb{C}[t]$ *preserving* $(t-1)\mathbb{C}[t]$, *so it has a one-dimensional module. Therefore* A *is not simple.*

Example. *The ring* $R = \mathrm{End}_{\mathbb{C}}(\mathbb{C}^\infty)$ *is primitive but not simple. It acts on* $M = \mathbb{C}^\infty$, *this is a faithful simple representation, so that* R *is primitve. But* R *has an ideal consisting of finite rank operators.*

Proposition 1.44. *An Artinian ring* R *is semi-primitive iff:*

$$R = \prod_{i=1}^{n} \mathrm{Mat}_{d_i}(D_i)$$

for D_i *skew-fields. It follows that artinian primitive rings are* $\mathrm{Mat}_n(D)$ *for* D *skew-field.*

Proof. By semi-primitivity, we know:

$$0 = \bigcap_{M \text{ max left}} M$$

By Artinian property, we know:

$$M_1 \cap \cdots M_n = 0$$

for some maximal left ideals. Therefore:

$$R \hookrightarrow \oplus R/M_i$$

Therefore we know R is semisimple. We thus conclude by classification of semisimple algebras. □

Corollary 1.45. *If R is Artinian, then a module M is semisimple iff $JM = 0$.*

Proof. The direction $\implies$ is clear. Conversely, if R is Artinian, M descends to a R/J-module. But R/J is semisimple. □

Remark. *Let $R = \mathbb{Z}$, then $J = 0$. But obviously not every module is semisimple.*

Corollary 1.46. *If R is Artinian, then:*

$$M \supset JM \supset J^2M \supset J^3M \supset \cdots$$

is the cosocle filtration;

$$0 \subset M^J \subset M^{J^2} \subset \cdots$$

is the socle filtration.

Example. *Let $R = \mathbb{C}[t]/t^N$, $M = V$ a finite dimensional vector space with an operator t with $t^N = 0$. So that the socle filtration:*

$$\ker t \subset \ker t^2 \subset \cdots$$

and cosocle filtration:

$$\text{im } t \supset \text{im } t^2 \supset \cdots$$

as we saw before.

Proposition 1.47 (Nakayama Lemma)**.** *Let R be any ring, M finitely generated. If $JM = M$, then $M = 0$.*

Proof. Since M is finitely generated, so the union of nested collection of proper submodules is a proper submodule. By Zorn's lemma, we can find $N \subset M$ maximal submodule, so that M/N simple. Therefore:

$$J(M/N) = 0.$$

This contradicts $JM = M$. □

Theorem 1.48 (Akizuki-Hopkins-Levitzky)**.** *If R is left Artinian, then R is finite length as R-module. In particular R is noetherian.*

Lemma 1.49. *The Jacobson radical J of a left Artinian ring R is a nilpotent ideal.*

Proof. Consider:

$$J \supset J^2 \supset \cdots$$

must stablize. Let's denote $J^n = J^{n+1} = I$. We know $I^2 = I$. Assume $I \neq 0$, we can find a minimal left ideal M such that $IM \neq 0$. Now pick $a \in M$ such that $Ia \neq 0 \implies I(Ia) = I^2a = Ia \neq 0$. By minimality we know $M = Ra = Ia$. Now apply Nakayama:

$$JM = JIa = Ia = M \implies M = 0$$

Contradiction. Therefore $I = 0$ and we're done. □

Proof. (of Akizuki...) We already know $J^n = 0$ for some n. It remains to show J^i/J^{i+1} has finite length. But J^i/J^{i+1} is semisimple, artinian R-module. This implies J^i/J^{i+1} has finite length. □

1.9 Projective Modules

Definition 1.50. A module P is projective if the functor $M \mapsto \mathrm{Hom}(P, M)$ is exact.

Lemma 1.51. *If $P_1 \oplus P_2 = P$ for P projective, then so is P_1 and P_2.*

Proof. Direct sum of two SES is exact iff both are exact. □

Proposition 1.52. *A module is projective iff it is a direct summand of a free module.*

Proof. It remains to see every projective module is a direct summand of a free module. Let P be projective. We pick:

$$F = \bigoplus_{i \in I} A \to P \to 0$$

There will be a splitting $P \to F$, this realizes P as a direct summand of F. □

Notice every module is a quotient of projective module. For an Artinian ring, there is a minimal way to do this.

Definition 1.53. A surjective map of modules $M \to N$ is essential surjection if for any $M' \subsetneq N$, then $M' \to N$ is not onto.

Proposition 1.54. *Let R be Artinian. Then:*

1. *$M \to N$ is onto iff $M/JM \to N/JN$ is onto.*
2. *$M \to N$ is an essential surjection iff $M/JM \xrightarrow{\sim} N/JN$.*

Proof. 1. $\implies$ is obvious. Conversely, consider the cosocle filtration $M^i = J^i M$, and consider the associated graded: $\mathrm{gr}(M) = \oplus M^i/M^{i+1}$. This is a module for $\mathrm{gr}(R) = \oplus J^i/J^{i+1}$. Now $\mathrm{gr}(M)$ is generated by $\mathrm{gr}_0(M)$ as a $\mathrm{gr}(R)$-module. Now we already know $\mathrm{gr}_0(M) \to \mathrm{gr}_0(N)$ is surjective, so that

$$\mathrm{gr}(M) \to \mathrm{gr}(N)$$

is surjection. Hence $M \to N$ is surjection, since our filtration is finite.

2. Now if $M/JM \simeq N/JN$, with $M' \subset M$ with $M' \to N \to 0$. Then we have:

$$M'/JM' \to N/JN \to 0 \implies M'/JM \to M/JM \to 0 \implies M' = M$$

Conversely, if $M/JM \to N/JN$ is not an isomorphism, it will have some kernel K. But M/JM is semisimple, so:

$$M/JM \simeq K \oplus N/JN$$

We just take $M' = \ker(M \to M/JM \to K)$, then $M' \to N$ is surjective.

□

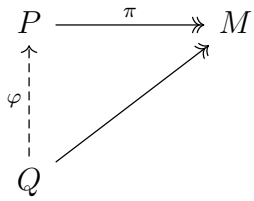
Remark. *The first part is also true for any R if M, N are finitely generated. But this is not true in general, consider:*

$$\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$$

Definition 1.55 (Projective Cover)**.** A projective cover of a module M an essential surjection $P \to M$ with P projective.

Lemma 1.56. *If a projective cover exists, it will be unique. Moreover, given a projective cover $\pi : P \to M$ and any projective $Q \to M \to 0$, we have $Q \simeq P \oplus Q'$ making the obvious diagram commute.*

Proof. Jut a diagram chase.



Since P is a projective cover, φ must be onto. But then φ must admit a spilt. □

Proposition 1.57. *Let R be Artinian.*

1. *Every module has a projective cover.*
2. *Irreducible R-modules up to isomorphism is in bijection with indecomposable projective modules up to isomorphism, via sending L to the projective cover of L.*

Before prove the theorem, we need:

Lemma 1.58 (Lifting of idempotent)**.** *Let $I \subset R$ with $I^2 = 0$, $\overline{R} = R/I$. Given $e \in \overline{R}$ with $e^2 = e$, there exists $\widetilde{e} \in R$ lifting e such that $\widetilde{e}^2 = \widetilde{e}$.*

Proof. Start from any lift $\widetilde{e}'$, and solve for x: $\widetilde{e} = \widetilde{e}' + x, x \in I$. Newton's method motivates me to set:

$$x = -(2\widetilde{e}' - 1)(\widetilde{e}'^2 - \widetilde{e}')$$

□

Remark. *This have a generalization. Consider $\overline{R} = R/I$ with $I^2 = 0$, and $x \in \overline{R}$. Suppose $P(x) = 0$ and $(P, P') = 1$, then we can find a lift $\widetilde{x}$ with $P(\widetilde{x}) = 0$.*

Proof. (of projective cover for Artinian rings).

1. Start by:
$$R/J = \oplus \mathrm{Mat}_{d_i}(D_i)$$
Let $\overline{e_i}$ be elementary matrix in $\mathrm{Mat}_{d_i}(D_i)$, then $L_i = (R/J)\overline{e_i}$ is a simple R-module. By previous lemma and induction, we can find $e_i \in R$ with $e_i^2 = e_i$ with $\overline{e_i} = e_i \bmod J$. Now we know:
$$P_i \coloneqq R \cdot e_i$$
is projective, since it's a direct summnad of free rank one module. But
$$P_i/JP_i = Re_i/Je_i = (R/J)\overline{e_i} = L_i$$

Hence $P_i \to L_i$ is an isomorphism modulo J, therefore this is an essential surjection by previous lemma and hence a projective cover. We thus constructed projective covers for simples L_i. Now for any module M, we write

$$M/JM = \bigoplus L_i$$

and set $P \coloneqq \bigoplus P_i$. Now if f_i are any lifts of $\overline{e_i} \in L_i$ in M, we have a map:

$$P \to M, \quad e_i \mapsto f_i$$

will be essential surjection.

2. For P indecomposable projective, we know $P/JP = \bigoplus L_i$. We then know:

$$\bigoplus P_i \to P \to 0$$

is the projective cover for P. But then the projective cover for P should just be P itself. But P is indecomposable, so that $P \simeq P_i$ for some i.

□

Remark. *P_L is unique up to a nonunique isomorphism. The projective cover for $M = \mathbb{C}[t]/t$ of $R = \mathbb{C}[t]/(t^2)$ is free rank 1-module R.*

Example. *Let R be an Artinian ring. General theory tells us we can write:*

$$R \simeq \bigoplus P_i^{\oplus m_i}$$

Here $m_i = \dim(L_i)$ as a vector space over $D_i = \mathrm{End}_R(L_i)$.

2 Categorical and Homological Methods

2.1 Morita Theory

Definition 2.1. We call two rings R, S Morita equivalent if there exists an equivalence of categories:

$$R\text{-mod} \simeq S\text{-mod}$$

We will write $R \sim S$.

Theorem 2.2. *$R \sim S \iff S = \mathrm{End}_R(P)^{\mathrm{op}}$ of a finitely generated projective generator P of R-mod. The equivalence is given by:*

$$M \mapsto \mathrm{Hom}(P, M)$$

Definition 2.3. A module M is a generator for R-mod if:

$$\mathrm{Hom}(M, -)$$

detects isomorphisms, i.e if I have $N_1 \to N_2$, then it is an isomorphism iff $\mathrm{Hom}(M, N_1) \xrightarrow{\sim} \mathrm{Hom}(M, N_2)$ is a bijection.

Lemma 2.4. *1. A projective module is a generator iff* $\mathrm{Hom}(P, M) \neq 0$ *for* $M \neq 0$. *We will write:*

$$P^{\perp} = \{M \mid \mathrm{Hom}(P, M) = 0\}$$

So the condition reads $P^{\perp} = 0$.

2. An arbitrary M *is a generator iff* R *is a direct summand in* $M^{\oplus n}$ *for some* n.

Proof. 1. $\Longrightarrow$ is clear. Now assume $P^{\perp} = 0$, given a non-isomorphism $\varphi : N_1 \to N_2$, I need to show $\mathrm{Hom}(P, N_1) \to \mathrm{Hom}(P, N_2)$ is not a bijection. Either $\ker(\varphi) \neq 0$, then I can find nonzero $P \to \ker \varphi \to N_1$, this will be sent to $0 \in \mathrm{Hom}(P, N_2)$. Hence $\mathrm{Hom}(P, N_1) \to \mathrm{Hom}(P, N_2)$ is not injective. On the other hand, assume $\mathrm{im}\,(\varphi) \subsetneq N_2$, we can pick $P \to N_2$ lifting a nonzero map $P \to N_2/\varphi(N_2)$, this is not in the image of $\mathrm{Hom}(P, N_1) \to \mathrm{Hom}(P, N_2)$. We're done.

2. If R is a direct summand of $M^{\oplus n}$, then

$$\mathrm{Hom}(M, N_1) \simeq \mathrm{Hom}(M, N_2) \implies \mathrm{Hom}(M^{\oplus n}, N_1) \simeq \mathrm{Hom}(M^{\oplus n}, N_2) \implies N_1 \simeq N_2$$

If M is a generator, then we can find a surjective homomorphism $M \twoheadrightarrow N$ for every N. Now specialize to $N = R$, we can find:

$$1 = \sum_{i=1}^{n} \varphi_i(m_i)$$

This provides a map:

$$M^{\oplus n} \to R$$

But R is projective, it will admit a splitting.

□

Example. *Let* $M = R^n$, *then*

$$\mathrm{End}(M)^{\mathrm{op}} = \mathrm{Mat}_n(R)$$

Hence Morita equivalence reads:

$$R\text{-mod} \simeq \mathrm{Mat}_n(R)\text{-mod}$$

Example. *Take* $P = R \cdot e$ *for* $e^2 = e$. *This is projective. Note that:*

$$\mathrm{Hom}(P, M) = eM$$

and therefore:

$$\mathrm{End}_R(P)^{\mathrm{op}} = eRe$$

this is a subring but not a sub-unital ring. Note that $P = R \cdot e$ *is a generator iff* $ReR = R$. *Indeed,* $R \cdot e$ *is a generator iff for all* $M \neq 0$, *we have* $\mathrm{Hom}(P, M) = eM \neq 0 \iff ReM = (ReR)M \neq 0$. *Therefore, if* $ReR = R$, *then obviously:*

$$(ReR)M = M$$

and it is a generator. Otherwise suppose $ReR \subsetneq R$*, then consider* $M = R/ReR \neq 0$*, it has property:*

$$\mathrm{Hom}(P, M) = eR/ReR = 0$$

So P *is not a generator. In summary if we have idemopotent* e *with* $ReR = R$*, then Morita theory tells me:*

$$R\text{-mod} \simeq eRe\text{-mod}$$

More generally, we can take $P = R^n \cdot e$ *for* $e \in \mathrm{Mat}_n(R)$ *idempotent. Then* P *is a projective generator iff* $\mathrm{Mat}_n(R)e\mathrm{Mat}_n(R) = \mathrm{Mat}_n(R)$*. In this case:*

$$\mathrm{End}_R(P)^{\mathrm{op}} = e\mathrm{Mat}_n(R)e \implies R\text{-mod} \simeq e\mathrm{Mat}_n(R)e\text{-mod}$$

Example. *Let* R *be Artinian, a projective:*

$$P = \bigoplus P_i^{\oplus d_i}$$

is a generator iff $d_i \geq 1$ *for all* i*. In particular we have a minimal choice* $P = \bigoplus P_i$*. Then by Morita equivalence we can consider:*

$$S = \mathrm{End}_R(\bigoplus P_i)^{\mathrm{op}}$$

This is a basic ring, namely every irreducible S*-module* L_i' *is one-dimensional over* $D_i = \mathrm{End}(L_i')$*. If* R *is finite dimensional over* $k = \overline{k}$*, then* $\dim_k(L_i') = 1$*.*

On the other hand, many things are Morita invariant.

Definition 2.5. Cartan matrix for an Artinian ring R is a $n \times n$ matrix C, with: $C_{ij} = [P_i : L_j]$ is the Jordan Holder multiplicities of L_j in P_i, for $L_1, \cdots, L_n$ are irreducibles and $P_1, \cdots, P_n$ are corresponding indecomposable projective modules. This is invariant under Morita equivalence.

Remark. *If* $R \sim S$*, then* $Z(R) \simeq Z(S)$*.*

Proof. (of Morita's theorem)

1. Let $S = \mathrm{End}_R(P)^{\mathrm{op}}$. We have a functor:

$$F : R\text{-mod} \to S\text{-mod}$$

$$M \mapsto \mathrm{Hom}(P, M)$$

Note that for any M, then:

$$\mathrm{Hom}_S(F(P), F(M)) = \mathrm{Hom}_S(S, F(M)) \simeq F(M) \simeq \mathrm{Hom}(P, M)$$

this is because $F(P) = S$ by definition. Now given any module N, using P is a generator, we can find:

$$P^{\oplus J} \to P^{\oplus I} \to N$$

We can then deduce

$$\mathrm{Hom}(N, M) \simeq \mathrm{Hom}(F(N), F(M))$$

In fact we will prove by:

(a) If the statement is true for N_1, N_2, then it is also true for N, where

$$N_1 \to N_2 \to N \to 0$$

Indeed we have

$$0 \to \mathrm{Hom}(N, M) \to \mathrm{Hom}(N_2, M) \to \mathrm{Hom}(N_1, M)$$

$$0 \to \mathrm{Hom}(F(N), F(M)) \to \mathrm{Hom}(F(N_2), F(M)) \to \mathrm{Hom}(F(N_1), F(M))$$

The first row is exact is clear. The second row is exact is because P is projective so that F is exact. Now the isomorphism:

$$\mathrm{Hom}(N_i, M) \simeq \mathrm{Hom}(F(N_i), F(M))$$

help me to deduce for $\mathrm{Hom}(N, M) \simeq \mathrm{Hom}(F(N), F(M))$.

(b) Now P is finitely genreated, so that F comutes with arbitrary direct sum:

$$F(P^{\oplus I}) = F(P)^{\oplus I}$$

Now that:

$$\mathrm{Hom}(P^{\oplus I}, M) = \prod_I \mathrm{Hom}(P, M) = \mathrm{Hom}(F(P)^{\oplus I}, F(M)) = \mathrm{Hom}(F(P^{\oplus I}), F(M))$$

Hence we prove fully faithfulness. Now deal with essential surjectivity. Given any $X \in S$-mod. We present:

$$\begin{array}{ccccccc} S^{\oplus J} & \longrightarrow & S^{\oplus I} & \longrightarrow & X & \longrightarrow & 0 \\ \uparrow{\scriptstyle =} & & \uparrow{\scriptstyle =} & & & & \\ F(P)^{\oplus J} & \longrightarrow & F(P)^{\oplus I} & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

where $M = P^{\oplus I}/P^{\oplus J}$. Then $X = F(M)$ will do the job since F is exact.

2. Conversely, assume $F : R$-mod $\simeq S$-mod. Let's consider $F^{-1}(S)$, where S is the free rank one module over S. First note that:

$$\mathrm{End}_R(P) = \mathrm{End}_S(S) = S^{\mathrm{op}}$$

It remains to check P is a finitely generated projective generator. In fact, since F^{-1} is an equivalence, it sends projective to projective and generators to generators. We're almost done since S is a projective generator. The remaining goal is to check P is finitely generated.

Lemma. *A projective module P is finitely generated iff* $\mathrm{Hom}(P, -)$ *commutes with arbitrary direct sum.*

To see this, note that $P \subset R^{\oplus I}$, so that $\mathrm{Hom}(P, R^{\oplus I}) = \mathrm{Hom}(P, R)^{\oplus I}$. Now that $i : P \hookrightarrow R^{\oplus I}$ corresponds to some element in $\mathrm{Hom}(P, R)^{\oplus I}$, so it must lies in $\mathrm{Hom}(P, R)^{\oplus n}$ for some finite n. This means that the image of P in $R^{\oplus I}$ must lie in $R^{\oplus n}$ for some finite n. Now assume $P \oplus Q = R^{\oplus I}$, then:

$$P \oplus (Q \cap R^{\oplus n}) = R^{\oplus n}$$

This proves P is finitely generated. The converse direction is clear since P is a direct summand of $R^{\oplus n}$ for some finite n.

From this description one sees F sends finitely generated projective to finitely generated projective. We're therefore done.

□

Remark. *The Lemma during the proof is false for general module. For example, we can take $R = \prod_S \mathbb{C}$ with S uncountable, and M be the ideal consists of those with nonzero entries at at most countable set of paces. Then clearly M is note finitely generated, but* $\mathrm{Hom}(M, -)$ *commutes with arbitrary direct sums.*

Proposition 2.6. *M is finitely generated iff $\bigcup_\alpha \mathrm{Hom}(M, N_\alpha) \xrightarrow{\sim} \mathrm{Hom}(M, N)$ iff $N = \bigcup N_\alpha$.*

Definition 2.7. The cocenter of R is $C(R) = R/\langle xy - yx \rangle$, where we quotienting out the abelian group generated by the commutators. So cocenter is merely an abelian group.

Proposition 2.8. *If $R \sim S$, then their center and cocenter are isomorphic.*

Proof. Our first claim is that:

$$Z(R) = \mathrm{End}(\mathrm{id}_{R\text{-mod}})$$

So the center can be recover from R-mod and is Morita invariant. For the second part, we need the following:

Definition 2.9. The **Trace functional** on an additive category with values in an abelian group A is a map:

$$(\varphi : M \to M) \mapsto \tau(M, \varphi) \in A$$

with the properties: for $a : M \to N, b : N \to M$, we have:

$$\tau(ab) = \tau(ba)$$

$$\tau(M \oplus N, a \oplus b) = \tau(M, a) + \tau(M, b)$$

$$\tau(a + b) = \tau(a) + \tau(b)$$

For the additive category $\mathrm{Proj}_R^{\mathrm{fin}}$ consists of finitely generated projective modules, there will be *universal trace* τ, whose abelian group A is the precisely the cocenter $C(R)$. □

A choice of an equivalence

$$F : R\text{-mod} \simeq S\text{-mod}$$

is the same as a choice of projective generator $P \in R$-mod with $\mathrm{End}_R(P)^{\mathrm{op}} \simeq S$. It has the advantage of only refering to the category structure on R-mod. But it breaks the symmetry between R and S.

Definition 2.10. By a R, S-bimodule, I mean an abelian group with a left R-action and a right S-action.

Construction: Given a bimodule B, we have a functor:

$$F_B : S\text{-mod} \to R\text{-mod}$$

$$M \mapsto B \otimes_S M$$

It has basic property:

$$F_{B \otimes_S B'} = F_B \circ F_{B'}$$

Remark. *This functor F_B is left adjoint to the functor:*

$$\mathrm{Hom}_R(B, -) : R\text{-mod} \to S\text{-mod}$$

Lemma 2.11. *The functor:*

$$(R, S)\text{-mod} \to \mathrm{Fun}(S\text{-mod}, R\text{-mod})$$

$$B \mapsto F_B$$

is fully faithful. In fact, its essential image consists of additive functors preserving arbitrary colimits.

Proof. Let P, Q be (R, S)-bimodules. Then we have a natural map:

$$\mathrm{Hom}(P, Q) \to \mathrm{Hom}(F_P, F_Q)$$

To construct the map from the other direction, note that $F_P(S) \simeq P$ as a (R, S)-bimodule. So that we can send a functor:

$$\eta : F_P \Rightarrow F_Q$$

to

$$\eta_S : F_P(S) = P \to F_Q(S) = Q$$

One check these are mutual inverse. Now let's check the second statement. It is clear that tensoring is always additive functor preserving arbitray colimits. Now let $F : S\text{-mod} \to R\text{-mod}$ be one of such functor. Note that $B \coloneqq F(S)$ is naturally a (R, S)-bimodule: the left R-module structure comes from definition, the right S-module structure comes from functoriality. So it suffices to prove that if F preserves arbitirary colimits, and $F \simeq F_B$.

1. For free S-module $S^{\oplus I}$, we have:

$$F(S^{\oplus I}) \simeq F(S)^{\oplus I} \simeq B \otimes_S S^{\oplus I} \simeq F_B(S^{\oplus I})$$

since F preserves arbitrary direct sum.

2. For an arbitrary S-module X, we choose a presentation:

$$S^{\oplus I} \to S^{\oplus J} \to X \to 0$$

But F preserves cokernel, so the following sequence is exact:

$$F(S^{\oplus I}) \to F(S^{\oplus J}) \to F(X) \to 0$$

But:

$$F_B(S^{\oplus I}) \to F_B(S^{\oplus J}) \to F_B(X) \to 0$$

is also exact with the first two terms agree. So that $F(X) \simeq F_B(X)$ as required.

□

Here is another version of Morita's theorem.

Proposition 2.12. *Let P be a finitely generated projective generator over R, and let $S = \mathrm{End}_R(P)^{\mathrm{op}}$. Then P is a generator for S^{op}-mod. Moreover, if P is faithful, then:*

$$R \simeq \mathrm{End}_{S^{\mathrm{op}}}(P)$$

Proof. Let's first prove the case $P = Re$ for some idempotent e. In this case $S = eRe$, and to check P is a generator for S^{op}-module we must see:

$$\mathrm{Hom}_{eRe}(M, N) \neq 0$$

for each $N \neq 0$, where we view M, N to be both right eRe-modules. But note that:

$$P = eRe \oplus (1 - e)Re$$

has a direct summand $S = eRe$, so of course the above condition holds. To check the next condition, let's consider:

$$R \hookrightarrow \mathrm{End}_{S^{\mathrm{op}}}(P) \hookrightarrow P$$

where the first map is given by:

$$r \mapsto (x \mapsto rx)$$

this is injective since P is faithful over R, and the second map is given by

$$\varphi \mapsto \varphi(e)$$

which is injective.

In general $P = R^n e$, with $e \in \mathrm{Mat}_n(R)$ an idempotent matrix. Now that:

$$S = e\mathrm{Mat}_n(R)e$$

is again a direct summand of $P^{\oplus n}$, and therefore P is a generator of S^{op}-module. The rest is similar to the previous part. □

For a module M over R, we define $M^\vee = \mathrm{Hom}_R(M, R)$ be a right R-module. This is not well-behaved in general. But if we restrict to finitely generated projective modules we have a good duality.

Lemma 2.13. *The functor:*

$$P \mapsto P^\vee$$

is an equivalence for

$$(\mathrm{Proj}_R^{\mathrm{fg}})^{\mathrm{op}} \simeq \mathrm{Proj}_{R^{\mathrm{op}}}^{\mathrm{fg}}$$

In fact:

$$P^{\vee\vee} = P.$$

Proof. This is clear for free modules, and then check for direct summands. □

Proof. (second proof of $C(R) \simeq C_{\mathrm{univ}}$: the cocenter of R equals the universal trance group for $\mathrm{Proj}_R^{\mathrm{fg}}$). Explicitly, given a universal trace τ, we defined:

$$\tau(P, \varphi) \in C(R), \quad P \in \mathrm{Proj}_R^{\mathrm{fg}}, \varphi \in \mathrm{End}(P)$$

Here is a construction. Consider the natural map:

$$P^\vee \otimes_R P \to \mathrm{End}(P)$$

It turns out that this is an isomorphism. On the other hand, I have:

$$P^\vee \otimes_{\mathbb{Z}} P \to R$$

the trace map. But I can write:

$$P^\vee \otimes_R P = (P^\vee \otimes_{\mathbb{Z}} P) \otimes_{R^{\mathrm{op}} \otimes R} R$$

Therefore I have a map:

$$\mathrm{End}(P) \to P^\vee \otimes_R P \to (P^\vee \otimes_{\mathbb{Z}} P) \otimes_{R^{\mathrm{op}} \otimes R} R \to R \otimes_{R^{\mathrm{op}} \otimes R} R$$

But the image is nothing but $C(R)$. □

The functor

$$F_P : M \mapsto \mathrm{Hom}(P, M) \simeq P^\vee \otimes_R M$$

sends R-mod $\to$ S-mod. We also have the functor S-mod $\to$ R-mod:

$$N \mapsto P \otimes_S N$$

Definition 2.14 (Morita Context)**.** Morita context is a pair of rings R, S, $P \in (R, S)$-bimodule, $Q \in (S, R)$-bimodule, together with:

$$\tau : P \otimes_S Q \to R$$

$$\eta : Q \otimes_R P \to S$$

compatible in the sense that the two maps:

$$P \otimes_S Q \otimes_R P \rightrightarrows P$$

coincide. And similarly for:

$$Q \otimes_R P \otimes_S Q \rightrightarrows Q$$

This is equivalent to giving an associative product on matrices:

$$\begin{pmatrix} r & p \\ q & s \end{pmatrix} \in R \oplus S \oplus P \oplus Q$$

Remark. *$R \sim S$ iff there exists a Morita context such that*

$$P \otimes_S Q \simeq R, \quad Q \otimes_R P \simeq S$$

are isomorphisms.

Example. *Let $R = k$ be a field, let $S = \mathrm{End}(V)$ for some k-vector space V. We thus have:*

$$R\text{-mod} \to S\text{-mod}, \quad W \mapsto V \otimes_k W$$

$$S\text{-mod} \to R\text{-mod}, \quad M \mapsto V^* \otimes_S M$$

Lemma 2.15. *In Morita context, then if τ is onto, then it's an isomorphism.*

Proof. If not, assume $0 \neq \sum p_i \otimes q_i P \otimes_R Q$ such that $\tau(\sum p_i \otimes q_i) = 0$. Let's pick p_i', q_i' such that $\tau(\sum p_i' \otimes q_i') = 1$. Let's consider:

$$\sum p_i \otimes q_i \otimes p_i' \otimes q_i' \in P \otimes Q \otimes P \otimes Q$$

Now I have:

$$P \otimes Q \otimes P \otimes Q \rightrightarrows P \otimes Q$$

given by $\tau \otimes 1, 1 \otimes \tau$. These two maps coincide. This is because:

$$\tau \otimes 1_{P \otimes Q} = 1_P \otimes \eta \otimes 1_Q = 1_{P \otimes Q} \otimes \tau$$

This gives a contradiction. □

Definition 2.16 (Derived Morita Context)**.** Start with the data $M \in S$-mod, and $R = \mathrm{End}(M)^{\mathrm{op}}$, $P = \mathrm{Hom}_S(M, S)$, $Q = M$. The the obvious Morita context we get from these data is called **derived Morita context**.

Remark. *This has nothing to do with derived cateogories.*

Lemma 2.17. *For a derived Morita context, the map:*

1. *$\eta : Q \otimes_R P \to S$ is onto iff M is a generator.*
2. *$\tau : P \otimes_S Q \to R$ is onto iff M is finitely generated projective.*

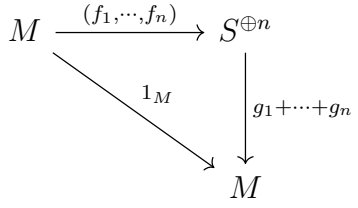
Proof. Unravel the definition.

$$Q \otimes_R P \to S \iff M \otimes_R \mathrm{Hom}_S(M,S) \to S, m \otimes f \mapsto f(m)$$

This is surjective precisely translates to M to be a generator. For the second part:

$$P \otimes_S Q = \mathrm{Hom}_S(M,S) \otimes_S M \to \mathrm{End}(M), f \otimes m \mapsto (m' \mapsto f(m')m)$$

This is onto iff $1_M = \sum g_i f_i$ for $f_i : M \to S, g_i : S \to M$. In other words, we have diagram:



This precisely translates to M is a direct summand of $S^{\oplus n}$, equivalently M is finitely generated projective. □

Now let's ask the following qustion: what is the relation between R-mod and S-mod for $S = \mathrm{End}_R(P)^{\mathrm{op}}$, for P not necessarily a generator?

In this case, we still have a functor:

$$R\text{-mod} \to S\text{-mod}$$

This is still exact, but send $P^\perp \mapsto 0$.

Proposition 2.18. *In this case R-mod$/P^\perp \simeq S$-mod is an equivalence.*

Remark. *The quotient we're taking in the proposition is* **Serre quotient***: the Serre quotient $\mathcal{A}/\mathcal{B}$ is the universal category with a functor from $\mathcal{A}$ sending $\mathcal{B}$ to 0.*

Example. *Let R be Artinian. Then:*

$$P = \bigoplus P_i$$

Then

$$P^\perp = \langle L_j \mid j \notin I \rangle$$

A consequence about the above equivalence is:

$$K(S\text{-mod}^{\mathrm{fg}}) = K(R\text{-mod}^{\mathrm{fg}})/\langle [L_j] \mid j \notin I \rangle$$

Example. *Let R be commutative noetherian, $\mathcal{A} = R$-mod^{fg}, $f \in R$, and define:*

$$\mathcal{B} = \{M \mid f^n M = 0 \text{ for some } n\}$$

Then $\mathcal{A}/\mathcal{B} = R_{(f)}$-mod.

Example. *Let $\mathcal{A} = \mathrm{Coh}(X)$, $\mathcal{B} = \mathrm{Coh}(Z)$ for some closed subscheme Z. Then:*

$$\mathcal{A}/\mathcal{B} \simeq \mathrm{Coh}(X \setminus Z)$$

Example. *Let R be a graded ring, $R_0 = k$. Let $\mathcal{A} = R$-mod$^{\mathrm{fg,gr}}$, and:*

$$\mathcal{B} = \{M \mid I^n M = 0 \text{ for some } n\}$$

where I is the irrelevant ideal. Then:

$$\mathcal{A}/\mathcal{B} = \mathrm{Coh}(\mathrm{Proj}(R))$$

2.2 Limits and Colimits

Let $\mathcal{C}$ be a category, a diagram of shape $\mathcal{D}$ is a functor $F : \mathcal{D} \to \mathcal{C}$. Let $\mathcal{C}_{\mathcal{D}}$ be all such diagrams.

Example. *Let Q be a quiver, let $\mathcal{D}$ be the vertices of Q, and $\mathrm{Hom}(v, w)$ be the collection of all paths starting at v and ending at w. Then $\mathcal{D}$. Let $\mathcal{C} = \mathrm{Vect}_k$, then $\mathcal{C}_{\mathcal{D}} = \mathrm{Rep}(Q)$.*

Note that we have an embedding functor $c : \mathcal{C} \hookrightarrow \mathcal{C}_{\mathcal{D}}$. The left adjoint of this functor is the **colimit**, the right adjoint of this functor is the **limit**.

Example. *Let $\mathcal{D}$ be two objects with no nontrivial morphisms. Then $\mathcal{C}_{\mathcal{D}} = \mathcal{C} \times \mathcal{C}$. This gives coproduct and product.*

Proposition 2.19. *Left adjoint commutes with colimits, right adjoint commutes with limits.*

Proof. Let's unpack what this means. Let $F : \mathcal{C} \to \mathcal{D}$, $G : \mathcal{D} \to \mathcal{C}$ be a pair of adjoint functors. Let $D : \mathcal{J} \to \mathcal{C}$ be a diagram of shape $\mathcal{J}$ in $\mathcal{C}$. Then

$$F \circ D : \mathcal{J} \to \mathcal{D}$$

is a diagram of shape $\mathcal{J}$ in $\mathcal{D}$. What we claim is that:

$$F(\mathrm{colim}(D)) = \mathrm{colim}(F \circ D)$$

To prove this, we use Yoneda lemma. Let $X \in \mathcal{D}$ be any object, we compute:

$$\begin{aligned}
\mathrm{Hom}_{\mathcal{D}}(F(\mathrm{colim}(D)), X) &\simeq \mathrm{Hom}_{\mathcal{C}}(\mathrm{colim}(D), G(X)) \\
&\simeq \lim_{j \in \mathcal{J}} \mathrm{Hom}_{\mathcal{C}}(D(j), G(X)) \\
&\simeq \lim_{j \in \mathcal{J}} \mathrm{Hom}_{\mathcal{D}}(F \circ D(j), X) \\
&\simeq \mathrm{Hom}_{\mathcal{D}}(\mathrm{colim}(F \circ D), X)
\end{aligned}$$

The proof for limits is similar. □

Proposition 2.20. *Pullback is a limit, pushout is a colimit.*

Proposition 2.21. *Equalizer is a limit, coequalizer is a colimit.*

2.3 Abelian Categories

First of all, let's recall:

Definition 2.22. An additive category $\mathcal{A}$ is a category where each Hom set has the structure of an abelian group such that the composition is bilinear and the following properties hold:

1. There exists $0_{\mathcal{A}} \in \mathcal{A}$ with $\mathrm{Hom}(0_{\mathcal{A}}, 0_{\mathcal{A}}) = 0$.
2. For every M_1, M_2, there exists $M_1 \oplus M_2$ with $p_i : S \to M_i$ and $\iota_i : M_i \to S$ such that $p_1\iota_2 = p_2\iota_1 = 0$, $p_1\iota_1 = \mathrm{id}_{M_1}$, $p_2\iota_2 = \mathrm{id}_{M_2}$, $\iota_1 p_1 + \iota_2 p_2 = \mathrm{id}_S$.

Definition 2.23 (Abelian Categories)**.** Let $\mathcal{A}$ be an additive category. Consider the following axioms.

1. (AB1) Kernel and cokernel of every arrow exists.
2. (AB2) We call kernel of cokernel to be the image, cokernel of the kernel to be coimage. We require they coincide. Equivalently, we require every monic to be a kernel and every epic to be a cokernel.
3. (AB3) Arbitrary coproducts exist.
4. (AB4) Exactness of coproducts.
5. (AB5) Can define subobjects and sums. We require if A_i is an increasing collection of subobjects, and B another subobject of A, we require:

$$(\sum A_i) \cap B = \sum (A_i \cap B)$$

   This is equivalent to say that taking filtered colimit is exact.

An abelian category is one satisfying (AB1)-(AB2). If the all five conditions are satisified, this is called a **Grothendieck category**.

Remark. *Note that (AB3)-(AB5) is not symmetric. We say* $(AB3^*)$ *holds if (AB3) holds for* A^{op}*. Notably,* $(AB3^*), (AB4^*)$ *holds for* R-mod *but not* $(AB5^*)$

Remark. *If (AB1)-(AB5) and* $(AB1^*)$ - $(AB5^*)$ *all hold, then* $\mathcal{A} = 0$*.*

Lemma 2.24. *Assume (AB1)-(AB3), all colimits exist.*

Proof. Given $X : \mathcal{D} \to \mathcal{C}$, we write X_i for each $i \in \mathrm{Ob}(\mathcal{D})$, and we have multiple edges $X_i \to X_j$. Let's consider:

$$\bigoplus_{e: i \to j} X_i \to \bigoplus_{i \in \mathcal{D}} X_i$$

For the edge $e : i \to j$:

$$X_i \to X_i \oplus X_j$$

is given by $\mathrm{id}_{X_i} - X(e)$. Then the cokernel will be the colimit of this diagram X. □

We call a poset **filtered** if for each i, i', there exists some j which receives an arrow from both i, i'. More generally, we call $\mathcal{D}$ **filtered** if for all i, i' there exists j receiving arrows from i, i'; and for all $e, e' : i \to j$, there exists $j \xrightarrow{f} j'$ such that $f \cdot e = f \cdot e'$.

Proposition 2.25. *Assume (AB1)-(AB5), taking filtered colimit is exact.*

Remark. *Every colimit is right exact.*

Our first proof of Morita's theorem also proves a more general theorem. Let $\mathcal{A}$ be an abelian category, we call P a projective generator if P is projective and:

$$P^{\perp} = \{M \mid \mathrm{Hom}(P, M) = 0\} = 0$$

Theorem 2.26. *Let $\mathcal{A}$ be an abelian category, and P is a projective generator such that* $\mathrm{Hom}(P,-)$ *commutes with arbitrary direct sum. Then:*

$$F : \mathcal{A} \xrightarrow{\sim} \mathrm{End}(P)^{\mathrm{op}}\text{-mod}$$

$$X \mapsto \mathrm{Hom}(P, X)$$

is an equivalence.

Remark. *Let P be a finitely generated projective object in an abelian category. Then* $\mathrm{Hom}(P,-)$ *commutes with filtered colimits.*

Definition 2.27. An object M is called **compact** if it commutes with filtered colimits.

Remark. *An element $M \in R$-*mod *is compact iff it's finitely presented. Notably a finitely generated projective module is finitely presented.*

Theorem 2.28 (Mitchell-Freyd Embedding)**.** *Any (small) abelian category admits an exact, fully faithful functor to R-*mod *for some R.*

2.4 Serre Quotient

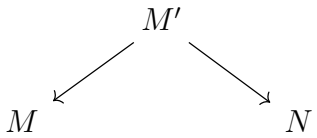
Let $\mathcal{A}$ be an abelian category, $\mathcal{B}$ a Serre subcategory(a full subcategory closed uner subquotients and extensions). We wish to construct a quotient $\mathcal{A} \to \mathcal{B}$, such that for any abelian category $\mathcal{C}$, we have:

$$\mathrm{Fun}(\mathcal{A}/\mathcal{B}, \mathcal{C}) \simeq \mathrm{Fun}(\mathcal{A}, \mathcal{C})_{\mathcal{B}\to 0}$$

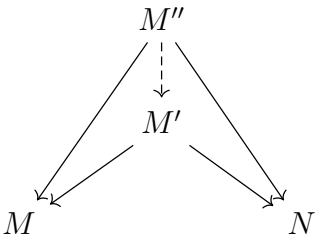
Construction: Let's set $\mathrm{Ob}(\mathcal{A}/\mathcal{B}) = \mathrm{Ob}(\mathcal{A})$. The morphisms

$$\mathrm{Hom}_{\mathcal{A}/\mathcal{B}}(M, N)$$

are in bijection with roof diagrams:



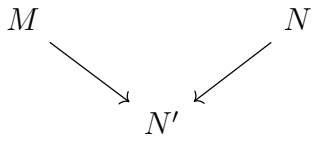
such that $M' \to M$ is mod $\mathcal{B}$-equivalent: namely kernel and cokernel lies in $\mathcal{B}$. We also need to modulo the equivalence relation where we can finding the dashed map:



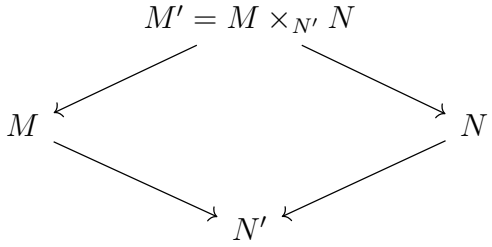
Another way to phrase this is:

$$\mathrm{Hom}_{\mathcal{A}/\mathcal{B}}(M, N) = \mathrm{colim}_{M'\to M}\mathrm{Hom}(M', N)$$

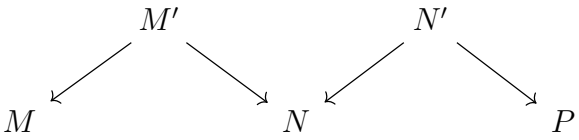
Remark. *One can also construct the morphisms* $\mathrm{Hom}_{\mathcal{A}/\mathcal{B}}$ *via "lower roofs":*



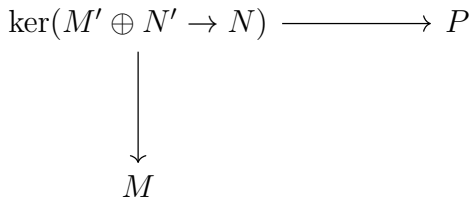
where $N \to N'$ *is an isomorphism modulo* B*. Note that we can reconstruct a upper roof by considering:*



The composition of two roof diagram



is given by diagram:



Proposition 2.29. *Let* X *be projective,* $X \in {}^{\perp}\mathcal{B}$ *(meaning that* $\mathrm{Hom}(X, N) = 0$ *for all* $N \in B$*); or* $Y \in \mathcal{B}^{\perp}$ *be injective. Then we always have:*

$$\mathrm{Hom}_{\mathcal{A}}(X, Y) = \mathrm{Hom}_{\mathcal{A}/\mathcal{B}}(X, Y) \quad (*)$$

We shall prove:

Lemma 2.30. *(*) holds if any of the following holds:*

1. $X \in {}^{\perp}\mathcal{B}, Y \in \mathcal{B}^{\perp}$.
2. $X \in {}^{\perp}\mathcal{B}$ *and* $\mathrm{Ext}^1(X, B) = 0$ *for all* $B \in \mathcal{B}$
3. $Y \in \mathcal{B}^{\perp}, \mathrm{Ext}^1(B, Y) = 0$ *for all* $B \in \mathcal{B}$.

The proof will be considering all possible roof diagrams. This in particular implies the above proposition.

Proposition 2.31. *Let $\mathcal{A} = R$-mod, P a finitely generated projective module. Let $\mathcal{B} = P^{\perp}$. Then:*

$$\mathcal{A}/\mathcal{B} \simeq \mathrm{End}(P)^{\mathrm{op}}\text{-mod}$$

Proof. It suffices to check P is a projective generator for $\mathcal{A}/\mathcal{B}$ and $\mathrm{Hom}_{\mathcal{A}/\mathcal{B}}(P, -)$ commutes with arbitrary coproduct, the result follows since we know:

$$\mathrm{End}_{\mathcal{A}/\mathcal{B}}(P) = \mathrm{End}_{\mathcal{A}}(P)$$

by the previous proposition. □

Remark. *This can be used to study double centralizer properties.*

2.5 Ext and Tor

Assume we have a functor $F : \mathcal{A} \to \mathcal{B}$ which are only exact on one side.

Example. *Tensor product is right exact,* $\mathrm{Hom}(M, N)$ *is right exact in M and left exact in N. More precisely, since* $\mathrm{Hom}(-, N)$ *is a contravariant functor, we're claiming that if:*

$$M_1 \to M_2 \to 0$$

then:

$$0 \to \mathrm{Hom}(M_2, N) \to \mathrm{Hom}(M_1, N)$$

Example. *Colimit is a left adjoint, so is right exact; limit is a right adjoint, so is left exact.*

For a left exact functor, we can define right derived functor $R^i F$; for a right exact functor, we can define left derived functor $L_i F$. These are the universal δ functors.

Theorem 2.32 (Grothendieck)**.** *Let $F : \mathcal{A} \to \mathcal{B}$ with $\mathcal{B} = R$-mod. A δ-functor is universal iff it's effacing: for all $x \in F^i(M)$, there exists $\varphi : M \hookrightarrow M'$ such that $F^i(M) \to F^i(M')$ with $x \mapsto 0$.*

Let's now construct the left derived functors for right exact functors (right derived functors are similar). Let $\mathcal{A}$ be an abelian category with enough projectives. We can thus consider $\mathcal{H}_0(\mathcal{A})$ the homotopy category for $\mathcal{A}$: this consists of complexes with terms in $\mathcal{A}$, with morphisms:

$$H^0(\mathcal{H}om^{\bullet}(X, Y))$$

where $\mathcal{H}om^{\bullet}(X, Y)$ is the chain complex whose j-th term is given by:

$$\prod_{p \in \mathbb{Z}} \mathrm{Hom}(X^p, Y^{p+j})$$

Now that by considering projective resolution we have a fully faithful embedding:

$$\mathcal{A} \to \mathcal{H}_0(\mathcal{A}), \quad M \mapsto P_M^{\bullet}$$

We can thus define:

$$L_i F(M) \coloneqq H^{-i}(F(P_M^{\bullet}))$$

We must check this is a universal δ functor.

Remark. *For the category R-mod, we have enough injectives. To see this, we can us dualtiy. Consider:*

$$R^{\mathrm{op}}\text{-mod} \to R\text{-mod}$$

$$M \to M^* = \mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z})$$

Then R^* will be an injective cogenerator for R-mod.

But for the category of finitely generated R-modules over a noetherian ring R, it has enough injectives but maybe not have enough injectives.

Example. *For $R = \mathbb{Z}$, we have injective: $\mathbb{Z}_{(p)}/\mathbb{Z}$ and $\mathbb{Z}/p \hookrightarrow \mathbb{Z}_{(p)}/\mathbb{Z}$.*

Example. *Let $R = \mathbb{C}[t]$, $I = \frac{\mathbb{C}[t,t^{-1}]}{t\mathbb{C}[t]}$, then:*

$$\mathbb{C}[t]/t \hookrightarrow I$$

Example.

$$\mathrm{Ext}^i_R(M, N) = R^i\mathrm{Hom}(M, -)(N) = L_i\mathrm{Hom}(-, N)(M)$$

Namely: take projective resolution on the first entry, take injective resolution on the second entry (sheaf cohomology corresponds to the second case). More explicilty:

$$\mathrm{Ext}^i_R(M, N) = H^i(\mathrm{Hom}^\bullet(P^\bullet_M, N)) = H^i(\mathrm{Hom}^\bullet(M, I^\bullet_N))$$

Example. *Similarly, for $M \in R^{\mathrm{op}}$-mod, $N \in R$-mod:*

$$\mathrm{Tor}^R_i(M, N) = L_i(M \otimes_R -)(N) = L_i(- \otimes_R N)(M)$$

Again we should they equal by comparing both with:

$$H^{-i}(P^\bullet_M \otimes_R P^\bullet_N)$$

In fact, given $\mathcal{A}$ an abelian category, we will define its derived category $D(\mathcal{A})$ and left (resp. right) exact functor $F : \mathcal{A} \to \mathcal{B}$ will be upgraded to:

$$RF : D(\mathcal{A}) \to D(\mathcal{B}) \text{ or } LF : D(\mathcal{A}) \to D(\mathcal{B})$$

A feature of derived category is that we have:

$$\mathcal{H}^-_0(\mathrm{Proj}_{\mathcal{A}}) \hookrightarrow D(\mathcal{A})$$

and

$$\mathcal{H}^+_0(\mathrm{Inj}_{\mathcal{A}}) \hookrightarrow D(\mathcal{A})$$

we will set these up more carefully in the future.

Now the Ext functor enables us to define cohomology theory in a lot of settings:

Example. *If R comes with augmentation (trivial module) L_0, then:*

$$H^i_R(M) \coloneqq \mathrm{Ext}^i_R(L_0, M)$$

For example, I can take $R = \mathbb{Z}[G]$, and $L_0 = \mathbb{Z}$, then I have the notion of **group cohomology**.

Example. *$H^1(G, M)$ is the set of M-torsors over G. More precisely, these are sets T with a free transitive M-action with a compatible action of G:*

$$g \cdot (m \cdot t) = g(m) \cdot gt$$

Indeed, $H^1(G, M)$ classifies short exact sequence of groups:

$$0 \to M \to \widetilde{M} \xrightarrow{\pi} \mathbb{Z} \to 0$$

Then $\pi^{-1}(1) \subset \widetilde{M}$ will be our torsor T.

Example. *Let $\mathfrak{g}$ be a Lie algebra, with enveloping algebra $A = U\mathfrak{g}$. It has trivial module k with trivial $\mathfrak{g}$-action. Then we can define the* **Lie algebra cohomology***:*

$$H^i(\mathfrak{g}, M) \coloneqq \operatorname{Ext}^i_{U\mathfrak{g}}(k, M)$$

Let A be an algebra over field k. View A as a $A \otimes A^{\mathrm{op}}$-module.

Definition 2.33. The **Hochschild cohomology** is defined to be:

$$\mathrm{HH}^i(A) \coloneqq \operatorname{Ext}^i_{A \otimes A^{\mathrm{op}}}(A, A)$$

The **Hochschild homology** is:

$$\mathrm{HH}_i(A) \coloneqq \operatorname{Tor}_i^{A \otimes A^{\mathrm{op}}}(A, A)$$

Remark. $\mathrm{HH}^0(A) = \operatorname{Hom}_{A \otimes A^{\mathrm{op}}}(A, A) = Z(A)$, $\mathrm{HH}_0(A) = C(A)$ *the cocenter of A.*

Remark. *There is a canonical map:*

$$\mathrm{HH}^i(A) \to \operatorname{Ext}^i(M, M)$$

Start with the equality $A \otimes_A M \simeq M$, we obtain:

$$\mathrm{HH}^i(A) = \operatorname{Hom}_{\mathcal{H}_0}(P_A, P_A[i]) \to \operatorname{Hom}_{\mathcal{H}_0}(P_A \otimes_A M, P_A \otimes_A M[i]) = \operatorname{Ext}^i(M, M)$$

2.6 Homological Dimension

Definition 2.34. We call $\mathcal{A}$ has **finite homological dimension** if for $n \gg 0$, then $\operatorname{Ext}^n_{\mathcal{A}}(X, Y) = 0$ for all $X, Y \in \mathcal{A}$. In this case, we will denoteminimal n is called $\operatorname{hdim}(\mathcal{A})$.

Example. *Let A be finite-dimensional algebra over k with finite homological dimension. Then we can define a pairing:*

$$K(A\text{-mod}) \times K(A\text{-mod}) \to \mathbb{Z}$$

$$[M] \times [N] \mapsto \sum_i (-1)^i \dim_k \operatorname{Ext}^i(M, N)$$

Note that $K(A$-mod$)$ has a basis $[L_i]$ of irreducibles, it has dual basis $[P_i]$ consisting of corresponding projective covers. Then:

$$\langle [P_i], [L_j] \rangle = \delta_{ij}$$

It is known that $[P_i]$ freely generate the lattice K. The Cartan matrix $C \in \mathrm{GL}_n(\mathbb{Z})$ has derterminant ± 1 (of course), it is an open conjecture that the Cartan matrix always has determinant 1.

Definition 2.35 (Projective Dimension)**.** Given a module M, we define the **projective dimension** $\mathrm{pdim}(M)$ to be the minimal length of a projective resolution of M. In particular $\mathrm{pdim}(P) = 0$ for projective P.

Lemma 2.36. $\mathrm{pdim}(M) = \max\{i \mid \mathrm{Ext}^i(M, N) \neq 0 \text{ for some } N\}$

Proof. LHS is zero iff RHS is zero. Indeed, given any M with RHS 0. Choose:

$$0 \to K \to P \to M \to 0$$

Note that $\mathrm{Ext}^1(M, K) = 0 \implies$ M is a direct summand of P so is projective. In general, we can induct on the RHS. □

Likewise, we can define the **injective dimension**:

$$\mathrm{idim}(M) = \max\{i \mid \mathrm{Ext}^i(N, M) \neq 0 \text{ for some } N\}$$

this is the minimal length of an injective resolution.

Lemma 2.37. *We have the equality:*

$$\mathrm{hdim}(R) = \max\{\mathrm{pdim}(M) \mid M \in R\text{-mod}\} = \max\{\mathrm{idim}(M) \mid M \in R\text{-mod}\}$$

Proof. This is tautological. □

We have the following:

Proposition 2.38 (Baer's criterion)**.** *M is injective iff $\mathrm{Ext}^1(R/I, M) = 0$ for any ideal I.*

Corollary 2.39. *If R is noetherian, then:*

$$\mathrm{hdim}(R) = \max\{\mathrm{pdim}(M) \mid M \text{ finitely generated}\}$$

If R is Artinian, then

$$\mathrm{hdim}(R) = \max\{\mathrm{pdim}(L) \mid L \text{ irreducible}\}$$

Proof. For the first claim, we will actually prove:

$$\mathrm{hdim}(R) = \max\{\mathrm{pdim}(R/I) \mid I \text{ left ideal}\}$$

Obviously $n = RHS \leq LHS$. We will show that $\mathrm{pdim}(M) \leq n$ for every module M. Let's take a partial injective resolution of length n:

$$0 \to M \to E_0 \to E_1 \to \cdots \to E_{n-1} \to K \to 0$$

We must show K is actually injective. Indeed:

$$\mathrm{Ext}^1(R/I, K) = \mathrm{Ext}^{n+1}(R/I, M) = 0$$

We're therefore done by Baer's theorem.

The second claim follows from the following fact. If I have short exact sequence $0 \to M \to N \to U \to 0$, then $\mathrm{pdim}(N) \leq \max(\mathrm{pdim}(M), \mathrm{pdim}(U))$. □

Let A be finite dimensinoal over k with finite homological dimension. Assume $k = \overline{k}$, then recall:

$$\mathrm{Hom}(P_i, L_j) = k^{\delta_{ij}}$$

for irreducible L_i and their projective covers P_i. The Cartan matrix $C_{ij} = \dim_k(\mathrm{Hom}(P_i, P_j))$

Recall that $K(A\text{-mod}_{\mathrm{fg}})$ is freely generated by $[L_i]$. Then K comes with a pairing:

$$\langle [M], [N] \rangle = \sum (-1)^i \dim_k \mathrm{Ext}^i(M, N)$$

Then P_i forms the dual basis of K to $[L_i]$ in this pairing. If we write:

$$C_{ij} = \langle [P_i], [P_j] \rangle, E_{ij} = \langle [L_i], [L_j] \rangle$$

then

$$E = C^{-1}$$

Example. *Let $A = k[t]/t^2$, $L_1 = k$, $P_1 = A$. Then: $C = (2) \notin \mathrm{GL}_1(\mathbb{Z})$. So tat A has inifinite homological dimension. In fact, using the resolution:*

$$\cdots \to A \xrightarrow{t} A \xrightarrow{t} A \to 0$$

we conclude: $\mathrm{Ext}^i(k, k) = k$ for all i.

Example. *Let $A \subset \mathrm{Mat}_n(k)$ be the upper triangular matrices. Let's try to compute the homological dimension of A. We have radical $J \subset A$ which are zero on the diagonal. So that:*

$$0 \to J \to A \to \prod_{i=1}^{n} k \to 0$$

We get irreducibles $L_1, \cdots, L_n$, where L_i corresponds to the matrix with the only entry at place (i, i). Let P_i be the projective cover for L_i (the i-th column of A), then we have:

$$0 \to P_{i-1} \to P_i \to L_i \to 0$$

It follows that $\mathrm{pdim}(L_i) = 2$ for each $i \geq 2$. Since A is Artinian, we conclude:

$$\mathrm{hdim}(A) = 1$$

2.7 Minimal Resolution

Let A be Artinian, M a module. Recall M has a projective cover $P \to M \to 0$ with $P/JM \xrightarrow{\sim} M/JM$. This is unique up to isomorphism. Now we use this to build **minimal resolution**:

$$\cdots \to P_2 \to P_1 \to P_0 \to M \to 0$$

where P_0 is the projective cover of M, and P_1 the projective cover of $\ker(P_0 \to M)$, P_2 the projective cover of $\ker(P_1 \to P_0)$ etc. This is a projective resolution such that:

$$P_i \to C_{i-1} = \ker(P_{i-1} \xrightarrow{d} P_{i-2})$$

induces isomorphism $P_i/JP_i \simeq C_i/JC_i$. So passing to quotient, we get:

$$\cdots \to P_i/JP_i \to P_{i-1}/JP_{i-1} \to \cdots$$

This is a complex with zero differential. It follows that

$$\mathcal{H}om(P_\bullet, L_i)$$

is a complex with zero differential, for $P_\bullet$ the minimal projective resolution.

Corollary 2.40. *Let $P_\bullet \to M$ be the minimal resolution of M. Then in the Krull-Schmidt decomposition of P_j, the multiplicity of Q_i (the projective cover of L_i) is precisely:*

$$\dim_{\mathrm{End}(L_i)}(\mathrm{Ext}^j(M, L_i))$$

Assuming A has finite homological dimension, we take projective resolution for each L_i:

$$0 \to P_{n,i} \to \cdots \to P_{0,i} \to 0$$

Then:

$$[L_i] = \sum (-1)^k [P_{k,i}] \in K(A\text{-mod})$$

By the above corollary, we know:

$$P_{k,i} = \sum Q_j^{\oplus \dim \mathrm{Ext}^k(L_i, L_j)}$$

This shows $C = E^{-1}$.

Another setting where we can form minimal resolution is for graded rings. Let $A = \bigoplus_{i \geq 0} A_i$ be a graded k-algebra. Let's assume A_0 semisimple and $\dim_k A_i < \infty$. Consider graded modules:

$$M = \bigoplus_{i \geq N} M_i, \quad \dim M_i < \infty$$

For such M we can form an A_0-module:

$$\overline{M} = M/A_+M$$

Lemma 2.41. *Let $N \to M$ be a morphism of graded modules. Then $N \to M \to 0$ iff $\overline{N} \to \overline{M} \to 0$.*

Now each irreducible A_0-module L_i has projective cover $P_i = Ae_i$. With $0 \to A_+P_i \to P_i \to L_i \to 0$. Also we have projective cover of any M. Indeed we write:

$$\overline{M} = \bigoplus L_{i_k}(m_k)$$

This upgrades to

$$\bigoplus P_{i_k}(m_k) \to M$$

We can therefore construct minimal projective resolution as before.

In this case we can consider the **Poincaré series**. For $A_0 = k$, the Poincaré series is:

$$f_M(t) = \sum_i \dim_k(M_i)t^i \in \mathbb{Z}((t))$$

If A_0 is semisimple with n irreducibles, we get a vector in $\mathbb{Z}((t))^{\oplus n}$. If $P_\bullet \to M$ is the minimal resolution, then we know:

$$f_M(t) = \sum (-1)^i f_{P_i}(t)$$

Moreover, we can still define the Cartan matrices.

Lemma 2.42. *Let R be an Artinian ring with Jacobson radical J. Let M be a R-module.*

1. *A projective resolution $P_\bullet$ is minimal iff differential is zero modulo $J \subset R$.*
2. *Any projective resolution contains the minimal one as a summand.*

Proof. Suppose $P_\bullet$ has zero differential module J. Note we have a map from the given resolution $P_\bullet$ to the minimal one $P_\bullet^{\min}$ and $P_\bullet^{\min} \to P_\bullet$. Both induces isomorphism $\overline{P}_i \simeq \overline{P}_i^{\min}$

If I have any resolution $P_\bullet$, look at $P_\bullet \to P_\bullet^{\min}$. We have induced surjection:

$$\overline{P}_i \twoheadrightarrow \overline{P}_i^{\min}$$

for $\overline{P}_i$ a complex of semisimple modules. Therefore:

$$P_i \twoheadrightarrow P_i^{\min}$$

This realizes $P_i^{\min}$ as a direct summand. □

Remark. *Let $A = \bigoplus_{i\geq 0} A_i$ with A_0 semisimple. If $M = \bigoplus_{i\geq 0} M_i$, then in the minimal resolution P_i has generators in degree $\geq i$. This is becasue $P \twoheadrightarrow M$ is an isomorphism in the bottom degree.*

3 More Advanced Topics

3.1 Koszul Duality

Assume $A_0 = k$ is a field, and each A_i is finite dimensional. Then:

$$\mathrm{Tor}_i^A(k,k) = \bigoplus \mathrm{Tor}_{i,j}(k,k)$$

has grading coming from the grading on A. Then $\mathrm{Tor}_{i,j} = 0$ for $j < i$. Moreover:

$$\mathrm{Ext}_A^i(k,k) = (\mathrm{Tor}_i^A(k,k))^* = \prod_{i,j} \mathrm{Tor}_{i,j}(k,k)^*$$

We may also denote:

$$\mathrm{Ext}^{i,j} = \mathrm{Tor}^*_{i,-j}$$

Definition 3.1. We call a graded algebra A **Koszul** if $\mathrm{Tor}_{i,j}(k,k) = 0$ for $i \neq j$.

Definition 3.2. We call a graded algebra A **quadratic** if it is generated by $A_1 = V$ and the ideal of relations has generators in degree 2. In this case, we can write:

$$A = k\langle V\rangle/(I), \quad I \subset V \otimes V$$

So A is given by $V = k^n$ and a subspace $I \subset V \otimes V$.

For any graded algebra, we can form its **quadratic part** A_q given by data $(V = A_1, I = \ker(A_1 \otimes A_1 \to A_2))$. So that we have $A_q \to A$ is an isomorphism in degree ≤ 2, and is the universal one.

For a quadratic algebra A, its **quadratic dual** $A^!$ is given by the data:

$$(V^! = V^*, I^! = I^\perp \subset V^* \otimes V^* = (V \otimes V)^*)$$

It is clear that $(A^!)^! = A$.

Theorem 3.3. *Let A be any graded algebra. Consider the graded algebra: $\mathrm{Ext}^\bullet_A(k,k)$. It always receives a map:*

$$A^!_q \to \mathrm{Ext}^\bullet_A(k,k)$$

inducing an isomorphism:

$$(A^!_q)_i \xrightarrow{\sim} \mathrm{Ext}^{i,-i}_A(k,k)$$

Corollary 3.4. *If A is Koszul, then $A^!_q \simeq \mathrm{Ext}^\bullet_A(k,k)$*

Proposition 3.5. *If A is Koszul, then A is quadratic.*

Theorem 3.6. *If A is quadratic, then A is Koszul iff $A^!$ is Koszul.*

Example. *Let $A = TV$ be the free algebra, and this is Koszul. Its quadratic dual is:*

$$(TV)^! = k\langle V^*\rangle/(V^* \otimes V^*) = k \oplus V^*$$

Assume $x_1, \cdots, x_n$ is a basis for V. We have a resolution for k as A-module:

$$0 \to A^{\oplus n} \to A \to k \to 0$$

$$(a_1, \cdots, a_n) \mapsto \sum a_i x_i$$

So that:

$$\mathrm{Ext}^i(k,k) = \begin{cases} k, & \text{if } i = 0; \\ V^*, & \text{if } i = 1; \\ 0, & \text{otherwise.} \end{cases}$$

This shows $\mathrm{Ext}^\bullet_A(k,k) = A^!$.

Example. *If $A = \mathrm{Sym}^\bullet(V)$ so that $I = \bigwedge^2 V$. Then we know $I^! = \mathrm{Sym}^2 V^*$ and $A^! = \bigwedge^\bullet(V^*)$*

Lemma 3.7. *If A is Koszul, then we have:*

$$f_A(t)f_{A^!}(-t) = 1$$

where f_A is the Poincaré series for algebra A.

Example. *If $A = TV$, then $f_A(t) = \frac{1}{1-nt}$. We computed $A^! = k \oplus V^*$, so that $f_{A^!}(-t) = 1 - nt$.*

Example. *If $A = \mathrm{Sym}^\bullet(V)$, then $f_A(t) = \frac{1}{(1-t)^n}$; and $A^! = \bigwedge^\bullet(V^*)$, so $f_{A^!}(-t) = (1-t)^n$.*

Proof. Last time we observed if P_i is a free resolution of M, then:

$$f_M(t) = \sum (-1)^i (\sum d_{i,m} t^m) f_A(t)$$

where $d_{i,m}$ is the number of generators for P_i in degree m. If $M = k$. then $f_M(t) = 1$. For the minimal resolution, we have:

$$d_{i,m} = \dim \mathrm{Tor}^A_{i,m}(k,k)$$

If A is Koszul, then only nonzero Tor is at $i = m$, and $\mathrm{Tor}^A_{m,m}(k,k) = (A^!_m)^*$. So RHS reads:

$$\sum (-1)^i (\dim(A^!_i) t^i) f_A(t) = f_{A^!}(-t) f_A(t)$$

□

Proof. (of Koszul implies quadratic). Let's first check A is generated in degree 1, which is equivalent to check $J = A_{>0}$ is generateed in degree 1. Look at:

$$0 \to J \to A \to k \to 0$$

Tensor with k, we get:

$$0 \to \mathrm{Tor}_1(k,k) \to k \otimes_A J \to k \xrightarrow{\sim} k \to 0$$

So: $\mathrm{Tor}_1(k,k) \simeq k \otimes_A J$. Since we know $\mathrm{Tor}_1(k,k)$ is concentrated in degree 1. By graded Nakayama, we know J is generated by A_1. To see the relations are in degree 2, consider $V = A_1$, and:

$$0 \to K \to V \otimes A \to J \to 0$$

Tensor with k:

$$0 \to \mathrm{Tor}_1(k,J) \to k \otimes K \to V \to k \otimes_A J \to 0$$

The last map is an isomorphism by the previous part. So that $\mathrm{Tor}_1(k,J) \simeq k \otimes K$. Moreover, the short exact sequence:

$$0 \to J \to A \to k \to 0$$

tells me $\mathrm{Tor}_2(k,k) \simeq \mathrm{Tor}_1(k,J)$, it follows that $k \otimes K = \mathrm{Tor}_2(k,k)$ is generated in degree 2. We are therefore done. □

Let's now introduce Koszul complex. Start from $A = TV/I$ a quadratic algebra, we have $A_n = V^{\otimes n}/\langle I\rangle_n$. Explicitly:

$$\langle I\rangle_n = I \otimes V^{\otimes n-2} + V \otimes I \otimes V^{\otimes n-3} + \cdots + V^{\otimes n-2} \otimes I$$

Then:

$$A_n^! = (V^*)^{\otimes n}/(I^\perp \otimes V^{*\otimes n-2} + \cdots + V^{*\otimes n-2} \otimes I^\perp) = (V^*)^{\otimes n}/(I \otimes V^{\otimes n-2} \cap \cdots \cap V^{\otimes n-2} \otimes I)^\perp$$

Hence:

$$(A_n^!)^* = I \otimes V^{\otimes n-2} \cap \cdots \cap V^{\otimes n-2} \otimes I$$

Definition 3.8. The **Koszul complex** is the following complex:

$$\cdots \to (A_i^!)^* \otimes A \to (A_{i-1}^!)^* \otimes A \to \cdots$$

$$(v_1 \otimes \cdots \otimes v_i) \otimes a \mapsto (v_1 \otimes \cdots \otimes v_{i-1}) \otimes (v_i a)$$

Theorem 3.9. *For any quadratic algebra A, the following are equivalent:*

1. *A is Koszul.*
2. *Koszul complex is a resolution of k.*
3. *For all n, the $(n-1)$ subspaces $V^{\otimes i} \otimes I \otimes V^{n-i-2}$ in $V^{\otimes n}$ form a distributive lattice.*

Example. *Let $A = k\langle V\rangle/(I)$ be a quadratic algebra. Assume $I \subset \bigwedge^2 V$, i.e. generating relations of form:*

$$\sum [x_i, y_i] = 0$$

So can consider Lie algebra L generated by V with the same relations (this is a Lie algebra with $\mathbb{Z}/2$-grading generated in degree 1), so that $A = U(L)$. Now that:

$$I^\perp \supset S^2 V^* \implies xy + yx = 0, \forall x, y \in V^*$$

So $A^!$ is super-commutative.

Conversely, if $A^!$ is commutative, then $I^\perp \supset \bigwedge^2 V^* \implies I \subset S^2 V$. This implies A is enveloping of a super Lie algebra:

$$\{a, b\} \coloneqq ab - (-1)^{\deg a \cdot \deg b} ba$$

Example. *Let $\mathfrak{g}$ be any finite dimensional Lie algebra. Consider:*

$$V = \mathfrak{g} \oplus k \cdot \hbar$$

Consider:

$$A = U_\hbar(\mathfrak{g}) = k\langle V\rangle/I$$

for

$$I = \langle [x, y] = \hbar[x, y], [h, x] = 0 \mid x, y \in \mathfrak{g}\rangle$$

Then:

$$A/\hbar = \mathrm{Sym}(\mathfrak{g}), \quad A/(\hbar - 1) = U\mathfrak{g}$$

Fact: *This A is a Koszul algebra. We wish to compute $A^!$:*

$$V^* = \mathfrak{g}^* \oplus k\partial, \quad I^\perp = \langle \partial^2, \xi_1\xi_2 = -\xi_2\xi_1, \partial\xi = \xi\partial = d(\xi) \in \bigwedge^2 \mathfrak{g}^* \rangle$$

where $d : \mathfrak{g}^* \to \bigwedge^2 \mathfrak{g}^*$ dual to $\bigwedge^2 \mathfrak{g} \to \mathfrak{g}$. This is close to **Chevalley complex**:

$$C(\mathfrak{g}) = \mathfrak{g}^* \xrightarrow{d} \bigwedge^2 \mathfrak{g}^* \to \bigwedge^3 \mathfrak{g}^* \to \cdots$$

with later differentials comes from Leibniz rule. Then:

$$H^*(C(\mathfrak{g})) = H^*(\mathfrak{g}) = \mathrm{Ext}^\bullet_{U\mathfrak{g}}(k, k)$$

In summary:

$$A^! = C(\mathfrak{g}) \oplus \partial C(\mathfrak{g})$$

where for $\alpha \in C^i(\mathfrak{g})$, $d(\alpha) = \{\partial, \alpha\}$

Our next goal is to prove Theorem 3.3. To do this we need to introduce **Bar resolution**.
Construction: Let A be an algebra over field k. Let's consider:

$$\cdots \to A \otimes A \otimes A \to A \otimes A \to A \to 0$$

For example the map between $A \otimes A \otimes A \to A \otimes A$

$$a \otimes b \otimes c \mapsto ab \otimes c - a \otimes bc$$

It's easy to imagine the general formula. This is a complex follows from the fact that A is associative.

Proposition 3.10. *This complex is exact, if A is unital.*

Proof. Let consider $h = a_0 \otimes \cdots \otimes a_n \mapsto 1 \otimes a_0 \otimes \cdots a_n$, then $\mathrm{id} = hd + dh$, so that the complex is homotopic to 0. □

Note that d is a map of A-bimodules. Therefore:

$$\cdots \to A \otimes A \otimes A \to A \otimes A \to 0$$

is a free resolution of the regular A-bimodule. Now assume A is augmented, i.e. fixing:

$$0 \to I \to A \to k \to 0$$

We have the **reduced bar complex**:

$$\overline{\mathrm{Bar}}(A) = \cdots \to A \otimes I \otimes \cdots \otimes I \otimes A \to \cdots \to A \otimes A \to 0$$

Lemma 3.11. $\mathrm{Bar}(A) = \overline{\mathrm{Bar}}(A) \oplus \mathrm{Bar}_1(A)$ *as complexes, for* $\mathrm{Bar}_1(A)$ *spanned of* $a_0 \cdots \otimes 1 \otimes \cdots \otimes a_n$.

Proof. This is a decomposition of vector spaces is clear. It is an easy computation to show both are actually subcomplexes. □

As a consequence, the reduced bar complex is again a free resolution for the regular bimodules. Now given an A-module M, then $\overline{\mathrm{Bar}} \otimes_A M$ is a resolution of M as an A-module:

$$\cdots \to A \otimes I \otimes I \otimes M \to A \otimes_k I \otimes_k M \to A \otimes_k M \to 0$$

Apply to the case when A is augmented and $M = k$, we get the Bar resolution for k:

$$\cdots \to A \otimes I \otimes I \to A \otimes I \to A \to 0$$

This is a free resolution of k over A. So $\mathrm{Tor}^A_\bullet(k,k)$ is computed by:

$$\cdots \to I \otimes I \to I \to k \to 0$$

and $\mathrm{Ext}^\bullet_A(k,k)$ is computed by:

$$0 \to k \to I^* \to I^* \otimes I^* \to \cdots$$

Proof. (of theorem 3.3) Let $A = \bigoplus_{i \geq 0} A_i$ as in the theorem, so that $I = A_1 \oplus A_2 \oplus \cdots$. Let's denote $V = A_1$. We have:

$$\mathrm{Ext}^n(k,k) = \frac{\ker((I^*)^{\otimes n} \to (I^*)^{\otimes n+1})}{\mathrm{im}\ ((I^*)^{\otimes n-1} \to (I^*)^{\otimes n})}$$

Note that the degree $-n$ part of the numerator is:

$$\ker((A_1^*)^{\otimes n} \to (I_1^*)^{\otimes n+1}) = (V^*)^{\otimes n}$$

This is because $(A_1^*)^n \to I_1^{\oplus n+1}$ is the zero map. The degree $-n$ part of the denominator is $d((I^*)^{\otimes n-1}) \cap (V^*)^{\otimes n}$. Now since d respects grading, so we have:

$$d((I^*)^{\otimes n-1}) \cap (V^*)^{\otimes n} = d(\bigoplus_i V^* \otimes \cdots \otimes A_2^* \otimes \cdots \otimes V^*)$$

where A_2^* is at i-th place (for i varies), and d is given by $\mathrm{id} \otimes \mu \otimes \mathrm{id}$ for the comultiplication $\mu : A_2^* \to A_1^* \otimes A_1^*$. Therefore the degree $-n$ part of the denominator is precisley:

$$\sum (V^*)^{\otimes k} \otimes \mu(A_2^*) \otimes (V^*)^{\otimes n-k-2}$$

This implies that as vector spaces:

$$(V^*)^{\otimes n}/\mathrm{im}\ (d) \simeq (A^!)_n$$

We need to check that multiplication in $A^!$ equals composition of Ext's. To do so, we need to consider the cobar complex:

$$B(A) = k \to I^* \to I^* \otimes I^* \to \cdots$$

This is a **differential graded algebra**. Moreover, $\mathrm{Ext}^\bullet(k,k)$ is another dga since:

Remark. *Let $C^\bullet$ be any complex, the internal Hom: $\mathcal{H}om(C^\bullet, C^\bullet)$ is a dga.*

Now $B(A)$ acts on the bar resolution for k via:

$$(\xi_1 \otimes \cdots \otimes \xi_m)(a_1 \otimes \cdots \otimes a_n) = \xi_1(a_1) \otimes \xi_m(a_m) \otimes a_{m+1} \otimes \cdots$$

This gives a homomorphism $B(A) \to \mathrm{Hom}(P^\bullet, P^\bullet)$ inducing isomorphism on H^* of dga. □

Remark. *In fact we constructed a homorphism of dga:*

$$B(A) \to A^!_q$$

where $A^!_q$ has zero differentials. If A is Koszul, then this map induces an isomorphism on cohomology (it's a quasi-isomorphism). We will call A is formal if A is equivalent to a dga with zero differentials. So we proved that if A is Koszul, then $B(A)$ is formal.

Our next goal is to prove that a quadratic algebra A is Koszul if the associated Koszul complex is exact, excpet for $H_0 = k$ (Theorem 3.9). Let's now start the proof.

Proof. If $K^\bullet$ is exact as above, it gives a free resolution of k, where the n-th term has generators in degree 1. This resolution gives $\mathrm{Tor}^A_n(k,k) = \mathrm{Tor}_{n,n} = (A^!_n)^*$. Conversely, if A is Koszul. Let's check by induction on n that $H_i(K) = 0$ for $0 < i < n$. Induction assumption is that the first $(n-1)$-terms of K are minimal resolutions of k, $0 < i < n$. Now let's look at:

$$\ker((A^!_n)^* \otimes A \xrightarrow{d_n} (A^!_{n-1})^* \otimes A)$$

It has space of generators $(A^!_{n+1})^*$ in degree $n+1$. But:

$$(A^!_{n+1})^* \otimes 1 \xhookrightarrow{d_{n+1}} \ker(d_n)$$

So its image contains all geneartors. □

Corollary 3.12. *A is Koszul implies that $A^!$ is Koszul.*

Proof. For the Koszul complex $K^\bullet_A$, we can write:

$$K^\bullet_A = \bigoplus_n K^\bullet(n)$$

$$K^\bullet(n) = (A^!_2)^* \otimes A_{n-2} \to (A^!_1)^* \otimes A_{n-1} \to A_n \to 0$$

But $K_{A^!}(n)$ is the dual complex to $K_A(n)$ for:

$$((A^!_i)^* \otimes A_{n-i})^* = A^!_i \otimes A^*_{n-i}$$

□

Let's now discuss the relation with distributive lattices.

Definition 3.13. A lattice is a poset L such that:

1. For all $x, y \in L$, there exists a smallest z with $z \geq x, z \geq y$.

2. For all $x, y \in L$, there exists a largest t with $t \leq x, t \leq y$.

Therefore there are operations:

$$+ : (x, y) \mapsto z, \quad \cap : (x, y) \mapsto t$$

Definition 3.14. We call a lattice distributive if $(x + y) \cap z = x \cap z + y \cap z$

Example. *Consider S any set, with L the lattice of all subsets. This is distributive.*

Example. *Consider V a vector space and L the lattice of its subspaces. Then L is not distributive. Subspaces $V_1, \cdots, V_n \subset V$ generated a distributive lattice if in a sublattice L which is distributive. In fact:*

Lemma. *This is equivalent to that there exists basis $e_1, \cdots, e_n \in V$ such that each V_i is spanned by a subset of that basis.*

Lemma 3.15. *Given $W_1, \cdots, W_n \subset V$ subspaces. We can form a complex:*

$$\cdots \to W_n/((W_1 + \cdots + W_{n-2}) \cap W_n) \to V/(W_1 + \cdots W_{n-1}) \to V/(W_1 + \cdots W_n) \to 0$$

The general term is:

$$W_n \cap \cdots \cap W_{i+1}/((W_{i-1} + \cdots + W_1) \cap W_n \cap \cdots \cap W_{i+1})$$

1. *If $W_1, \cdots, W_n$ generates a distributive lattice, then the complex is exact.*
2. *Suppose that for all i, $W_1, \cdots, W_{i-1}, W_{i+1}, \cdots, W_n$ generate a distributive lattice, and the complex is exact, then $W_1, \cdots, W_n$ also generates distributive lattice.*

This lemma will imply the last part of Theorem 3.9.

Remark. *We assumed $A_0 = k$ to ease notation. All generalizes to A_0 semisimple.*

3.2 Hochschild Cohomology and Deformation Theory

For any algebra A, the bar resolution can be used to compute Hochschild homology for a bimodule M:

$$\mathrm{HH}_*(A, M)$$

Indeed, for M, we have a free resolution:

$$M \otimes_{A \otimes A^{\mathrm{op}}} A^{\otimes m} \to M \otimes_{A \otimes A^{\mathrm{op}}} A^{\otimes m-1}$$

$$d(m \otimes a_1 \otimes \cdots \otimes a_m) = ma_1 \otimes a_2 \otimes \cdots a_n - m \otimes a_1 a_2 \otimes \cdots \pm a_m m \otimes a_2 \cdots$$

Example. *Let $A = \mathbb{C}[x_1, \cdots, x_n]$, then $A \otimes A^{\mathrm{op}} = \mathbb{C}[x_i, y_i] \simeq \mathbb{C}[x_i, z_i]$ for $z_i = x_i - y_i$. Then:*

$$\mathrm{HH}_*(A) = \mathrm{Tor}_*^{\mathbb{C}[x_i, z_i]}(\mathbb{C}[x_i, z_i]/(z_i), \mathbb{C}[x_i, z_i]/(z_i)) = \mathbb{C}[x_i] \otimes \mathrm{Tor}_*^{\mathbb{C}[z_i]}(\mathbb{C}, \mathbb{C}) \simeq \mathbb{C}[x_i] \otimes \bigwedge^{\bullet}(\mathbb{C} \cdot \{z_i\})$$

More canonically, if A is commutative, $\mathrm{HH}_1(A) = I/I^2$ is the module of Kähler differentials. More generally, $\mathrm{HH}_k(A)$ is the module of k-forms. And for an A-module M, we have:

$$\mathrm{HH}_i(M) = \Omega^i(M)$$

Let M be an A-bimodule.

Lemma 3.16. $\mathrm{HH}^2(M)$ *classifies the pair:* $(\widetilde{A}, I, \varphi)$ *such that:*

1. $I^2 = 0$ *and* $\widetilde{A}/I = A$.
2. *So* I *is an* A*-module, and* $\varphi : I \simeq M$.

Definition 3.17. An **order** n **deformation of** A is an algebra $\widetilde{A}$ free as a module over $k[\hbar]/\hbar^{n+1}$ with an isomorphism:

$$\widetilde{A}/\hbar\widetilde{A} \simeq A$$

Given an n-th order deformation, the obstruction for extending it to order $n+1$-th order deformation lies in $\mathrm{HH}^3(A)$: an expression in terms of the multiplication on $\widetilde{A}$ must vanish on $\mathrm{HH}^3(A)$. In particular, if $\mathrm{HH}^3(A) = 0$ then any deformation can be extended, and the set of all such extensions biject with $\mathrm{HH}^2(A)$.

3.3 Brauer Group of a Field

Let k be a commutative field.

Definition 3.18. A **central simple algebra** (CSA) over k is a k-algebra A, such that A is simple and $Z(A) = k$. If not stated otherwise, we assume $\dim_k(A) < \infty$.

Note a central simple algebra must take form:

$$A = \mathrm{Mat}_n(D)$$

for some D skew field over k, with $Z(D) = k$.

Definition 3.19. The **Brauer group** of k is defined to be: $\mathrm{Br}\ k = \{\mathrm{CSA}/k\}\ \sim$, modulo the Morita equivalence.

Note that we always have $D \sim \mathrm{Mat}_n(D)$. Therefore as a set, we can identify

$$\mathrm{Br}\ k = \{\text{skew fields } D \mid Z(D) = k, \dim_k(D) < \infty\}$$

Example. *If* k *is algbebraically closed, then* $\mathrm{Br}\ k = 1$.

Example. *Let* $k = \mathbb{R}$*, then* $\mathrm{Br}\ k = \mathbb{Z}/2\mathbb{Z}$ *generated by* $\mathbb{H}$.

Lemma 3.20. *Let* A *be a CSA over* k*, then*

$$A_e \coloneqq A \otimes_k A^{\mathrm{op}} \simeq \mathrm{End}_k(A)$$

Proof. By assumption A is a simple A_e algebra, and $Z(A) = \mathrm{End}_{A_e}(A) = k$. By density theorem, we know $A_e \twoheadrightarrow \mathrm{End}_k(A)$. We're done by comparing dimensions. □

Lemma 3.21. *Let* A *be a central simple* k*-algebra, and* B *any* k*-algebra. Then two-sided ideals in* $A \otimes_k B$ *biject with two-sided ideals in* B*, given by:*

$$A \otimes I \longleftrightarrow I$$

Proof. We need to describe $A_e \otimes B_e$-submodules of $A \otimes B$. Consider $A \otimes B$ as an A_e-module first, then its submodule must take form $A \otimes I$ for some $I \subset B$ vector subspaces (using A is simple A_e-module). It follows that I must be a two-sided ideal in B by looking at the B_e-actions. □

Lemma 3.22. *Let A, B be any k-algebras, then $Z(A \otimes_k B) = Z(A) \otimes_k Z(B)$.*

Proof. This is clear. □

Corollary 3.23. *If A and B are CSAs, then $A \otimes_k B$ is also a CSA.*

As a consequence, we get a binary operatrions on the set of CSAs. It is clear that it descends to a binary operation on Br k. This is actually a group operation, according to the following lemma.

Lemma 3.24. *Let A be a CSA over k, M a finitely generated A-module. Let's take $E = \mathrm{End}_A(M)$. Then E is also a CSA, and:*

$$A \otimes_k E \xrightarrow{\sim} \mathrm{End}_k(M)$$

Proof. WLOG we can assume $A = D$ is a skew field, and $M = D^{\oplus n}$. So that $E = \mathrm{Mat}_n(D^{\mathrm{op}})$. It follows that:

$$A \otimes_k E \to \mathrm{End}_k(M)$$

is a homomorphism of central simple algebras. One sees this is an isomorphism by just counting dimension over k. □

Corollary 3.25. *Let A be a CSA over k, then:*

$$A \otimes_k A^{\mathrm{op}} \simeq \mathrm{End}_k(A) = \mathrm{Mat}_n(k) \sim k$$

Proof. We apply the above lemma to $M = A$. □

Therefore Br k is a group, with $[A]^{-1} = [A^{\mathrm{op}}]$.

Corollary 3.26. *Let F/k be a field extension. Let A/k be an CSA, then $F \otimes_k A$ is a CSA over F. Therefore I have a group homomorphism:*

$$\mathrm{Br}\ k \to \mathrm{Br}\ F$$

Example. *Take $F = \overline{k}$. It follows that $A \otimes_k \overline{k} \simeq \mathrm{Mat}_d(\overline{k})$ for some d. Therefore $\dim_k(A) = d^2$. It's clear that we can arrange $A \otimes_k F \simeq \mathrm{Mat}_d(F)$ for some finite extension F/k. In this case, we call F the* **splitting field** *for the CSA A.*

Proposition 3.27. *Let F/k be a finite field extension. Let A be a CSA over F, and $A \supset E \supset F$ for E some commutative subfield. Set $B = Z_A(E)$, so that we have:*

$$A \supset B \supset E \supset F \supset k$$

Then:

$$\dim_k B = \frac{\dim_k A}{[E : F]}$$

Moreover, B is a CSA over E and $[B] = [A_E] \in \mathrm{Br}\ E$.

Proof. We know $A \otimes_F E$ is a CSA over E. But $A \otimes_F E$ act on A via:

$$(a \otimes e) \cdot x = axe$$

Now $B^{\text{op}} = \text{End}_{A \otimes_F E}(A)$. It follows that B is a CSA and $[B] = [A \otimes_F E]$. The dimension formula comes from:

$$\text{End}_k(A) = B^{\text{op}} \otimes_k (A \otimes_F E)$$

and then count dimension. □

Corollary 3.28. *If $A = D$ is a central division algebra with degree d over F (i.e. $\dim_F(D) = d^2$). Then it splits over a degree d extension.*

Proof. Let E be a maximal commutative subfield in D:

$$D \supset E \supset F$$

We claim that $[E : F] = d$ and A splits over E. Set $B = Z_A(E) \supset E$. We claim that $B = E$. If not, we can take $x \in B, x \notin E$ so that $E' = E(x) \supsetneq E$, contradiction. Hence:

$$[A_E] = [B] = [E] = 1_{\text{Br } k}$$

So A splits completely over E. Moreover previous proposition tells me $[D : E] = [E : F]$. But $[D : E] \cdot [E : F] = [D : F] = d^2 \implies [E : F] = d^2$. □

Example. *We will prove* $\text{Br } \mathbb{Q}_p = \mathbb{Q}/\mathbb{Z}$. *An order n element splits in many degree n extensions.*

Theorem 3.29 (Skolem-Noether)**.** *Let A be simple Artinian algebra over k with center k (need not be finite dimensional). Let B be a simple finite dimensional algebra over k (need not be central). Then: any two homomorphisms of k-algebras $B \to A$ are conjugate.*

Remark. *Typically, we will take A/k be a CSA, and $B = F/k$ some field extension. For example let's take:*

$$k = \mathbb{R}, A = \mathbb{H}, B = \mathbb{C}$$

We have two embeddings:

$$\varphi : i \mapsto i, \quad \psi : i \mapsto -i$$

Then they're conjugate by j.

Proof. Note given $\varphi : B \to A$, we can make A into a (B, A)-bimodule, denote it as M_φ. We claim that φ is conjugate to ψ iff $M_\varphi \simeq M_\psi$. Indeed, an isomorphism must given by left multiplication by some $a \in A^\times$, with the property that $a^{-1}\varphi a = \psi$. We know $C = B \otimes A^{\text{op}}$ is simple and finite rank over A^{op}, so is Artinian. Therefore C has unique simple module L, and every finitely generated moudle $M \simeq L^{\oplus n}$. But then $M_\varphi \simeq L^{\oplus n}, M_\psi \simeq L^{\oplus m}$ for some m, n. But $n = m$ because they're free A-modules of the same rank. □

Let $\text{char}(F) \neq 2$. Let's discuss degree two central division algebras D over F. Note we have:

$$D \supset E = F(\sqrt{\alpha}), \quad \alpha \in F$$

Let $\sigma \in \text{Gal}(E/F)$ be the nontrivial element. Then we have:

$$i, i \circ \sigma : E \hookrightarrow D$$

By Skolem-Noether, we know they must be conjugate by some element in D. In other words, there exists $y \in D$ such that $yxy^{-1} = \sigma(x) = -x$, for $x = \sqrt{\alpha}$. Now $y \notin E$ so that it must generate D as an E-algebra. Now its easy to check that y^2 commutes with x, y, so that $y^2 \in F$. So:

$$D = D_{\alpha,\beta} \simeq F\langle x, y\rangle/(x^2 = \alpha, y^2 = \beta, yx = -xy)$$

for some $\alpha, \beta \in F$.

Theorem 3.30 (Wedderburn-Artin Theorem)**.** $\text{Br}(\mathbb{F}_q) = \{1\}$. *Namely, every finite skewe field over a finite field is in fact commutative.*

Lemma 3.31. *If G is a finite group, and $H \subset G$ is a subgroup and $G = \bigcup_{x \in G} xHx^{-1} \implies G = H$.*

Proof. Write $G = \coprod C_i$ into conjugacy classes. By assumption, we know $C_i \cap H \neq \emptyset$ for each i. We claim that:

$$\frac{|C_i \cap H|}{|C_i|} \geq [G : H]^{-1}$$

Now choose $g_i \in C_i \cap H$, we have:

$$|C_i \cap H| \geq \overline{|H|} Z_H(g_i) \geq \frac{|H|}{Z_G(g_i)} = [G : H]^{-1} \cdot |C_i|$$

If $[G : H] > 1$, the equality is strict for some C_i (for example $C_i = \{1\}$). Now:

$$|G| = \sum |C_i| < \sum |C_i \cap H| \cdot [G : H] = |G|$$

A contradiction. □

Proof. (first proof of Wedderburn-Artin) Let D be a finite skew field, $Z(D) = \mathbb{F}_q$. Let $E = \mathbb{F}_{q^d}$ be a maximal commutative subfield. Then we know $\dim_{\mathbb{F}_q} D = d^2$. Now we only need to check that every $x \in D$ is conjugate to an element in E, then apply the lemma to $G = D^\times, H = E^\times$ we get $d = 1$, and we're done.

Now let $E' = \mathbb{F}_q(x) \simeq \mathbb{F}_{q^{d'}}$, then $d' \mid d$ because:

$$d' = [E' : \mathbb{F}_q] = [D : Z(E')]$$

So $d^2 = (d')^2 \dim_{E'}(Z(E'))$. So there exists field embedding $E' \hookrightarrow E$. We're therefore done by Skolem-Noether. □

Recall last time, given F a field, we have a division algebra:

$$D_{a,b} = F\langle x, y\rangle/(x^2 = a, y^2 = b, xy = -yx)$$

Note $D_{a,b}$ is always a CSA (but it can be just $\mathrm{Mat}_2(F)$). We wish to see when is this a skew field. Let's assume a, b are neither a square in F. Set $E = F\langle x\rangle/(x^2 - a)$.

Lemma 3.32. *$D_{a,b}$ is a skew field iff $b \notin \mathrm{Nm}(E)$.*

Proof. We have $D_{a,b}$ is either $\mathrm{Mat}_2(F)$ or a skew field D. If $b = \mathrm{Nm}(E)$, we may assume $b = 1$. Then we know $(y + 1)(y - 1) = 0$, so that D is not a skew field. On the other hand, if $A \simeq \mathrm{End}(V)$ for $V \simeq F^2$, so we have an algebra homomorphism:

$$E \hookrightarrow \mathrm{End}(V)$$

realizing V as an E-module. But then $\dim_E(V) = 1$, so we can write $V = E$ as an E-module. Now we need to specify the y-action. If $y(1) = z \in E$, then:

$$y(t) = (yt)(1) = (yty^{-1})y(1) = \sigma(t)z \implies y^2(1) = y(z) = \sigma(z)z \in \mathrm{Nm}(E)$$

□

Example. *Let $F = \mathbb{Q}, a = -1, b = 3$. Note that $b \notin \mathrm{Nm}(\mathbb{Q}(i))$. So that $D_{-1,3}$ is a skew field, split both by $\mathbb{Q}(\sqrt{-1}), \mathbb{Q}(\sqrt{3})$.*

Let's now discuss a more general construction. Let E/F be Galois extension, and $\Gamma = \mathrm{Gal}(E/F)$. Let's construct A as a twist of $\Gamma\#E$.

Suppose A is a CSA over F of degree d, and $E \subset A$ is a subfield such that $[E : F] = d$ and is Galois (so E is a maximal subfield, $Z_A(E) = E$). Then by Skolem-Noether for every $g \in \Gamma$, there exsists y_g such that:

$$g(x) = y_g x y_g^{-1}, \forall g \in E$$

Now y_g is unique up to left multiplication by some element in E. Then $A = \bigoplus E \cdot y_g$. This is because by dimension count, we only need to see y_g are independent over E: this is just linear independence of characters. Clearly the structure of A is determined by $c_{g,h} \in E$ such that $y_g y_h = c_{g,h} y_{gh}$.

Conversely, if we have $A = \oplus E \cdot y_g$, with multiplication:

$$y_g x y_g^{-1} = g(x), \quad y_g y_h = c_{g,h} y_{gh}$$

for $c_{g,h} \in E^\times$. Moreover, we must have:

$$(y_g y_h)y_k = y_g(y_h y_k) \iff c_{g,h}(y_{gh} y_k) = y_g(c_{hk} y_{hk}) \iff c_{g,h} c_{gh,k} = g(c_{h,k}) c_{g,hk}$$

This is the 2-cycle condition. It is an exercise to check that for these $c_{g,h}$ satisfying the cocycle condition, A is a CSA over F.

Example. *Let $\Gamma = \langle\gamma\rangle \simeq \mathbb{Z}/n\mathbb{Z}$ be cyclic.Let E/F be a galois extension with galois group Γ. For exmaple one can take $F = \mathbb{Q}$, $E = \mathbb{Q}(X)/X^3 - 3X + 1$, and then $\mathrm{Gal}(E/F) = \mathbb{Z}/3\mathbb{Z}$ (since the discriminant of $X^3 - 3X + 1$ is $81 = 9^2$).*

Let's choose $y_{\gamma^k} = y_\gamma^k$ for $0 \leq k \leq n-1$. Also note $y_\gamma^n = b \in F^\times$. Therefore:

$$c_{\gamma^i,\gamma^j} = \begin{cases} 1, & \text{if } i+j < n; \\ b, & \text{otherwise.} \end{cases}$$

It's straightforward to satisfy this satisfies the cocycle condition. THerefore one can form:

$$A = \langle E, y\rangle/(yxy^{-1} = g(x), y^n = b \in F^\times)$$

This will be a CSA that splits iff $b \in \mathrm{Nm}(E/F)$.

The above constructoin is an instance of group cohomology. Recall that we defined group cohomology to be:

$$H^i(G, M) = \mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, M)$$

This can be computed using bar resolution. Very concretely, set $A = \mathbb{Z}[G]$, we have:

$$A \otimes_{\mathbb{Z}} A \otimes_{\mathbb{Z}} \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} A = \mathbb{Z}[G^n]$$

Apply $\otimes_{\mathbb{Z}[G]}\mathbb{Z}$ to the bar complex, we get:

$$\cdots \to \mathbb{Z}[G^2] \to \mathbb{Z}[G] \to 0$$

a resolution for $\mathbb{Z}$ by free $\mathbb{Z}[G]$-module. Now n-th term is a free module, with set of generators G^n. So taking $\mathrm{Hom}_G(-, M)$, we get a complex:

$$M \to \mathrm{Map}(G, M) \to \mathrm{Map}(G^2, M) \to \cdots$$

given by:

$$df(g_0, \cdots, g_n) = g_0 f(g_1, \cdots, g_n) + \cdots + (-1)^i f(g_0, \cdots, g_i g_{i+1}, \cdots g_n) + \cdots + (-1)^n f(g_0, \cdots, g_{n-1})$$

Remark. *In the case $G = \mathbb{Z}/n = \langle g\rangle$, then we get a much smaller resolution:*

$$\cdots \to \mathbb{Z}[G] \xrightarrow{1-g} \mathbb{Z}[G] \xrightarrow{1+g+\cdots+g^{n-1}} \mathbb{Z}[G] \xrightarrow{1-g} \mathbb{Z}[G] \to 0$$

Apply $\mathrm{Hom}_{\mathbb{Z}[G]}(-, M)$, we get:

$$0 \to M \xrightarrow{1-g} M \xrightarrow{1+g+\cdots+g^{n-1}} M \xrightarrow{1-g} \cdots$$

One can compute the group cohomology from this.

Example. *Let E/F be cyclic galois extension, with $G = \mathrm{Gal}(E/F) = \mathbb{Z}/n$. Then:*

$$H^2(G, E^\times) = F^\times/\mathrm{Nm}(E^\times)$$

Recall that $H^1(G,M)$ classifies G-torsors over M, and $H^2(G,M)$ classifies the **cross-product extensions of G by M.**

Definition 3.33. A cross-product extension of G by M is an exact sequence of groups:

$$0 \to M \to \widetilde{G} \to G \to 0$$

such that G-action on M coming from this sequence coincides with our prescribed action.

Proposition 3.34. *$H^2(G,M)$ classifies cross-product extensions of G by M.*

Proof. Given $\widetilde{G} \to G$, pick $s : G \to \widetilde{G}$, and set:

$$c_{g,h} = s(gh)s(h)^{-1}s(g)^{-1}$$

One can check that $\widetilde{G}$ being a group implies $c_{g,h}$ is a cocycle. We may replace s by s' with $s'(g) = \alpha(g)s(g)$, from this we see:

$$c'_{g,h} = d\alpha \cdot c_{g,h}$$

Therefore we get a cohomology class from a cross-product extension. Conversely, given a class c, let $\widetilde{G} = G \times M$ with multiplication:

$$(g,m)(g',m') = (gg', (g')^{-1}(m)m'c_{g,g'})$$

□

Remark. *There is a coordinate free way to approach this. Let R be a ring with M, N any modules, we can identify $\mathrm{Ext}^2_R(N,M)$ as objects $C^\bullet$ in $\mathcal{H}_0(\mathrm{Proj}_R)$ with $H^0(C^\bullet) = N, H^{-1}(C^\bullet) = M$. In our case, givne $\widetilde{G}$, pick a resolution $P^\bullet$ for $\mathbb{Z}$ over $\widetilde{G}$, and consider:*

$$C^\bullet = P^\bullet \otimes_{\mathbb{Z}[\widetilde{G}]} \mathbb{Z}[G]$$

Then $H^0(C^\bullet) = \mathbb{Z}, H^{-1}(C) = M$. So the desired complex is $C^{\bullet,\geq -1}$

Theorem 3.35. *Let E/F be finite Galois extension. Then:*

$$\mathrm{Br}\ (E/F) \simeq H^2(G, E^\times)$$

Proof. We first need a lemma.

Lemma. *For a finite field extension E/F, and A a CSA over F. TFAE:*

1. *$A \in \mathrm{Br}\ (E/F)$.*
2. *$A \sim A'$ for A' a degreee $d = [E:F]$ CSA such that $A' \supset E$.*

Proof. (of the lemma). (2) implies (1) is done before: $A_E \sim Z_A(E)$ if $E \subset A$. But $\deg A = [E:F] \implies E = Z_A(E)$. Conversely, assume A_E splits. Let's pick $A' \sim A$ minimal such that $A' \supset E$. Then consider $Z_{A'}(E) \simeq \mathrm{Mat}_m(E)$ for some m. However, if $Z_{A'}(E)$ contains an idempotent $e \neq 0, 1$, then $eA'e \supset E$. But $eA'e \sim A'$, contradicting the minimality of A'. Now $m = 1$, so $Z_{A'}(E) = E$. It follows that $\deg(A') = [E:F]$. □

Now from the lemma, we see Br (E/F) bijects with isomorphism classes of CSAs of $\deg d = [E : F]$ containing E. Such an A is a cross product algebra. Indeed by Noether-Skolem, we can write:

$$A = \oplus E \cdot \gamma_g$$

where $\gamma_g x \gamma_g^{-1} = g(x)$, $g \in G$. Now we can construct the cocycle from:

$$\gamma_g \gamma_h = c_{g,h} \gamma_{gh}$$

Also without using cocycles, we have $\widetilde{G} = N_{A^\times}(E^\times)$ fitting into:

$$0 \to E^\times \to \widetilde{G} \to G \to 0$$

We omit the constructing of the inverse map. To check this is a group isomorphism, we first interpret the group structure on $H^2(G, M)$ on the level of cross-product extensions: given

$$M \to \widetilde{G}' \to G$$

$$M \to \widetilde{G}'' \to G$$

we're getting $\widetilde{G}' \times_G \widetilde{G}''/m' - m''$ another cross-product extension. Now $A' \otimes A'' \supset E \otimes_F E = \oplus_{g \in G} E_g$. Now let c_1, c_2 be two cocycles. We need to check:

$$[A_{c_1} \otimes_F A_{c_2}] = [A_{c_1 c_2}]$$

But:

$$A_{c_1} \otimes A_{c_2} \supset E \otimes_F E = \bigoplus_{g \in G} E_g$$

Let's pick $e = 1_{g=1} \in E \otimes_F E$. Let's define $B \coloneqq A_{c_1} \otimes A_{c_2} = \bigoplus_{g_1, g_2} (E \otimes_F E) \gamma_{g_1} \otimes \gamma'_{g_2}$, so that:

$$B \sim eBe = \mathrm{End}_{B^{\mathrm{op}}}(eB) \implies \dim(eBe) = d^2$$

Moreover, one can check:

$$eBe = \bigoplus E_1 \gamma_g \otimes \gamma'_g \simeq A_{c_1 c_2}$$

□

Corollary 3.36. Br $(F) = H^2(\mathrm{Gal}(F), F_{\mathrm{sep}}^\times)$

Remark. *People sometimes just write* $H^*(F^\times) = H^*(\mathrm{Gal}(F), F_{\mathrm{sep}}^\times)$

Example. *Let* E/F *be Galois cyclic, we have* $H^2(E/F) = F^\times/\mathrm{Nm}(E^\times)$.

second proof of Weddernburn-Artin. We need to prove Br $(\mathbb{F}_{q^n}/\mathbb{F}_q) = 0$ for all n. But $\mathbb{F}_{q^n}/\mathbb{F}_q$ is cyclic, and $\mathrm{Nm} : \mathbb{F}_{q^n} \to \mathbb{F}_q$ is surjective. □

Proposition 3.37. *Any CSA* A/F *splits by a finite separable extension.*

Proof. It's enough to show that A contains $x \notin F$, and x is separable over F. Then $A' = Z_A(F(x)) \sim A_{F(x)}$, we're done by induction on $\deg A$. We thus only need to prove the claim. If the claim is false, then all $F \subset E \subset A$ are purely inseparable. Therefore, for $x^{p^n} \in F$ for some n.

Lemma. *Let $A/\mathbb{F}_p$ be an algebra, and* $\mathrm{ad}(a) : x \mapsto ax - xa$. *Then:*

$$\mathrm{ad}(a)^p = \mathrm{ad}(a^p)$$

The key observation for proving the lemma is that left multiplication commutes with right multiplication, so $\mathrm{ad}(a) = L_a - R_a$ for L_a, R_a commutes. Now pick $x \in A$, $x^p \in F, x \notin F$. Therefore: $\mathrm{ad}(x)^p = \mathrm{ad}(x^p) = 0$. Now if $E = F(x)$, then $[E : F] = p$, and $\dim_F Z_A(E) = \frac{\dim A}{p} = \frac{d^2}{p}$. Now $\mathrm{ad}(x) : A \to A$ is a nilpotent operator (and $\mathrm{ad}(x)^p = 0$), with $\frac{d^2}{p}$ Jordan blocks, with each Jordan block has size $\leq p$. It follows that each block must have size precisley p. Therefore:

$$\ker(\mathrm{ad}(x)) \subset \mathrm{im}\ (\mathrm{ad}(x))$$

In particular, $x = [x, y]$ for some y. But then $\mathrm{ad}(y)(x) = -x$, and therefore $\mathrm{ad}(y)$ is not nilpotent. Contradiction. □

Remark. *We can avoid the trick by invoking Engel's theorem.*

Let's now discuss Galois forms. Let E/F be Galois. Suppose I want to classify all CSA A over F such that $A_E \simeq \mathrm{Mat}_d(F)$. The general procedure tells me I want to consider:

$$H^1(\mathrm{Gal}(E/F), \mathrm{Aut}(\mathrm{Mat}_d(E))) \simeq H^1(\mathrm{Gal}(E/F), \mathrm{PGL}_d(E))$$

This will calssify all degree d CSA's over F splitting over E, for E/F Galois.

Remark. *Suppose I want to classify X/F, with $X_E \simeq (X_0)_E$. Then we need to look at $H^1(G, \mathrm{Aut}(X_0)_E)$. To see this, given $X_E \simeq (X_0)_E$, we consider:*

$$T = \{\text{isomorphisms } X_E \simeq (X_0)_E\}$$

Hence on T we have a simply transitive action by $\mathrm{Aut}(X_0)_E$. Note G also acts on this, so T is an $\mathrm{Aut}(X_0)_E$-torsor over G.

Now if

$$A \mapsto [T_A] \in H^1(G, \mathrm{PGL}_d(E))$$

then $[A] \in H^2(G, E^\times)$ is obtained from:

$$1 \to E^\times \to \mathrm{GL}_d(E^\times) \to \mathrm{PGL}_d(E) \to 1$$

Remark. *A posteriori we know: $H^1(G, \mathrm{PGL}_d(E)) \hookrightarrow H^2(G, E^\times)$. This is from Hilbert theorem 90, which asserts that there is no Galois forms of vector spaces. In other words, we need to show:*

Lemma. *For $V \simeq E^n$ withan action of $G = \mathrm{Gal}(E/F)$ such that $g(\lambda v) = g(\lambda)g(v)$, then V has a G-invariant basis. In other words, $V = E \otimes V'$ for some F-vector space V'.*

Proof. We can check that V is spanned by G-invariant vectors. Think of V as a F-vector space, and use that for an F-vector space W with a G-action:

$$(E \otimes_F W)^G = E \otimes_F W^G$$

Now $E \otimes_F V = \bigoplus_g V_g$. From this we see $E \otimes V$ is generated by $(E \otimes V)^G$ over $E \otimes_F E$, and therefore V is generated by V^G over E. Hence we have:

$$E \otimes_F V^G \twoheadrightarrow V$$

It's kernel K must satisfies $K = E \cdot K^G = 0$. □

Remark. *Many questions about cross product algebras are hard.*

Definition 3.38. Let $\alpha \in \text{Br}\ (F)$. Let's define the **index** of α to be $\deg(D_\alpha) = \min\{[E : F] \mid \alpha \in \text{Br}\ (E/F)\}$, where D_α is the division algebra representing α. The **period** of α is the minimum of n such that $n\alpha = 0$

Proposition 3.39. $\text{ind}(\alpha)\alpha = 0$. *In other words, period divides index.*

Proof. For $H \subset G$ be subgroups, $n = [G : H]$, one can form:

$$H^i(G, M) \xrightarrow{\text{res}} H^i(H, M) \xrightarrow{\text{infl}} H^i(G, M)$$

The composition of these is just multiplication by n. Now let D represents α, and $E \subset D$ is a maximal commutative subfield. Let's take K be the Galois closure of E. Then we're done by taking $G = \text{Gal}(K/F), H = \text{Gal}(K/E)$. □

Let's now do some computations.

Definition 3.40. A field is quasi-closed (c_1) if a homogeneous polynomial of degree d in n-variables has a root if $n > d$. A field is calle c_k if we have a root for $n > d^k$.

Remark. *If F is c_k , then $F(t)$ is c_{k+1}. Artin conjectured that $\mathbb{Q}_p$ is c_2, but this is wrong.*

Proposition 3.41. *If F is c_1, then* $\text{Br}\ (F) = 0$

Theorem 3.42 (Chevalley-Warning)**.** *A finite field is c_1. Hence Brauer group of finite field is zero.*

Proposition 3.43. *Let E/F is a finite extension. Then F is c_1 implies E is c_1.*

Proof. Let $P \in E[x_1, \cdots, x_n]$, and pick a basis $\alpha_1, \cdots, \alpha_d$ for E/F. Let's introduce new variables y_{ij}, and we will set:

$$x_i = \sum_{j=1}^{d} y_{ij}\alpha_j$$

Now $P(x_1, \cdots, x_n) = 0$ becomes:

$$\text{Nm}_{E/F}(P(\sum y_{ij}\alpha_j)) = 0$$

but this just multiplies both the degree and the variables by d. □

Proof. (of Chevalley Warning) It suffices to show for $F = \mathbb{F}_p$. We claim the number of zeroes is 0 modulo p, and this will imply our results because we have the trivial zero $(0, 0, \cdots, 0)$. But now the number of zeroes equals:

$$\sum_{(a_1,\cdots,a_n)\in\mathbb{F}_p^n} (1 - p(a_1, \cdots, a_n)^{p-1})$$

Every monomials in this sum will have at least one variable whose exponent is less than $(p-1)$, because this monomial has degree $d(p-1) < n(p-1)$. Summing over that variable and using $\sum_a a^m = 0$ for $0 \leq m \leq p-1$, we are done. □

Theorem 3.44 (Tsen)**.** *$k(t)$ is c_1 if k is algebraically closed.*

Lemma 3.45. *Let A/F be a degree n CSA. Then there exists unique trace: $\tau : A \to F$ such that:*

$$\tau_E \simeq \mathrm{Mat}_n(E) \to E$$

is the usual trace for any E/F splitting field. There exists unique determinant $\delta : A \to F$ such that:

$$\delta_E \simeq \mathrm{Mat}_n(E) \to E$$

is the usual determinant for E/F splitting field.

Proof. Since $\det, \mathrm{Tr}$ are invariant under $\mathrm{PGL}_n(E)$-action for E/F Galois splitting field, it descends. □

Example. *Let's take $A = \mathbb{H}$, $F = \mathbb{R}$. Then:*

$$\delta(x) = x\overline{x}, \tau(x) = x + \overline{x}$$

Proof. (c_1 implies vanishing of Brauer group). If D/F is a skew field of degree d. Then $\delta(x)$ is a degree d polynomial with d^2 variables. If $d > 1$ then we know it has a root since the field is c_1. However, we know:

$$\delta(D^\times) = F^\times$$

this is a contradiction. □

Let's now discuss Brauer group of local fields.

Theorem 3.46. *Let F be a local field, then* $\mathrm{Br}\ F = \mathbb{Q}/\mathbb{Z}$.

Let E/F be unramified, so that E/F is Galois with:

$$\mathrm{Gal}(E/F) \simeq \mathrm{Gal}(k_E/k_F) \simeq \mathbb{Z}/n$$

In this case $\mathrm{Br}\ (E/F) = F^\times/\mathrm{Nm}(E^\times)$. We also know: $\mathrm{Nm}(E^\times) = \{x \in F^\times : n \mid v(x)\}$. Therefore:

$$\mathrm{Br}\ (E/F) \simeq \mathbb{Z}/n$$

Proposition 3.47. *Let A/F be a CSA over a local field. Then there exists a splitting field E with E/F unramified.*

Remark. *In fact the maximal unramified extension F^{ur} is a c_1 field, and the reslut follows.*

Proof. (Sketched). Let's give a proof not using the above fact. Let D/F be a central skew field of degree n. We have an "evaluation map":

$$v_D : D^\times \to \mathbb{Z}$$

We can similarly define $\mathcal{O}_D = \{0\} \cup \{x \in D^\times \mid v(x) \geq 0\}$. Now:

$$D \supset \mathcal{O}_D \supset \mathfrak{m}_D$$

Now $k_D = \mathcal{O}_D/\mathfrak{m}_D$ is a finite skew field over finite field k_F, hence k_D is in fact a a finite field. Let $d = [k_D : k_F]$ and $d' = v_D(\varpi_F)$. Now we need the following:

Lemma. *We have $d, d' \leq n$ and $d \cdot d' = n$*

If this is true, then $d = n$ and we get $E \subset D$ unramified over F of degree n which splits D. Now let's prove the lemma.

For every $x \in D$. Consider $F(x)/F$ be a field extension of degree $\leq n$. We choose x such that $v_D(x) = 1$, we see that $d' \leq n$. We choose $x \in \mathcal{O}_D$ and $\overline{x}$ generates k_D, we get $d \leq n$. So that $\mathcal{O}_D$ is free over $\mathcal{O}_F$ of rank $n = dd'$. □

As a consequence:

$$\text{Br } (F) = \text{Br } (F^{\text{ur}}/F) = \varinjlim \mathbb{Z}/n = \mathbb{Q}/\mathbb{Z}$$

Now let F be a global field. We have a natural map:

$$\text{Br } (F) \to \bigoplus_v \text{Br } (F_v)$$

Theorem 3.48. *We have a short exact sequence:*

$$0 \to \text{Br } (F) \to \bigoplus_v \text{Br } (F_v) \xrightarrow{\sum} \mathbb{Q}/\mathbb{Z} \to 0$$

Example. *Let $\mathbb{H} = \mathbb{Q}[i,j]/(i^2 = p, j^2 = q, ij = -ji)$. Then $\sum_v [H]_v = 0$ is precisely the statement of quadratic reciprocity.*

Remark. *We used cross product algebras to describe Brauer groups. A natural question to ask is that are there CSA not cross products? The answer is yes, the first counterexample is given by Amitsur, 1972. There also exists one for $F = \mathbb{Q}(t), \mathbb{Q}((t))$, 1995.*

Let's briefly discuss Brauer groups for rings. Let R be a commutative ring. We can also define Br R. Let's assume R to be Noetherian for simplicity.

Definition 3.49. We call A/R is an **Azumaya algebra**, if A as a module is projective, faithful, and finitely generated over R, together with one of the following (equivalent) conditions:

1. $A \otimes_R A^{\text{op}} \simeq \text{End}_R(A)$

2. For all $R \to k$, we have A_k is a CSA over k.

3. For all $R \to k$ with $k = \overline{k}$, we have $A_k \simeq \mathrm{Mat}_n(k)$.

Example. *Consider $\mathbb{H} = \mathbb{Z}[i,j]/(i^2 = j^2 = -1, ij = -ji)$. This is not an Azumaya algebra over $\mathbb{Z}$. Since it will not be a CSA when base change to $\mathbb{F}_2$. But $\mathbb{H}[\frac{1}{2}]$ is an Azumaya algebra over $\mathbb{Z}[\frac{1}{2}]$.*

There is a cohomological interpretation for this. There will be an injection:

$$\mathrm{Br}\ (R) \hookrightarrow H^2_{\text{ét}}(\mathrm{Spec}\ (R), \mathcal{O}^\times)$$

This can be computed using **Amitsur complex**. Let $R \to S$ be an extension of commtutative rings. Let's consider:

$$C_{S/R} \coloneqq (0 \to S^\times \to (S \otimes S)^\times \to (S \otimes S \otimes S)^\times \to \cdots)$$

For example, if I take $R = F$ be a field, E/F a Galois extension with $S = E$. Then the complex reads:

$$0 \to E^\times \to \prod_G E^\times \to \prod_{G^2} E^\times \to \cdots$$

In general:

$$\varinjlim_{R \to S \text{ faithfully flat}} H^2(C_{S/R}) = H^2_{\text{ét}}(\mathrm{Spec}\ (R), \mathcal{O}^\times)$$

3.4 Noncommutative Localization

Let R be a noncommutative ring, $S \subset R$ a multiplicative subset. We want to build $R \to R_S$ with obvious universal property. Obviously I should choose:

$$R_S \coloneqq R\langle i_s \mid s \in S\rangle/(i_s s = s i_s = 1)$$

Remark. *If ab is a unit in R, this doesn't imple a, b are both invertible. But if we assume ab is a unit, then: a being invertible is equivalent to b being invertible. Therefore if $as = t$ (or $sa = t$) and $s, t \in S$, we can add a to S without changing R_S. But $ab \in S$ doesn't imply we can add a, b to S.*

Definition 3.50. We call S saturated if $sa = t$ and $s, t \in S \implies a \in S$. From the above remark, we know for each S, we can consider its saturation and leave R_S unchanged.

Definition 3.51. We call S is a **right denominator set** (Ore set) if:

1. (O1) For every $a \in R, s \in S$, we have $aS \cap sR \neq \emptyset$.

2. (O2) For all $a \in R, s \in S$ with $sa = 0$, there exists $t \in S$ such that $at = 0$.

Note if O1 holds, then we can find $t \in S, b \in R$ with $at = sb$, and therefore:

$$s^{-1}a = bt^{-1}$$

This is saying "we can pull all denominators to the right".

Proposition 3.52. *If $S \subset R$ is a right Ore set. Then:*

$$R_S = \{(a,s) \in R \times S\}/((a,s) \sim (a',s'), a' = at, s' = st \in S, t \in R)$$

with natural addition and multiplication by pulling all denominators to the right.

Assume (O12), we can write:

$$R_S = \bigcup_{s \in S} Rs^{-1} = \text{colim}_S R$$

This is a filtered colimit. Here we make S as a diagram as follows: we connect an edge called u from t to s if $u \in R$ with $tu = s$. We thus get a direct system of R-modules: $M_s = R$ for all $s \in S$, and the arrow $u : s \to t$ is right multiplication by u. Let's check this forms a *filtered* direct system.

1. Given t, t' we need to find s such that:

   $$s = tu = t'u'$$

   This is a direct consequence of O1.

2. We also need: if

   $$u, u' : t \to s$$

   we must find t' with $t' : s \to st'$ such that:

   $$t'(u - u') = 0$$

   This is guaranteed by O2.

Corollary 3.53. *We have:*

$$\ker(R \to R_S) = \{x \mid xs = 0, \text{ for some } s \in S\}$$

Let's discuss localization of modules. For a module M over R with $S \subset R$, the map:

$$M \to M_S$$

shoudl be the universal one sending $S \to \text{Aut}(M_S)$. One can check that for left modules:

$$M_S = R_S \otimes_R M$$

and for right modules:

$$M_S = M \otimes_R R_S$$

Under the assumption O12, we have for right modules:

$$M_S = M \otimes_S (\text{colim}_S R) = \text{colim}_S M$$

This is a filtered colimit, so is exact.

Definition 3.54. We call $x \in R$ is regular if it's not a left nor right zero divisor. We call R a domain if $ab = 0 \implies a = 0, b = 0$. We call R an Ore domain if $R \setminus \{0\}$ satisfies O1.

Remark. *If R is an Ore domain, and set $S = R \setminus \{0\}$, we have R_S a skew field.*

Example. *If R is a filtered ring, with $\mathrm{gr}(R)$ a commutative domain, then R is a domain. A basic example for this is $R = U(\mathfrak{g})$.*

Example. *If $A = k\langle x_1, \cdots, x_n\rangle$ for $n > 1$, then A is not an Ore domain. Indeed:*

$$x_1A \cap x_2A = \{0\} \implies x_1S \cap x_2S = \emptyset$$

for $S = A \setminus \{0\}$.

Proposition 3.55. *Let R/k be a domain. If R is not an Ore domain, then $R \supset k\langle x, y\rangle$. In particular, it has exponential growth.*

Proof. Indeed, there must exist $x, y \in R$ with $x, y \neq 0$ such that:

$$xA \cap yA = \{0\}$$

We claim that there is no relation between x, y. If not, let's pick a noncommutative polynomial of smallest degree n given the relation: $P(x, y) = 0$. Let's write:

$$P(x, y) = a + xf(x, y) + yg(x, y)$$

Then $\deg f < n, \deg g < n$. If $f \equiv 0$, then g is not the zero polynomial. We also need $a \neq 0$ since R is a domain. Without loss we may assume $a = 1$, and therefore y is invertible, this implies $xA \cap yA \neq \{0\}$. Contradiction. Therefore we know $f \not\equiv 0$ and similarly $g \not\equiv 0$. The minimality tells us $f(x, y), g(x, y) \neq 0$. If $a = 0$ then our assumption is violated. So Let's assume $a = 1$. We have:

$$xf(x, y)y + (a + yg(x, y))y = x(f(x, y)y) + y(a + g(x, y)y) = 0$$

This again violates O1 since $f(x, y)y \neq 0$. □

Let's now have a digression on the discussion of growth of algebras. Let A/k be a finitely generated algebra over a field. Let V be the finite-dimensional vector space spanned by generators of A, so that we have a surjection:

$$d_V : TV \to A$$

Let $A_n = d_V(T^{\leq n}V)$. Let's set $d_V(n) = \dim(A_n)$.

Now given another finite dimensional vector space of generateros $V' \subset A$, we can similarly define $d_{V'}(n)$. Note that $V' \subset A_m$ for some m. Therefore there exists m such that:

$$d_{V'}(n) \leq d_V(mn)$$

$$d_V(n) \leq d_{V'}(mn)$$

Now let's view $d_V, d_{V'}$ both as monotone functions $\mathbb{N} \to \mathbb{N}$. We call $f \leq g$ if there exists m such that $f(n) \leq g(mn)$. This will give a partial ordering, and defines an equivalence relation. It is clear that $d_V \sim d_{V'}$, so we can talk about the growth rate of A as the equivalence class that d_V belongs to (independent of V).

Definition 3.56. A has exponential growth if there exists $c > 1$ such that:

$$d_V(n) \geq \text{const} \cdot c^n$$

Lemma 3.57. *1. Free algebra in $n > 1$ variables has exponential growth.*

2. Every algebra containing free algebra also has exponential growth.

Corollary 3.58. *If A/k is a domain of subexponential growth, then A is an Ore domain.*

Definition 3.59. We call A has polynomial growth if $d_V(n) = O(n^\delta)$ for some δ. In fact this δ is uniquely defined. In this case, we call δ the **Gelfand-Kirillov dimension** of A.

Example. *Let $A = k[x_1, \cdots, x_n]$. Then A has GK dimension n.*

Example. *For $A = U\mathfrak{g}$, the GK dimension is* $\dim \mathfrak{g}$.

Example. *For $A = \mathcal{D}(\mathbb{A}^n)$, the GK dimension is $2n$.*

Remark. *If A is a commutative algebra, then this notion coincides with other definition of other dimensions.*

We can also define similar notion of growth for modules. If M is a finitely generated A-module, and W is the space of generators for M. We can also define:

$$d_{V,W}(n) = \dim(A_n \cdot W)$$

It is clear for other choice of W' we have:

$$d_{V,W'}(n) \leq d_{V,W}(n+m)$$

for some m such that $W' \subset A_m \cdot W$. Therefore, it makes sense to call M has polynomial growth if $d_{V,W} = O(n^\delta)$. In this case,

$$\delta \coloneqq \text{GKdim}(M) \leq \text{GKdim}(A)$$

Remark. *Either A is finite dimensional, or $d_V(n) \geq n$.*

Remark. *There is a similar growth theory for groups, instead of algebras.*

Let's see how the discussion of growth interacts with Ore localizations.

Proposition 3.60. *Let R be a domain but not a (right) Ore-domian, then R contains an infinite rank free right ideal.*

Proof. Let $a, b \in R$ with $a, b \neq 0$. Suppose:

$$aR \cap bR = 0$$

Let's consider:

$$a, ba, b^2a, \cdots$$

We claim this generates a free right ideal. We omit the details for this verification. □

Corollary 3.61. *If R is a domain and right noetherian, then R is a right Ore domain.*

Remark. *Ore like conditions are interesting for a set S of morphisms in a category. Here are two examples:*

1. *Take $\mathcal{C} = \mathcal{H}o(\mathcal{A})$ the homotopy category of some abelian category, and S be the subset of morphisms inducing isomorphisms on cohomology. Then S satisfies Ore condition. We get derived category $D(\mathcal{A})$ as localization.*
2. *For $\mathcal{A}$ an abelian category, with $\mathcal{B}$ some subcategory. Let $S = \{s : M \to N \mid \ker(s), \text{coker } (s) \in \mathcal{B}\}$. Then localizing at S gives me $\mathcal{A}/\mathcal{B}$*

3.5 Prime ring and Goldie's Theorem

Definition 3.62. We call R a **prime ring** if:

$$I \neq 0, J \neq 0 \implies IJ \neq 0$$

for I, J two-sided-ideals. We call R a **semiprime ring** if

$$I \neq 0 \implies I^2 \neq 0$$

Lemma 3.63. *If R is semiprimitive ($J(R) = 0$), then R is semiprime.*

Proof. Let R be semiprimitive, then there exists a semisimple faithful module $M = \oplus L_i$. Now let $I \neq 0$, then there exists simple L such that $IL \neq 0$. By density theorem, we pick $v \in L$, $x \in I$ such that $xv \neq 0$. So there exists $y \in R$ such that $yxv = v$. It follows that $(yx)^n v \neq 0$ for all n, therefore $I^n \neq 0$. □

The aim for this section is to prove the following.

Theorem 3.64 (Goldie's theorem)**.** *Let R be semiprime and right noetherian, then the set S consisting of regular elements is an Ore set and R_S is an Artinian semisimple ring.*

Corollary 3.65. *For any right (or left) noetherian ring R, there exists a homorphism:*

$$R \to \text{Mat}_n(D)$$

In particular,R has IBN property.

Proof. Consider:

$$R \to R/J(R) = \overline{R} \to \overline{R}_S$$

□

Remark. *Let's appreciate Goldie's theorem for commuatative ring R. In this case R is a reduced ring, so that we can write:*

$$0 = \cap_{i=1}^n \mathfrak{p}_i$$

It follows: $R_S = \prod k_i$ for k_i the residue field of generic points of each component.

The rough idea to prove Goldie's theorem is that $aR \cap sR \neq 0$ because sR is big and hard to miss. To begin with, we need a measure of "size".

Definition 3.66. Let $0 \neq M$ be a R-module, a submodule $N \subset M$ is called **essential** if $N \neq 0$ and for each $0 \neq T \subset M$, we have $T \cap N \neq 0$.

Example. *Let M be a finite length module. Then N is essential iff N contains the socle of M.*

Lemma 3.67. *1. If $M'' \subset M' \subset M$ with M'' essential in M' and M' essential in M, then M'' is essential in M.*

2. Given $M \to N$. Let N' be essential in N, then its preimage in M is essential in M.

3. Suppose $M' \subset M$, $N' \subset N$ are essential, then $M' \oplus N' \subset M \oplus N$ is essential.

Proof. For part(c), apply (a) to:

$$M' \oplus N' \subset M' \oplus N \subset M \oplus N$$

□

Lemma 3.68. *(Existence of essential complement). Given $N \subset M$, then we can find $N' \subset M$ such that $N \oplus N' \subset M$ is essential.*

Proof. Choose N' to be maximal submodule such that $N' \cap N$. □

Corollary 3.69. *If M has no proper essential submodules iff M is semisimple.*

Proof. Recall if M is semisimple iff for all $M' \subset M$, we can find a complement M'', hence $\Longleftarrow$. Conversely, if M is not semisimple, choose M' with no complement. But by lemma we can find an essential complement, which gives a direct complment in this case. Contradiction. □

Definition 3.70. A module is called **uniform** if every nonzero submodule is essential.

Remark. *M is uniform iff there is no:*

$$M' \oplus M'' \subset M$$

for M', M'' nonzero.

Example. *If M has finite length, then M is uniform is equivalent to saying M has simple socle.*

Definition 3.71. The **Goldie rank** of a module M is the maximal n such that:

$$M \supset \bigoplus_{i=1}^{n} N_i$$

for $N_i \neq 0$.

Proposition 3.72. *Let M be a noetherian module.*

1. *There exists uniform submodules $M_1, \cdots, M_n$ such that $M_1 \oplus \cdots \oplus M_n \subset M$ is essential.*

2. *The number n equals the Goldie rank of M.*

3. *Any submodule of M has Goldie rank $\leq n$; a submodule of Goldie rank n is essential.*

Proof. 1. Suppose not. In particular M is not uniform, so I can take $M' \subset M$ not essential with essential complement M''. If M', M'' contains essential finite sum of uniform submodules, then we get a contradiction by taking direct sum of these. Otherwise there must be at least one of M', M'' to violate the condition. Without loss M'' violate this condition and so I can (use the same argument) to write:

$$M'' \supset N \oplus N'$$

So that I have:

$$M' \oplus N \oplus N' \subset M$$

Continue this process indefinitely we get an increasing chain of submodules, contradicting the noetherian property of M.

2. Suppose not, then there exists

$$N = N_1 \oplus \cdots \oplus N_m \subset M$$

for $m > n$. Let

$$N' = N_2 \oplus \cdots \oplus N_m$$

This is not essential in M (since $N' \cap N_1 = 0$).

<u>*Claim:*</u> At least one of $N' \cap M_1, N' \cap M_2, \cdots N' \cap M_n$ equals zero.

If we have all nonzero $N' \cap M_i$, then $N' \cap M_i \subset \oplus M_j$ are all essential (since $\oplus M_j$ is uniform). But $\oplus M_j \subset M$ is essential, by transitivity we know: $N' \cap M_i \subset M$ is essential. It follows that:

$$N' \supset \oplus N' \cap M_i$$

is essential in M. Contradiction. Therefore we proved the claim. Now we may assume $N' \cap M_1 = 0$, therefore:

$$M \supset M_1 \oplus N_2 \oplus \cdots \oplus N_m$$

Continuing inductively, and since $m > n$ we can finally get:

$$M \supset M_1 \oplus M_2 \oplus \cdots \oplus M_n \oplus N_{n+1} \oplus \cdots \oplus N_m$$

Contradicting that $\oplus M_i$ is essential.

3. Submodules have smaller Goldie rank is clear. Now suppose $N \subset M$ has Goldie rank n, then I can find:

$$N_1 \oplus \cdots \oplus N_n \subset N \subset M$$

Note that $N_1 \oplus \cdots \oplus N_n$ is already essential in M, since otherwise the essential complement gives another summand. It follows that N is essential. In fact, in this case each N_i is automatically uniform.

□

Let's denote $\mathrm{rAnn}, \mathrm{lAnn}$ for right annhilator and left annhiltor. The key fact for the proof of Goldie's theorem is the following:

Proposition 3.73. *Let R be right Noetherian semiprime, with $I \subset R$ a right ideal. Then there exists $x \in I$ such that:*

$$\mathrm{rAnn}(x) \cap I = 0$$

We also need:

Lemma 3.74. *If R is right noetherian semiprime, $I \subset R$ is essential right ideal. Then $\mathrm{lAnn}(I) = 0$.*

Proof. Assume for contradiction $J = \mathrm{lAnn}(I) \neq 0$. Consider the set:

$$S = \{I' \supset I \text{ right ideal} \mid \mathrm{lAnn}(I') \neq 0\}$$

We may replace I by the maximal element in S (since R is noetherian) which is still essential, and at the same type update J. Note that $J^2 \neq 0$. Otherwise $(JR)^2 = 0$ for the two-sided ideal JR, contradicting the semiprime condition of R. Therefore, we can find $x, y \in J$ with $xy \neq 0$. Now since I is essential, so that:

$$yR \cap I \neq 0$$

So we can find $r \in R$ such that $yr \neq 0 \in I$ and therefore $xyr = 0$. It follows that:

$$r \in \mathrm{rAnn}(xy), r \notin \mathrm{rAnn}(y)$$

So that:

$$\mathrm{rAnn}(xy) \supsetneq \mathrm{rAnn}(y) \supset I$$

But $xy \in \mathrm{lAnn}(\mathrm{rAnn}(xy))$, contradicting the maximality of I. □

Corollary 3.75. *Let R be right noetherian semiprime ring. An essential right ideal I contains a regular element.*

Proof. By the above proposition we can find x such that $\mathrm{rAnn}(x) \cap I = 0$. But I is essential, this means $\mathrm{rAnn}(x) = 0$. This means that xR is a rank 1 free right R-module. It follows that $xR \subset R$ is essential since they have the same Goldie rank. Then by the previous lemma we know that $\mathrm{lAnn}(x) = 0$. The result follows. □

Now the corollary implies Goldie's theorem.

Proof. (of Goldie's theorem). Let S be the collection of regular elements. Let's check O1:

$$aS \cap sR \neq \emptyset, \quad a \in R, s \in S$$

Now $sR \simeq R$ as right R-module, so Goldie rank of sR equals the Goldie rank of R, therefore sR is essential right ideal. Now consider:

$$L_a : R \to R$$

The preimage of essential ideal sR is also an essential ideal, i.e. $\{x \mid ax \in sR\}$ is essential. By corollary we know it contains a regular element t and thus $at \in sR$ as desired.

Now we need to check that R_S is Artinian semisimple. Since R_S is noetherian, it suffices to see R_S is semisimple as a right module over itself. But this is equivalent to saying that R_S has no essential right ideals except R_S. If $I \subset R_S$ is essential, then $I \cap R$ is essential in R, so it contains an element in S. Therefore $I = R_S$. □

It remains to prove Proposition 3.73.

Proof. (of Proposition 3.73) Let's first prove the result when I is uniform. Note that $I^2 \neq 0$, so we can pick $x, y \in I$ such that $xy \neq 0$. In this case we claim

$$\mathrm{rAnn}(x) \cap I = 0$$

Otherwise $\mathrm{rAnn}(x) \cap I$ is essential in I. Now consider:

$$L_y : R \to I, r \mapsto yr$$

Then the preimage of $\mathrm{rAnn}(x) \cap I$ is essential in R. So:

$$\{r \in R \mid xyr = 0\} = \mathrm{rAnn}(xy)$$

is essential in R. Now left annhilator of any right essential ideal is 0, so that:

$$xy \in \mathrm{lAnn}(\mathrm{rAnn}(xy)) = 0$$

Contradiction.

Now let's turn to the general case. Let's pick $J \subset I$ maximal such that there exists $v \in J$ with $\mathrm{rAnn}(v) \cap J = 0$. We pick such v. If $\mathrm{rAnn}(v) \cap I = 0$ then we're done. So we may asssume $\mathrm{rAnn}(v) \cap I \neq 0$. Let's choose $U \subset \mathrm{rAnn}(v) \cap I$ a uniform ideal (this choice can be made since every has finite Goldie rank in our setting). Then by previous discussion we know there exists $u \in U$ such that $\mathrm{rAnn}(u) \cap U = 0$. We claim that:

$$\mathrm{rAnn}(u + v) \cap U \oplus J = 0$$

and this will contradict the maximality of J. So it suffices to show this claim. The first observation is that $U \cap J = 0$, so that if:

$$x \in \mathrm{rAnn}(u + v) \implies ux + vx = 0 \implies ux = vx = 0 \implies x \in \mathrm{rAnn}(u) \cap \mathrm{rAnn}(v)$$

Now suppose I have $x = u' + v' \in \mathrm{rAnn}(u + v) \cap U \oplus J$. Note that:

$$0 = vx = vu' + vv' = vv'$$

because $u' \in U \subset \mathrm{rAnn}(v)$. Hence $v' \in J \cap \mathrm{rAnn}(v) = 0$. So $x = u'$, and I must have:

$$0 = uu' \implies u' \in U \cap \mathrm{Ann}(u) = 0$$

This justifies $x = 0$, we're done. □

Lemma 3.76. *Let R be right noetherian and semiprime. If $I \subset R$ is a uniform ideal, then $I_S \subset R_S$ is simple.*

Proof. One can check that uniformality is preserved under localization. So that I_S is uniform semisimple, hence is simple. □

Remark. *If R is a prime ring, then $R_S = \mathrm{Mat}_n(D)$. In this case n is the Goldie rank of R.*

Let's conclude by mentioning two important applications of Goldie rank.

1. Let $\mathfrak{g}$ be a simple Lie algebra. Suppose one want to classify primitive ideals of $U(\mathfrak{g})$. One can consider the ring $R = U(\mathfrak{g})/I$ and its associated invariants. They're dependent on something called "Goldie rank polynomial".

2. Gelfand-Kirillov conjecture. Let $\mathfrak{g}$ be a finite dimensional Lie algebra over $\mathbb{C}$. Denote $D(\mathfrak{g})$ as the fraction field of $U(\mathfrak{g})$. On the other hand, let

$$W_{n,k} = \mathbb{C}(z_1, \cdots, z_k)\langle x_i, y_j \mid i, j \leq n\rangle/([x_i, y_j] = \delta_{ij} \cdots)$$

be the Weyl algebra of n variables over the a field of pure transcendence degree k over $\mathbb{C}$. Denote $D_{n,k}$ as the fraction field of $W_{n,k}$.

Conjecture 3.77. *For each reductive Lie algebra $\mathfrak{g}$, we can find n, k such that:*

$$D(\mathfrak{g}) \simeq D_{n,k}$$

Gelfand and Kirillov proved the conjecture for $\mathfrak{gl}_n$ and $\mathfrak{sl}_n$, but it is false in general. However, if we replace $D(\mathfrak{g})$ by the fraction field of:

$$U(\mathfrak{g}) \otimes_{Z(U_\mathfrak{g})} \mathbb{C}_\psi$$

for some central character ψ, then the statement is true for each type (including exceptional types).

3.6 PI Algebras

Definition 3.78. A ring R is a **polynomial identity (PI)** ring if there exists nonzero:

$$P \in \mathbb{Z}\langle x_1, \cdots, x_n\rangle$$

such that $P(r_1, \cdots, r_n) = 0$ for all $r_1, \cdots, r_n \in R$. Likewise, we can talk about PI algebras over a field k.

Example. *Commutative rings are PI: $P = xy - yx$.*

Example. *Boolean algebras are PI: $P = x^2 - x$.*

Example. *If A is a finite dimensional k-algebra of dimension n, then A is PI. It satisfies:*

$$S_{n+1} = P(x_1, \cdots, x_{n+1}) = \sum_{\sigma \in \Sigma_{n+1}} (-1)^{|\sigma|} x_{\sigma_1} \cdots x_{\sigma_{n+1}}$$

Lemma 3.79. *1. If A is PI ring, then it satisfies a multilinear identity.*

2. If A/k is a PI algebra satisfying $P = \sum P_i$ for P_i homogeneous of degree i, then P_i is also an identity.

Proof. 1. Given $P(x_1, \cdots, x_n)$ such that A satisfies, we do double integration on the top degree of P in each variable and the number of variables in which it has that degree. Suppose $r > 1$ is the top degree, without loss we may assume P has degree r in x_1, then let's consider:

$$Q(x_0, \cdots, x_n) = P(x_0 + x_1, \cdots, x_n) - P(x_0, x_2, \cdots, x_n) - P(x_1, x_2, \cdots, x_n)$$

Now Q holds in our ring and that it has in both x_0, x_1 smaller than r; the degree of other variables will not increase. One can also check that Q is not the zero polynomial. Therefore by induction we can find an identity P such that A satisfies with each variable has degree ≤ 1. Assume P is not linear in some variable x_i, we can consider:

$$P(x_1, \cdots, x_n) - P(x_1, \cdots, x_{i-1}, 0, x_{i+1}, \cdots, x_n)$$

and again do induction.

2. Rescale x by λ, then P_i is rescaled to $\lambda^i P_i$. Now span of these contain P_i using Vendermont determinant.

□

Theorem 3.80 (Amitsur-Levitzki)**.** *Let $A = \mathrm{Mat}_n(k)$, then the identity S_{2n} holds. Moreover, it has satisfies no nonzero homogeneous polynomials with lower degree.*

Proof. 1. Let's first try to prove the second statement. By lemma above we only need to consider multilinear identities. Let's find $a_1, \cdots, a_{2n-1} \in A$ such that:

$$a_1 \cdots a_{2n-1} \neq 0$$

but

$$a_{\sigma(1)} \cdots a_{\sigma(n)} = 0, \forall \sigma \neq \mathrm{id}$$

Indeed we can take elementary matrices:

$$E_{11}, E_{12}, E_{22}, E_{23}, \cdots, E_{nn}$$

Moreover the first r matrices in this collection satisfies the same property. Now give any homogeneous polynomial of degree $r < 2n$, and without loss let's assume it contains a monomial $x_1 \cdots x_r$. Let's plug in:

$$x_1 = a_1, \cdots, x_r = a_r$$

and the polynomial will not vanish.

2. Before giving a proof for the other direction, let's first introduce a conceptual proof for Cayley Hamilton (for matrices over an arbitrary *commutative* ring R). Let $x \in A = \mathrm{Mat}_n(R)$, then $A[t]$ acts on A by letting t acts by right multiplication by x. Then $(x-t)(I_n) = 0$. We view $x - t \in \mathrm{Mat}(R[t])$, and then it has determinant $P_x(t)$ the caracteristic polynomial. Therefore it has an adjoint y such that:

$$y(x-t) = (x-t)y = \det(x-t) = P_x(t) \cdot I_n$$

It follows that:

$$0 = y(x-t)(I_n) = P_x(t) \cdot I_n$$

Therefore $P_x(t) = 0$.

3. Let's now turn to the proof of the the harder direction of the theroem. We may assume $k = \mathbb{Q}$. Now let's introduce the auxillary ring:

$$\Lambda = \Lambda^\bullet(\mathbb{Q}^{2n})$$

Let $\varepsilon_1, \cdots, \varepsilon_{2n}$ be a basis for $\mathbb{Q}^{2n}$. For $x_1, \cdots, x_{2n} \in \mathrm{Mat}_n(\mathbb{Q})$, let's consider:

$$x = \varepsilon_1 x_1 + \cdots + \varepsilon_{2n} x_{2n} \in \mathrm{Mat}_n(\mathbb{Q}) \otimes \Lambda = \mathrm{Mat}_n(\Lambda)$$

Note that:

$$x^{2n} = S_{2n}(x_1, \cdots, x_{2n}) \varepsilon_1 \wedge \varepsilon_2 \wedge \cdots \wedge \varepsilon_{2n}$$

Therefore it suffices to prove that $x^{2n} = 0$. In order to use Cayley Hamilton, we must restrict ourselves to the commutative ring:

$$\Lambda^{\mathrm{even}} = \mathbb{Q} \oplus \Lambda^2 \mathbb{Q} \oplus \cdots$$

and check that $x^2 \in \mathrm{Mat}_n(\Lambda^{\mathrm{even}})$ has characteristic polynomial t^n. Indeed, we only need to check that:

$$\mathrm{Tr}(x^{2i}) = 0$$

But note that:

$$x^{2i} = \sum_{\sigma:[2i]\to[2n]} x_{\sigma(1)} \cdots x_{\sigma(2i)} \varepsilon_{\sigma(1)} \wedge \cdots \wedge \varepsilon_{\sigma(2i)}$$

By considering a Σ_{2i}-symmetry we see that $\mathrm{Tr}(x^{2i}) = 0$ as desired.

4. Let's give another proof using Lie algebra cohomology. We need the following facts:

   (a) Let $\mathfrak{g} = \mathfrak{gl}_n$. Recall that $H^*(\mathfrak{g}) = \mathrm{Ext}_{U\mathfrak{g}}(\mathbb{C}, \mathbb{C})$. From Chevalley complex we can see:

   $$H^i(\mathfrak{g}) = (\bigwedge^i \mathfrak{g}^*)^G$$

   (b) I have isomorphisms:

   $$H^*(\mathfrak{g}) \simeq H^*(U(n), \mathbb{C}) = H^*(\mathrm{GL}_n(\mathbb{C}), \mathbb{C})$$

   Using the fibration:

   $$U(n-1) \to U(n) \to S^{2n-1}$$

   From this we see that $H^*(U(n))$ is freely generated by $\xi_1, \cdots, \xi_n$ for $\deg(\xi_i) = 2i - 1$.

Note that $S_n(x_1, \cdots, x_{2n}) = 0 \iff$ for all x_0 we have $\mathrm{Tr}(S_{2n+1}(x_0, x_1, \cdots, x_{2n})) = 0$. This is because:

$$\mathrm{Tr}(S_{2n+1}(x_0, x_1, \cdots, x_{2n})) = (2n+1)\mathrm{Tr}(x_0 \cdot S_n(x_1, \cdots, x_n))$$

and trace pairing is perfect. Now let $\tau_{2n+1}(x_0, \cdots, x_{2n}) = \mathrm{Tr}(S_{2n+1}(x_0, \cdots, x_n))$, we only need to show that $\tau_{2n+1} \mid_{\mathrm{Mat}_n} \equiv 0$. However, note that:

$$\tau_{2n+1} \in H^{2n+1}(\mathfrak{g})$$

By induction we may assume $\tau_{2n-1} \mid_{\mathfrak{gl}_{n-1}} \equiv 0 \implies \tau_{2n+1} \mid_{\mathfrak{gl}_{n+1}} \equiv 0$. Therefore:

$$\tau_{2n+1} \in \ker(H^{2n+1}(\mathfrak{gl}_n) \to H^{2n+1}(\mathfrak{gl}_{n-1})) = 0$$

□

Example. *Let $\mathfrak{g}$ be a finite dimensional reductive Lie algebra over k, then $U(\mathfrak{g})$ is PI iff* $\mathrm{char}(k) \neq 0$. *Indeed, over $\mathbb{C}$ there are finite dimensional irreducible representation of arbitrarily large dimension d. Then I have:*

$$U\mathfrak{g} \twoheadrightarrow \mathrm{Mat}_d(\mathbb{C})$$

for arbitrarily large d. By Amitsur-Levitzki, we know $U\mathfrak{g}$ is not PI. In characteristic p, $U\mathfrak{g}$ is a (finite) Azumaya algebra over its center $Z(U\mathfrak{g})$. Then we just apply AL to $\mathrm{Frac}(Z(U\mathfrak{g}))$.

Let's list a few theorems on PI algebras.

Theorem 3.81 (Kaplansky)**.** *A primitive PI algebra over a field k is a CSA over (another) field K.*

Theorem 3.82 (Posner)**.** *Let A be a prime PI algebra A with center Z a domain. Then:*

$$A \otimes_Z \mathrm{Frac}(Z)$$

is a CSA.

Theorem 3.83 (Rowen)**.** *Let A be a semiprime PI algebra, then every nonzero ideal meets the center.*

The proof of Rowen's theorem relies on the existence of **central polynomials**: for all n, there exists a noncommutative multilinear polynomial in $N = 2n^2 + 2$ variables such that the map:

$$\mathrm{Mat}_n(k)^N \to \mathrm{Mat}_n(k)$$

is nonzero and lands in scalar matrices.

3.7 Growth of Algebra

Theorem 3.84. *Warfield For any $d \geq 2$, there exists an algebra A with* $\mathrm{GKdim}(A) = d$.

Lemma 3.85. $\mathrm{GKdim}(A[t]) = \mathrm{GKdim}(A) + 1$.

Proof. By the lemma, it suffices to take $d \in (2, 3)$. Given an increasing sequence γ_n, define:

$$I \subset k\langle x, y\rangle$$

spanned by all monomials ≥ 3 in y and monomials:

$$x^i y x^j y x^k, j < \gamma_n, n = i + j + k$$

The quotient $k\langle x, y\rangle / I$ is a graded algebra by degree in x. Moreover:

$$\dim A_n = 1 + n + \binom{n + 2 - \gamma_n}{2}$$

If we take $q = (0, 1)$ and $\gamma_n = n - [n^q]$, then one can compute:

$$\mathrm{GKdim}(A) = \max(2, 2q + 1)$$

We're done. □

Theorem 3.86 (Belov-Ivanov)**.** *For any $\alpha \in \mathbb{Q}$ and $\alpha > 5$, there exists a finitely presented algebra of GK dimension α.*

Theorem 3.87 (Bergman gap theorem)**.** *There is no finitely generated algebra with GK dimension $\in (1, 2)$.*

Proof. (sketched) Without loss we may assume A is graded generated in degree 1. The key propostion is the following:

Proposition. *If* $\dim A_d < d$ *for some d, then there exists N such that* $\dim A_k < N$ *for all k. It follows* $\dim A = 0$ or 1.

It is clear that the proposition implies the theorem. Without loss we may assume all relations are monomials of degree d. The rest is a combinaotrial argument. □

Theorem 3.88. *(A. Smoktunowicz) A graded domain cannot have GK dimension in $(2, 3)$.*

Theorem 3.89. *(Berele). A finitely generated PI algbera has finite GK dimension.*

Remark. *There are finitely presented algebras of intermediate growth: namely it's sub-exponential and sup-polynomial. For groups, there are finitely generated groups of intermediate growth; the existence of finitely presented group of intermediate growth is open.*

Example. *Consider:*

$$A = k[x_1, x_2, \cdots]$$

with $\deg(x_i) = i$. *Then the Poincaré series reads:*

$$P_A(t) = \frac{1}{(1-t)(1-t^2)\cdots}$$

It converges for $|t| < 1$. *This shows that* A *has subexponential growth. On the other hand:*

$$\limsup(\dim A_n)^{\frac{1}{n}} = 2 > 1$$

because the radius of convergence of $P_A(t)$ *is* $\frac{1}{2}$. *This shows* A *has sup-polynomial growth.*

Example. *Let's present examples of finitely presented algebras with intermediate growth. We claim that if* L *is an infinite dimensional Lie algebra with polynomial growth, then* $U(L)$ *must have intermediate growth. Indeed, using PBW we see that its envelopoing has growth equals that of* $k[x_1, x_2, \cdots]$ *such that the number of generators of* $\deg i = \dim L_i/L_{i-1}$.

Concretely we can take: $L = \mathrm{Vect}_{\mathrm{pol}}(\mathbb{C}^*) = \langle v_i \mid i \in \mathbb{Z} \rangle$, *where* $v_i = z^{i+1}\partial_z$ *so that* $[v_j, v_i] = (j-i)v_{i+j}$. *It's easy to show that* L *is finitely presented as a Lie algebra (it is generated by* $v_i, -2 \leq i \leq 2$ *with obvious relations), and has linear growth. This implies* $U(L)$ *is finitely presented with intermediate growth.*

3.8 Noncommutative Geometry

Recall that commutative algebra is closed related to algebraic geometry. The goal of this section is to give an overview about the connection between geometry and noncommutative algebras. Let's discuss several different point of views.

Many interesting noncommutative algebras are defined by deformation / quantization of commutative algebras. Let A/k be a commutative algebra, recall the order n-deformation of A is a an associative algebra $\widetilde{A}_{\hbar,n}$ over $k[\hbar]/(\hbar^{n+1})$, which is free as a module, together with the data of an isomorphism:

$$\widetilde{A}_{\hbar,n}/\hbar \simeq A$$

where we take the convention $A_{\hbar,0} = A$. Let's consider:

$$A_\hbar = \varprojlim A_{\hbar,n}$$

Remark. *Given* $A_\hbar$, *fix* $S \subset A$ *be a multiplicative set, then:*

$$\widetilde{S} = \{\widetilde{a} \mid \widetilde{a} \bmod \hbar \in S\}$$

is an Ore subset of $A_\hbar$.

Therefore $(A_\hbar)_{\widetilde{S}}$ is a quantization of A_S. It means that, given an algebraic variety $(X, \mathcal{O}_X)$, one might be able to deform $\mathcal{O}_X$ into a sheaf flat $k[[\hbar]]$-algebra $\mathcal{O}_{X,\hbar}$ and do geometry.

Recall that if A has a quantization $A_\hbar$, this makes A a Poisson algebra via:

$$\{f,g\} = \frac{\widetilde{f}\widetilde{g} - \widetilde{g}\widetilde{f}}{\hbar} \bmod \hbar$$

A natural question to ask is that if I start from some Poisson algebra $(A, \{-,-\})$(or Poisson variety), must it come from some deformation of A? The answer is positive in many cases.

Theorem 3.90. *(Kontsevich) Let $A = \mathbb{C}[x_1, \cdots, x_n]$ with any Poisson bracket, one can construct a canonical quantization $A_\hbar = A[[\hbar]]$ with multiplication:*

$$f * g = fg + \sum_n \hbar^n \sum_\Gamma w_\Gamma D_\Gamma(f \otimes g)$$

where Γ runs through certain graphs, $w_\Gamma \in \mathbb{C}$, and D_Γ are some differential operators.

Remark. *Similar expressions appear in QFT (e.g. Feynmann graphs), and w_Γ is defined via some complicated integral.*

In practice, many interesting examples of noncommutative algebra arise as quantization of certain Poisson algebra A and specialzing at $\hbar = 1$.

Example. *Let X be an affine algebraic variety and*

$$A = \mathcal{O}(T^*X) = \mathbb{C}[f, v \mid f \in \mathcal{O}(X), v \in Der(\mathcal{O}_X)]$$

This is a Poisson algebra (since T^*X is symplectic). We can define its quantization:

$$A_\hbar = \mathbb{C}[[h]]\langle f, v\rangle / ([v, f] = \hbar v(f))$$

Specializing to $\hbar = 1$ gives me $A_1 = \mathcal{D}(X)$ the algebra of differential operators on X. Indeed, this construction can be done for any algebraic variety and define a sheaf of differential operators on X.

Example. *Let $\mathfrak{g}$ be a Lie algebra so that $\mathfrak{g}^*$ is a Poisson variety. We wish to define a quantization of $A = \mathbb{C}[\mathfrak{g}^*]$. Indeed:*

$$A_\hbar = T^\bullet \mathfrak{g}[[h]] / (xy - yx = \hbar[x, y])$$

Specializing to $\hbar = 1$ gives me $U(\mathfrak{g})$ the universal everloping algebra of the Lie algebra.

Let's assume $\mathfrak{g}$ is reductive and set:

$$U_\hbar(\mathfrak{g}) = \mathbb{C}\langle \mathfrak{g} \rangle / (xy - yx = \hbar[x, y])$$

and

$$\overline{U}_\hbar(\mathfrak{g}) = U_\hbar \otimes_{Z(U_\hbar)} k[\hbar]$$

Then $\overline{U}_0 = \mathcal{O}(\mathcal{N}) = \mathcal{O}(T^*\mathcal{B})$ is just the function on the nilpotent cone of $\mathfrak{g}$.

Noncommutative algebras also arise in the study of algberaic geometric objects. Morita equivalence can be generalized to *derived Moritat equivalence*: we say A, B are derived Morita equivalent if:

$$D^b(A\text{-mod}) \simeq D^b(B\text{-mod})$$

Remark. *If A, B are derived Morita-equivalent, then $K(A\text{-mod}) \simeq K(B\text{-mod})$. The change of basis matrix of this isomorphism is very interesting.*

This can be generalized in the context of algebraic geoemtry. Let X be an algebraic variety, we consider $D^b(X) \coloneqq D^b(\mathrm{Coh}(X))$. We call X, X' are derived equivalent if $D^b(X) \simeq D^b(X')$. In this case write $X \sim X'$.

Theorem 3.91. *If $X \sim X'$ are smooth projective varieties, then $H^*(X) \simeq H^*(X')$.*

We call a variety X derived affine if there exists an algebra A such that:

$$D^b(X) \simeq D^b(A)$$

A striking fact is that many varieties are derived affine.

Example. *Let $X = \mathbb{P}^n$, then $\mathrm{Coh}(X) = k[x_0, \cdots, x_n]\text{-mod}_{\mathrm{fg,gr}}/k[x_0, \cdots, x_n]\text{-mod}_{\mathrm{gr}}^{\mathrm{f.dim}}$. Let's set*

$$\mathcal{E} = \mathcal{O} \oplus \mathcal{O}(1) \oplus \cdots \oplus \mathcal{O}(n)$$

and $A = \mathrm{End}(\mathcal{E})^{\mathrm{op}}$. This algebra can be understood using the formula:

$$\mathrm{Hom}(\mathcal{O}(i), \mathcal{O}(j)) = \mathrm{Sym}^{j-i}(\langle x_0, \cdots, x_n \rangle)$$

Then $D^b(X) \simeq D^b(A)$.

Example. *Let $X = \mathrm{Hilb}^n(\mathbb{C}^2)$ be the Hilbert scheme of n points on $\mathbb{C}^2$. This scheme is important because it sits inside the resolution of singularity:*

$$X \to \mathbb{C}^{2n}/S_n$$

Then X is derived affine:

$$D^b(A) \simeq D^b(X)$$

for $A = S_n \# \mathbb{C}[x_1, \cdots, x_n, y_1, \cdots, y_n]$.

Let us also discuss some noncommutative differential geometry. Let X be a smooth affine variety, we proved that:

$$\mathrm{HH}_i(\mathcal{O}(X)) \simeq \Omega_X^i$$

Recall that HH_i is Morita-invariant, so it actually makes sense to talk about $\mathrm{HH}_i(\mathcal{A})$ for some abelian category $\mathcal{A}$. For X smooth and projective, we have:

$$\mathrm{HH}_i(\mathrm{Coh}(X)) = \bigoplus_{q+p=i} H^q(X, \Omega^p) \simeq H^i_{\mathrm{DR}}(X)$$

where the first equivalence is the Hochschild-Kostant-Rosenberg isomorphism, and the second equivalence comes from Hodge theory.

Tor recover H^*_{DR} for nonprojective X, we can use *cyclic homology*. The bar complex for HH_* has cyclic symmetry:

$$C : a_0 \otimes \cdots \otimes a_n \to (-1)^n a_n \otimes a_0 \otimes \cdots \otimes a_{n-1}$$

Then $\mathrm{Bar}/(C - \mathrm{id})\mathrm{Bar}$ inherits the differential form from the bar complex, and it has cohomology:

$$\mathrm{HC}_n = \Omega^n / d\Omega^{n-1} \oplus \bigoplus_{i \geq 1} H^{n-2i}_{\mathrm{DR}}(X)$$

We also define:

$$\mathrm{HC}^{\mathrm{per}}_n = \varinjlim \mathrm{HC}_{n+2i} = \bigoplus H^{n+2i}_{\mathrm{DR}}(X)$$

There will be a spectral sequence converges to $\mathrm{HC}^{\mathrm{per}}_*$ whose first page is Hochschild cohomology. Hodge decomposition can be phrase as the spectral sequence degenerates after the first page.

Note that if X smooth projective, $D^b(X)$ is a dg category with finite homological dimension. We have the following vast generalization of Hodge theory.

Conjecture 3.92 (Kontsevich)**.** *Let $\mathcal{A}$ be a dg category with finite homological dimension. Then Hodge theory holds.*

This is partly proved by Dmitry Kaledin and Akhil Mathew.

A Homework 1

Exercise A.1. *Show the converse of Schur lemma doesn't hold by presenting a three dimensional algbera $A/\mathbb{C}$ with a reducible two-dimensional module M with $\mathrm{End}_A(M) = \mathbb{C}$. Can a module whose endomorphism form a division ring be decomposable?*

Proof. Consider:

$$A = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \subset \mathrm{Mat}_2(\mathbb{C})$$

and M is the natural 2-dimension module of A. Note that M is reducible since it has a submodule $k \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Let $f : M \to M$ be any endomorphism. Note that $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ generates M, so that f is determined by

$$f(\begin{pmatrix} 0 \\ 1 \end{pmatrix}) = \begin{pmatrix} x \\ y \end{pmatrix} \implies f(\begin{pmatrix} a & b \\ & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}) = \begin{pmatrix} a & b \\ & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \iff f(\begin{pmatrix} b \\ 0 \end{pmatrix}) = \begin{pmatrix} ax + by \\ 0 \end{pmatrix}$$

This must holds for all choice a, b. So that we must have $x = 0$. Therefore, $f(\begin{pmatrix} 0 \\ 1 \end{pmatrix}) = \begin{pmatrix} 0 \\ y \end{pmatrix}$. It follows that $f(m) = ym$ for each $m \in M$ since $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ generates. For the second part, the answer is no. Suppose $M = N \oplus P$ is decomposable. We can pick:

$$f = \mathrm{id}_N \oplus 0_P \in \mathrm{End}_A(M), \quad g = 0_N \oplus \mathrm{id}_P \in \mathrm{End}_A(M)$$

but it follows that $f \circ g = 0$ with $f, g \neq 0$. Contradiction. □

Exercise A.2. *Consider*

$$R = \{(a_{ij}) \in \mathrm{Mat}_2(\mathbb{R}) \mid a_{21} = 0, a_{22} \in \mathbb{Q}\}$$

Show that R is left noetherian and Artinian, but is neither right noetherian nor Artinian. Deduce that R is not isomorphic to R^{op}

Proof. Let's classify the left ideals of R. The basic compuation is:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & q \end{pmatrix} = \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & q \end{pmatrix} = \begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & q \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & q \end{pmatrix}$$

Therefore if there exists $\begin{pmatrix} a & b \\ 0 & q \end{pmatrix} \in I$ with $q \neq 0$, then:

1. If $a \neq 0$, then this ideal must be the whole ring R.
2. If $a = 0$ for all elements in this left ideal, then:

$$I = \begin{pmatrix} 0 & * \\ 0 & * \end{pmatrix}$$

Now let's assume for all elements in I we have $q = a_{22} = 0$, so:

$$I \subset \begin{pmatrix} * & * \\ 0 & 0 \end{pmatrix}$$

In fact then the three elementary matrix above acts as: $(1, 0, 0)$. So that subideals correspond to $\mathbb{R}$-sub-vector spaces of I. In particular, from this classification we see any chain of left ideals have finite length (length ≤ 3). Next we see that there are infiintely long ascending and descending chain of right ideals. In fact consider:

$$I = \{\begin{pmatrix} 0 & b \\ 0 & q \end{pmatrix} \mid b \in \mathbb{R}, q \in \mathbb{Q}\}$$

we compute:

$$\begin{pmatrix} 0 & b \\ 0 & q \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = 0, \begin{pmatrix} 0 & b \\ 0 & q \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = 0, \begin{pmatrix} 0 & b \\ 0 & q \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & b \\ 0 & q \end{pmatrix}$$

So that subideals of I correspond to $\mathbb{Q}$-subspaces of I. Since I as a $\mathbb{Q}$-vector space is not finite-dimensional, there obviously exists arbitrarily long ascending or descending chains. $\square$

Exercise A.3. *Let k be a field and $I \subset k[x, y]$ be the ideal generated by (x, y). Let $A = k[x, y]/I^2$. Show that $M_n = I^n/I^{n+2}$ is an indecomposable module. Construct an infinite dimensional indecomposable A-module.*

Proof. Suppose I have a decomposition $M_n = P \oplus Q$. Let $\overline{P}$ be the image of the map:

$$P \hookrightarrow M \to M/I^{n+1}$$

and similarly define $\overline{Q}$. We can view $\overline{P}, \overline{Q} \subset k[x, y]_n$. It is clear that $\dim \overline{P} + \dim \overline{Q} \geq n + 1$. Now if $\overline{P}, \overline{Q} \neq 0$. We claim that $\dim k\{x, y\} \cdot \overline{P} \geq \dim \overline{P} + 1$. To see this, we first observe that $x \cdot \overline{P} \neq y \cdot \overline{P}$. Since otherwise:

$$x \cdot \overline{P} = y \cdot \overline{P} \implies xy \mid \overline{P}$$

so that we can write $\overline{P} = xy\overline{P}'$ for $\overline{P}' \subset k[x, y]_{n-2}$. So that:

$$x^2 y\overline{P}' = xy^2\overline{P}' \implies x\overline{P}' = y \cdot \overline{P}'$$

since multiplication by xy is an injective map. But then we derive a contradiction via induction. So that we know $x \cdot \overline{P} \neq y \cdot \overline{P}$, therefore:

$$\dim k\{x, y\} \cdot \overline{P} \geq \dim \overline{P} + 1$$

Similarly

$$\dim k\{x, y\} \cdot \overline{Q} \geq \dim \overline{Q} + 1$$

Consider the map:

$$k\{x, y\} \cdot \overline{P} \oplus k\{x, y\} \cdot \overline{Q} \to k[x, y]_{n+1}$$

Note that LHS has $\dim \geq \dim \overline{P} + \dim \overline{Q} + 2 \geq n+3$, but RHS has dimension $n+2$. So this map must have a kernel. Therefore there exists $\overline{f} \in \overline{P}, \overline{g} \in \overline{Q}$ such that:

$$p(x,y)\overline{f} + q(x,y)\overline{g} = 0$$

where $p, q \in k\{x,y\}$ are linear maps. It follows that for any lift f, g of $\overline{f}, \overline{g}$, we have:

$$pf + qg = 0 \in I/I^2$$

this is a contradiction. Therefore, we must have $\overline{P} = 0$ or $\overline{Q} = 0$. Say $\overline{Q} = 0$ so that $Q \subset I^{n+1}/I^{n+2}$. But then there must be elements:

$$f_i = x^i y^{n-i} + g_i \in P$$

with $\deg g_i = n+1$, for all $i = 0, 1, \cdots, n$. It follows that:

$$xf_i = x^{i+1}y^{n-i} + xg_i = x^{i+1}y^{n-i} \in P$$

Similarly

$$yf_i = x^i y^{n-i+1} \in P$$

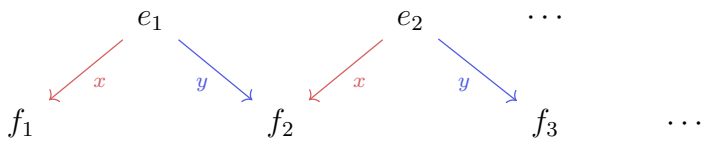
In particular $P \supset I^{n+1}/I^{n+2} \supset Q$. But then since $P \cap Q = 0 \implies Q = 0$. This proves our module is indecomposable. Now consider the vector space:

$$M = \mathbb{C}\{e_i, f_j \mid i, j \in \mathbb{Z}_{\geq 1}\}$$

with basis e_i, f_j, equipped with the A-action via:

$$x \cdot e_i = f_i, y \cdot e_i = f_{i+1}, x \cdot f_i = y \cdot f_i = 0$$

This is clearly an infinite dimensional A-module. It remains to see it is indecomposable. The following diagram shows explicit the module structue:



This is the "infinite analogue" of our previous construction. As before suppose I have $P \oplus Q = M$. As before we set $\overline{P}$ to be the image of the map:

$$P \to M \to M/(f_1, f_2, \cdots)$$

and similarly define $\overline{Q}$. We still can view $\overline{P}, \overline{Q} \subset \mathbb{C}\{e_1, e_2, \cdots\}$ as before. We will prove one of them is zero. Obviously $\overline{P} + \overline{Q} = \mathbb{C}\{e_1, e_2, \cdots\}$. Let's still consider the map:

$$k\{x,y\} \cdot \overline{P} + k\{x,y\} \cdot \overline{Q} \to \mathbb{C}\{f_1, f_2, \cdots\}$$

Now observe that $y \cdot \overline{P} + y \cdot \overline{Q} \to \mathbb{C}\{f_1, f_2, \cdots\}$ has image of codimension 1. If we have $\overline{P}, \overline{Q} \neq 0$, then $x \cdot \overline{P} \not\subseteq y \cdot \overline{P}$, $x \cdot \overline{Q} \not\subseteq y \cdot \overline{Q}$ by looking at where the first f_i appears. So that $k\{x,y\} \cdot \overline{P} \supsetneq y \cdot \overline{P}, k\{x,y\} \cdot \overline{Q} \supsetneq y \cdot \overline{Q}$. It follows that the above map has a kernel. So that we can argue as in the finite case to derive a contradiction. Therefore in fact we must have one of $\overline{P}, \overline{Q}$ equals 0, say $\overline{Q}$. Now it follows that $Q \subset (f_1, f_2, \cdots) = x \cdot \overline{P} \subset P$. This forces $Q = 0$. $\square$

Exercise A.4. *Compute the socle and co-socle filtration of the free rank one module for the following rings:*

1. $R = \mathbb{Z}/72$
2. $R = k[D_4]$, *where* k *is a field of characteristic two, and* D_4 *is the dihedral group of order* 8.

Proof. 1. Note that a R-module is simple if it is simple as a $\mathbb{Z}$-modules. But a $\mathbb{Z}$-module is simple iff it has form $\mathbb{Z}/p\mathbb{Z}$ for some prime p. It follows that the simple submodules of R are:

$$36\mathbb{Z}/72, 24\mathbb{Z}/72$$

it follows that $\mathrm{soc}(R) = 12\mathbb{Z}/72$. Now that $R/\mathrm{soc}(R) \simeq \mathbb{Z}/12\mathbb{Z}$, it has socle $2\mathbb{Z}/12\mathbb{Z}$. So the socle filtration looks like:

$$0 \subset 12\mathbb{Z}/72 \subset 2\mathbb{Z}/72 \subset \mathbb{Z}/72 = R$$

To specify cosocle filtration, we need to consecutively find Jacobson radical of modules. Start from $\mathbb{Z}/72\mathbb{Z}$, it has Jacobson radical $R_1 = 6\mathbb{Z}/72 \simeq \mathbb{Z}/12$. Now that the Jacobson radical of R_1 equals $R_2 = 36\mathbb{Z}/72 \simeq 6\mathbb{Z}/12$. Therefore the cosocle filtration is:

$$R = \mathbb{Z}/72 \supset 6\mathbb{Z}/72 \supset 36\mathbb{Z}/72 \supset 0$$

2. Note that D_4 has order group is a p-group for $p = 2$, so its only simple module is the trivial module. Now that:

$$M_1 = \mathrm{soc}(k[D_4]) = k \cdot \sum_{g \in D_4} g$$

Let M_2 be the second term in the socle filtration, we have:

$$M_2/M_1 = (M/M_1)^{D_4}$$

If we give D_4 the presentation:

$$\langle r, s \mid r^4 = s^2 = 1, srs = r^{-1} \rangle$$

then: if $\overline{m} = \overline{\sum_{i=0}^{3} a_i r^i + b_i r^i s} \in (M/M_1)^{D_4}$, we have:

$$rm - m \in D_4, sm - m \in D_4$$

It is evident that:

$$1 + r + r^2 + r^3, 1 + r^2 + s + r^2 s \in M_2$$

and they will be a basis for M_2/M_1. So that:

$$M_2 = k\{1 + r + r^2 + r^3, s + rs + r^2 s + r^3 s, 1 + r^2 + s + r^2 s\}$$

Again $M_3/M_2 \simeq (M/M_2)^{D_4}$, so that:

$$1 + r^2, s + r^2 s, 1 + r + s + rs \in M_3$$

It follows that:

$$M_3 = k\{1 + r^2, r + r^3, s + r^2 s, rs + r^3 s, 1 + r + s + rs\}$$

Similar computation shows:

$$M_4 = k\{1 + r, 1 + r^2, 1 + r^3, s + rs, s + r^2 s, s + r^3 s, 1 + s\}$$

and $M_5 = k[D_4]$. So the cosocle fitration is:

$$0 \subset M_1 \subset M_2 \subset M_3 \subset M_4 \subset M_5 = k[D_4]$$

for:

$$M_1 = k \cdot \sum_{g \in D_4} g$$

$$M_2 = k\{1 + r + r^2 + r^3, s + rs + r^2 s + r^3 s, 1 + r^2 + s + r^2 s\}$$

$$M_3 = k\{1 + r^2, r + r^3, s + r^2 s, rs + r^3 s, 1 + r + s + rs\}$$

$$M_4 = k\{1 + r, 1 + r^2, 1 + r^3, s + rs, s + r^2 s, s + r^3 s, 1 + s\}$$

note $\dim M_i = 2i - 1$ for $1 \leq i \leq 4$. Now let's compute the cosocle filtration.

$$M^1 = \{gx - x \mid g, x \in k[D_4]\} = k \cdot \{g - 1 \mid g \in D_4\} = M_4$$

Note that $M^1/M_3 \simeq k \oplus k$ is semisimple, $M^2 \subset M_3$. But:

$$1 + r^2 = r(r - 1) - (r - 1) \in M^2$$

$$1 + r + s + rs = r(s - 1) - s - 1 \in M^2$$

and these two elements generate M_3 so that $M^2 = M_3$. Follow the same reasoning we know $M^3 \subset M_2$, and:

$$1 + r + r^2 + r^3 = r(1 + r^2) - (1 + r^2) \in M^3$$

$$1 + r^2 + s + r^2 s = r(1 + r + s + rs) - (1 + r + s + rs) \in M^3$$

So that $M^3 = M_2$. Similarly we can compute $M^4 = M_1$. So the cosocle filtration is:

$$k[D_4] = M^0 \supset M^1 \supset M^2 \supset M^3 \supset M^r \supset M^5 = 0$$

and $M_i = M^{5-i}$.

□

Exercise A.5. *Let L be a k-algebra:*

$$k\langle x, y, a, b\rangle/(xa + yb = ax = by = 1, bx = ay = 0)$$

Show that $L \simeq L \oplus L$. Also, given a ring R construct a bijection between $\mathrm{Hom}(L, R)$ *and isomorphisms $R \simeq R \oplus R$. In particular, L admits a nonzero homomorphism into* $\mathrm{End}(V)$ *for each infinite-dimensional V, so that L is a finitely generated nonzero ring has no IBN property.*

Proof. The defining relation is nothing but:

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 1, \quad \begin{pmatrix} a \\ b \end{pmatrix} \begin{pmatrix} x & y \end{pmatrix} = I_2$$

Now I have a L-linear map:

$$\varphi : L \to L^2, u \mapsto \begin{pmatrix} a \\ b \end{pmatrix} u$$

with inverse:

$$L^2 \to L, \begin{pmatrix} s \\ t \end{pmatrix} \mapsto \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} s \\ t \end{pmatrix}$$

So φ is an isomorphism. Now the data of an isomorphism: $R \simeq R \oplus R$ is given by $a', b', x', y' \in R$ such that:

1. $R \mapsto R^2$ is given by $r \mapsto \begin{pmatrix} a \\ b \end{pmatrix} r$
2. $R^2 \to R$ is given by $\begin{pmatrix} r_1 \\ r_2 \end{pmatrix} \mapsto \begin{pmatrix} x' & y' \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$
3. the choice a', b', x', d' satisifes the relations as in L.

In other words, the data is given by a homomorphism $L \to R$. We're therefore done. □

Exercise A.6. *The goal of this exercise is to show Krull-Schmidt fails without finiteness conditions.*

1. *Let R be a Dedekind domain which is not a PID. Show that the conclusion of Krull-Schmidt fails for finitely generated projective modules over R.*
2. *Let A be the algebra of smooth real functions on the real line such that $a(x) = a(x+1)$. Let M be the A-module of smooth functions on the line such that $b(x + 1) = -b(x)$. Show that M is indecomposable and not isomorphic to A, but $M \oplus M \simeq A \oplus A$ as left A-modules. Thus this fails Krull-Schmidt.*

Proof. 1. Since R is not a PID, there exists an ideal I such that $I \not\simeq R$ as an R-module. Now choose J be an ideal which is the inverse of I in the class group, then follow the hint we have:

$$I \oplus J \simeq R \oplus IJ \simeq R \oplus R$$

This will fail Krull-Schmidt if we can prove R and any ideal I is indecomposable. But this is easy, suppose:

$$R = M \oplus N$$

as R-modules. Since M, N are submodules of R so that they're ideals. It follows that:

$$R \simeq M \oplus N \simeq R \oplus MN$$

But this is absurd since the later module cannot generated by one element. For ideal $I \not\simeq R$ the argument is similar, assume:

$$I = M \oplus N \simeq R \oplus MN$$

Now pick J such that IJ is principal, we get:

$$IJ \simeq J \oplus JMN$$

but the later is again not principal.

2. Note that $A = C^\infty(S^1)$, M is the global(smooth) sections of the Mobius bundle for S^1. We need to prove the following:

   (a) Free rank 1 module over A is indecomposable. Indeed $\mathrm{End}_A(A) \simeq A$, and note that the only idempotents in A are constant functions, so it is indecomposable.

   (b) A-module M is indecomposable. Suppose I have decomposition $M = P \oplus Q$. Now choose $b \in M$ such that its vanshing locus is $1 \in S^1$ a single point. Now we can find $p \in P$ and $q \in Q$ with $p + q = b$. We know that:

   $$\{p = 0\} \cap \{q = 0\} \subset \{1\}$$

   Set $V_1 = \{p = 0\}, V_2 = \{q = 0\}$ be the vanishing locus of p, q. If $V_1 \cup V_2 \subsetneq S^1$, then we can find an open subset $U \subset S^1$ such that $f \neq 0, g \neq 0$. Choose a nonzero smooth function h supported in a closed subset of U we see that h is multiples of both f and g, so that: $h \in P \cap Q$, contradiction. Therefore we must have $V_1 \cup V_2 = S^1$. But remove the point 1 we get two closed subsets covering $S^1 \backslash \{1\}$, therefore by connectedness one of them must be empty. Without loss let's assume $V_1 \subset \{1\}$ so that $V_2 = S^1$ since $V_1 \cup V_2 = S^1$. In other words $q = 0$. In other words, any $b \in M$ with zero locus only at a single point lie in one of P or Q. But these elements generate b, so it follows that any element in M lies in one of P or Q. But this is possible when one of P and Q is zero. This proves M is decomposable.

   (c) $A \not\simeq M$. Note A is generated by one element $1 \in A$, but M is not generated by any element $b \in M$. In fact $A \cdot b$ only consists of functions with vanishing locus containing the vanishing locus of b, but $V(b)$ is nonempty by intermediate value theorem. Choose any $x \in V(b)$, there will be $b' \in M$ such that $b'(x) \neq 0$. It follows $b' \notin A \cdot b$.

(d) $A \oplus A \simeq M \oplus M$. Consider the A-linear map:

$$\varphi : A \oplus A \to M \oplus M$$

$$\begin{pmatrix} a_1 & a_2 \end{pmatrix} \mapsto \begin{pmatrix} a_1 & a_2 \end{pmatrix} \cdot B$$

for certain $B \in \mathrm{Mat}_2(M)$. This map will be injective if we impose the condition that $\det B = 1$. This motivates the natural choice:

$$B = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

The map is also surjective since we have the map from the other direction by right multiplying by B^{-1}. This establishes that φ is an isomorphism.

□

B Homework 2

Exercise B.1. *Is it true that every indecomposable module over an Artinian ring is a quotient of an indecomposable projective module? Prove or give a counterexample.*

Proof. No. Recall that we have classification of indecomposable projective modules over an Artinian ring, in particular they must be generated by one element. So any quotient of them must be generated by one element. Therefore it suffices to construct an Artinian ring R with an indecomposable module which is not cyclic. Let's consider the example from last time:

$$A = k[x, y]/I^2, \quad I = (x, y)$$

and

$$M = I/I^3$$

We showed last time M is indecomposable, but it cannot be generated by one element. □

Exercise B.2. *1. Let R be a ring and $I \subset R$ a two-sided nilpotent ideal. It was proved in class that every idempotent $e \in R/I$ admits a lifting to an idempotent $\tilde{e} \in R$.*

2. Let R be an Artinian ring. Prove that the set of conjugacy classes of idempotents in R is finite and give a formula for the cardinality of that finite set in terms of dimensions of irreducible representations of R as vector spaces over their respective skew fields of endomorphisms.

3. Show that the following rings have no idempotents except for the unit element.

(a) $k[G]$ for $char(k) = p$ and G a p-group.

(b) $k[G]$ for G a torsion free group.

Proof. 1. Suppose we have two lifts $\tilde{e}, \tilde{f}$. The goal is to find $r \in 1 + I$ such that:

$$r\tilde{e} = \tilde{f}r$$

Observe that:

$$\widetilde{f}\widetilde{e}, (1 - \widetilde{f})(1 - \widetilde{e})$$

both satisifies the above equation. We will pick:

$$r = \widetilde{f}\widetilde{e} + (1 - \widetilde{f})(1 - \widetilde{e})$$

It remains to show $r \in 1 + I$. We compute:

$$r - 1 = 2\widetilde{f}\widetilde{e} - \widetilde{e} - \widetilde{f} \equiv 2e - 2e = 0 \bmod I$$

We're done.

2. Recall $J(R)$ is nilpotent since R is artinian, and we have structure theorem:

$$R/J(R) \simeq \bigoplus \mathrm{Mat}_{n_i}(D_i)$$

The irreducible representations are just: $D_i^{n_i}$. We have a map:

$$\text{idempotent in } R/(\text{conjugacy}) \to \text{idempotent in } \frac{R}{J(R)}/(\text{conjugacy})$$

This is obviously surjective. It is also injective since each fibre is a single object by the first part. Therefore it suffices to repace R by $R/J(R)$. Now a matrix is idempotent iff its conjugate to a diagonal matrix with only 1 and 0 on the diagonals. Therefore in $\mathrm{Mat}_{n_i}(D_i)$ there are $n_i + 1$ idempotents up to conjugation. Therefore the answer is:

$$\prod (n_i + 1)$$

3. (a) It suffices to $k[G]$ is indecomposable. Note that:

$$\mathrm{Hom}_G(\mathsf{triv}, k[G]) = \mathrm{Hom}_k(\mathsf{triv}, \mathsf{triv}) = k$$

is one-dimensional by induction-restriction adjunction. Now if $k[G] = M \oplus N$, then both M and N will have a copy of triv since it's the only irreducible for $k[G]$, unless one of them is zero. Therefore $k[G]$ is indecomposable.

□

Exercise B.3. *Show that a map from a projective $\mathbb{Z}$-module to $M = \mathbb{Z}/p\mathbb{Z}$ can not be an essential surjection.*

Proof. Suppose I have a surjection $P \to M = \mathbb{Z}/p\mathbb{Z}$. We must show it cannot be essential surjection. Assume it is, we will have a lifted:

$$\begin{array}{ccccc} \mathbb{Z} & \longrightarrow & \mathbb{Z}/n & \longrightarrow & 0 \\ {\scriptstyle\varphi}\uparrow & \nearrow & & & \\ P & & & & \end{array}$$

Now assume the image of φ is $m\mathbb{Z} \subset \mathbb{Z}$, we have:

$$\begin{array}{ccccc} & & 0 & & \\ & & \uparrow & & \\ m\mathbb{Z} & \longrightarrow & \mathbb{Z}/n & \longrightarrow & 0 \\ {\scriptstyle\varphi}\uparrow & \nearrow & & & \\ P & & & & \end{array}$$

But then since $m\mathbb{Z}$ is free, φ will have a splitting, so that we can find a copy of $\mathbb{Z} \subset P$ that surjects onto $\mathbb{Z}/n\mathbb{Z}$. Therefore by essential surjectivity, we must have $P \simeq \mathbb{Z}$. It remains to show every map $\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ cannot be essential surjection. Indeed any surjection map:

$$\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$$

is given by

$$1 \mapsto t$$

for some $1 \leq t \leq p-1$. Then it is obvious that $(p+1)\mathbb{Z} \subset \mathbb{Z}$ already surjects to $\mathbb{Z}/p\mathbb{Z}$, contradiction. Therefore our assumption that $P \to M$ is an essential surjection is wrong. $\square$

Exercise B.4. *Let R be an Artinian ring. For a module M, let $C(M)$ be the set of cyclic elements in M. Let M be such that $C(M) \neq \emptyset$. Show the following are equivalent:*

1. *The complement $M \backslash C(M)$ is a submodule.*
2. *M is a quotient of an indecomposable projective module.*

Proof. Let's first show (2) $\implies$ (1). Suppose M is the quotient of P_L, for P_L the projective cover for simple module L. Then we must have:

$$P_L \twoheadrightarrow M \twoheadrightarrow L$$

Since L is the simple quotient for P_L. Now $M \backslash C(M)$ is clearly stable under scalar multiplication by R. We only need to show its stable under addition. Given $m_1, m_2 \notin C(M)$, we must show $m_1 + m_2 \notin C(M)$. We will show something stronger: if $N_1 \subsetneq M, N_2 \subsetneq M$, then $N_1 + N_2 \subsetneq M$. If so take $N_1 = R \cdot m_1, N_2 = R \cdot m_2$ we get the desired result. But since $P_L \to L$ is essential surjection, then so is $M \to L$. So both N_1, N_2 won't surject onto L. But L is simple, so that N_1, N_2 has image 0 in L. Therefore $N_1 + N_2$ has image 0 in L and thus we must have $N_1 + N_2 \subsetneq M$.

Now let's show (1) $\implies$ (2). We will first show that M/JM is irreducible. Indeed, since M/JM is a module for semisimple ring R/J, it suffices to show it is indecomposable. If not, say $M/JM = Q_1 \oplus Q_2$ for $Q_1, Q_2 \neq 0$. Now since we assume M is cyclic, say generated by $m \in M$. Let $\overline{m} \in M/JM$ be its image in M/JM, and then we can write:

$$\overline{m} = \overline{e_1} + \overline{e_2}, \quad \overline{e_i} \in Q_i$$

so that $M/JM = R\overline{m}$. Let e_1, e_2 be any lifts of $\overline{e_1}, \overline{e_2}$. Then clearly $e_1 \notin C(M), e_2 \notin C(M)$. But $e_1 + e_2 \in C(M)$. Indeed, the quotient map:

$$M \to M/JM$$

is an essential surjection, and $R \cdot (e_1 + e_2) \subset M$ has image all of M/JM. It follows that we must have $R \cdot (e_1 + e_2) = M$. This contradicts our assumption that $M \backslash C(M)$ is a submodule. Hence this shows M/JM must be an irreducible L. But from our construction in lecture we see that M thus must have projective cover P_L as desired. $\square$

Exercise B.5. *An Artinian ring is called self-injective if the free module is injective. A finite dimensional algebra A over a field k is called Frobenius if there exists a linear functional τ on A such that the bilinear pairing:*

$$A \times A \to k$$

$$(x, y) \mapsto \tau(xy)$$

is nondegenerate.

1. *Show that for a finite group G the group algebra $k[G]$ is Frobenius.*

2. *Prove that a Frobenius algebra is self-injective.*

3. *Let A be a finite dimensional algebra over a field. Assume that every simple A-module L the multiplicity of L in the co-socle of A viewed as a left A-module equals the multiplicity of L^* in the socle of A viewed as a right A-module. Show that A is self-injective.*

Proof. 1. Set

$$\tau : k[G] \to k$$

$$\sum_{g\in G} a_g g \mapsto a_1$$

Then the pairing will be nondegerate. Since for each $\sum_{g\in G} a_g g \in k[G]$ nonzeor, there exists $h \in G$ such that $a_h \neq 0$, then:

$$\tau(\sum_{g\in G} a_g g, h^{-1}) = a_h \neq 0$$

2. Essentially, the Frobenius pairing provides me with an isomorphism of right A-modules:

$$A \simeq A^* = \mathrm{Hom}_k(A, k)$$

$$a \mapsto (\varphi_a : b \mapsto \varphi(ab))$$

Now we wish to find a lift:

$$\begin{array}{ccccc} 0 & \longrightarrow & M & \xrightarrow{\lambda} & N \\ & & & \searrow^{f} & \downarrow \\ & & & & A \end{array}$$

We're given:

$$M \xrightarrow{f} A$$

Take k-linear dual, we're having a map:

$$A \simeq A^* \to M^*$$

of right A-modules. Now since taking k-linear dual is exact, so that we have:

$$N^* \to M^* \to 0$$

But A as a right A-module is projective, so that we can find a lift:

$$A \to N^*$$

of right A-modules. Again taking k-linear dual we have:

$$N \to A^* \simeq A$$

of left A-modules. This is the desired lift that we want. To be more precise, let's spell out all maps that we get:

$$s : A \to M^*, \quad a \mapsto (s_a : m \mapsto \tau(af(m)))$$

This is right A-lienar since for every $b \in A$:

$$(s_a \cdot b)(m) = s_a(bm) = \tau(a\varphi(bm)) = \tau(ab\varphi(m)) = s_{ab}(m)$$

Now choose $t : A \to N^*$ to be a lift of s (here involves a choice), namely:

$$s_a = t_a \circ \lambda$$

for every $a \in A$. Then the corresponding $\mu : N \to A$ is given by:

$$n \mapsto \mu_n$$

such that $\tau(a\mu_n) = t_a(n)$. Now we must check $\mu \circ \lambda = f$. Indeed, take any $m \in M$, we have:

$$\tau(a\mu(\lambda(m))) = t_a(\lambda(m)) = s_a(m) = \tau(af(m))$$

Since the pairing is perfect, one sees indeed $\mu \circ \lambda(m) = f(m)$ as desired.

3. Note if we can prove $A \simeq A^*$, then the image of $1 \in A$ in A^* will be the desired Frobenius functional, giving A the Frobenius algebra structure. Therefore it suffices to prove $A \simeq A^*$ as right A-modules. Note that we can write:

$$A = \bigoplus P_{L_i}^{\oplus m_i}$$

for P_{L_i} be the projective cover for simple L_i. Now that:

$$A^* = \bigoplus (P_{L_i}^*)^{\oplus m_i}$$

Note that $P_{L_i}^*$ is injective by duality, and we have $L_i^* \hookrightarrow P_{L_i}^*$. Therefore, by injectivity of A^* and assumption, we obtain:

$$\begin{array}{ccccc} 0 & \longrightarrow & \mathrm{soc}_r(A) & \longrightarrow & A \\ & & \big\downarrow & \swarrow & \\ & & A^* & & \end{array}$$

where everything are right modules. In fact, we constructed:

$$\begin{array}{ccccc} 0 & \longrightarrow & \mathrm{soc}_r(A) & \longrightarrow & A \\ & & \big\downarrow \sim & & \big\downarrow \\ 0 & \longrightarrow & \mathrm{soc}_r(A^*) & \longrightarrow & A^* \end{array}$$

Dualizing, we get a diagram for left modules:

$$\begin{array}{ccccc} 0 & \longleftarrow & \mathrm{csoc}_l(A^*) & \longleftarrow & A^* \\ & & \sim\uparrow & & \uparrow \\ 0 & \longleftarrow & \mathrm{csoc}_l(A) & \longleftarrow & A \end{array}$$

But now Nakayama's lemma helps me to conclude $A \to A^*$ is actually an isomorphism of left mdoules. We're therefore done.

□

Exercise B.6. *Let $R = \mathbb{C}[S]$, where S is the semigroup of 2×2 integer matrices of the following form:*

$$\begin{pmatrix} 1 & b \\ 0 & 2^a \end{pmatrix}$$

with $a, b \in \mathbb{Z}, a \geq 0$

1. *Describe an isomorphism:*

$$R \simeq \mathbb{C}\langle s, t, t^{-1}\rangle/(ts = st^2)$$

2. *Define an action of R on $M = \mathbb{C}[z, z^{-1}]$ where $P \in \mathbb{C}[z, z^{-1}]$:*

$$t \cdot P(z) = zP(z)$$

$$s(P(z^2)) = P(z), s(P) = 0 \text{ for odd polynomial}$$

Therefore:

$$s(P)(\lambda^2) = \frac{1}{2}(P(\lambda) + P(-\lambda))$$

Show that M is a faithful simple module.

3. *Now consider $R^{\mathrm{op}} = \mathbb{C}\langle s', t', (t')^{-1}\rangle/(s't' = (t')^2 s')$. Define an action of R^{op} on $M' = \mathbb{C}[z, z^{-1}]$ by:*

$$t'(P)(z) = zP(z)$$

$$s'(P)(z) = P(z^2)$$

Note that for every n the ideal $M_n = (z^n - 1)$ is a submodule. For which n the quotient M'/M_n is semisimple? For such n explicitly describe its decomposition into simple modules.

Proof. 1. Consider:

$$\mathbb{C}\langle s, t, t^{-1}\rangle/(ts = st^2) \to \mathbb{C}[S]$$

$$s \mapsto \begin{pmatrix} 1 & \\ & 2 \end{pmatrix}, \quad t \mapsto \begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix}$$

It is easy to check this is well-defined. It is an isomorphism since left hand side has an explicit basis:

$$s^i t^j, \quad i \geq 0, j \in \mathbb{Z}$$

which corresponds to the basis:

$$\begin{pmatrix} 1 & j \\ & 2^i \end{pmatrix}$$

of RHS.

2. Let N be a nonzero submodule of $\mathbb{C}[z, z^{-1}]$. Only using the t-action we see N must be an ideal inside $\mathbb{C}[z, z^{-1}]$. But $\mathbb{C}[z, z^{-1}]$ is a PID, there must exists $0 \neq P \in \mathbb{C}[z, z^{-1}]$ such that:

$$N = \mathbb{C}[z, z^{-1}] \cdot P$$

We may also adjust P so that $P \in \mathbb{C}[z]$ with nonzero constant term. It is clear that $N \cap \mathbb{C}[z] = \mathbb{C}[z] \cdot P$. Now $s(P) \cdot \mathbb{C}[z] \subset N \cap \mathbb{C}[z] = \mathbb{C}[z] \cdot P$, therefore:

$$P \mid s(P)$$

But $\deg P > \deg s(P)$ unless P is constant. This shows P is constant and therefore $N = M$. So M is simple. Now to show M is faithful, it suffices for each $r \in R$, we construct an element $m \in M$ with $rm \neq 0$. Given:

$$r = \sum_{i=0}^{N} s^i P_i(t)$$

with $P_N \neq 0$. Suppose t^k appears in $P_N(t)$, we choose:

$$m = z^{2^n - k}$$

for n sufficiently large, we compute:

$$(s^i t^j)(z^{2^n-k}) = \begin{cases} z^{\frac{2^n-k+j}{2^i}}, & \text{if } \dfrac{2^n-k+j}{2^i} \in \mathbb{Z}; \\ 0, & \text{otherwise.} \end{cases}$$

In particular, the only contribution of the term $z^{2^{n-N}}$ is from $t^N s^k$. Therefore $rm \neq 0$. This proves M is faithful.

3. Consider:

$$\mathbb{C}[z]/(\mathbb{C}[z] \cap M_n) \to \mathbb{C}[z, z^{-1}]/M_n$$

It is injective by design and is also clearly surjective. Therefore:

$$M'/M_n = \mathbb{C}\{1, z, z^2, \cdots, z^{n-1}\}$$

is n-dimensional. With the action:

$$t' \cdot z^i = z^{i+1}$$

$$s' \cdot z^i = z^{2i}$$

with the indexing to be understood up to modulo n. Consider:

$$N = \{\sum_{i=0}^{n-1} a_i z^i \mid \sum a_i = 0\}$$

This is clearly a submodule of M'/M_n. Therefore if M'/M_n it must have a complement, which necessarily have dimension 1. Now if I have:

$$0 \neq v = \sum b_i z^i$$

generates the complement. Then there exists $\lambda \in \mathbb{C}$ such that:

$$t'v = \lambda v$$

But note that $t'^n v = v \implies \lambda^n = 1$. In particular, $\lambda \neq 0$. Now $b_{i+1} = \lambda b_i$ so that each coefficient in v is nonzero. But if n is even, then:

$$s'v = \sum b_i z^{2i}$$

So that the coefficent before z is zero. But $s'v \in \mathbb{C} \cdot v \implies s'v = 0$. But $s'v$ has coefficient $b_0 \neq 0$ for the constant term, this is a contradiction. Therefore for n even M'/M_n cannot be semisimple.

For n odd, we just pick:

$$v = \sum_{i=0}^{n-1} z^i$$

This is clearly a submodule of M'/N. So that:

$$M'/M_n = N \oplus \mathbb{C} \cdot v$$

Suppose t' acting on N with $\rho(t')$, then evidently we know:

$$\rho(t')^n = 1$$

Therefore we can decompose N into eigenspaces of t':

$$N = \mathbb{C}\{v_1, \cdots, v_{n-1}\}$$

so that $t' \cdot v_i = \zeta^i v_i$ for ζ a primitive n-th root of unity. Explicitly:

$$v_i = 1 + \zeta^{-i} z + \zeta^{-2i} z^2 + \cdots$$

Now recall the relation:

$$s't' = (t')^2 s' \implies s't'v_i = (t')^2 s'v_i \implies \zeta^i s'v_i = (t')^2 s'v_i$$

So $s'v_i$ is an eigenvector of $(t')^2$ with eigenvalue ζ^i. But now that $(t')^2$ is also diagonalizable on N with distinct eigenvalues, so that $s'v_i$ is actually an eigenvector for t' with

eigenvalue $\zeta^{\frac{i}{2}}$, where we understand: $\frac{i}{2} \in \mathbb{Z}/n$. Therefore: $s'v_i = \lambda v_{\frac{i}{2}}$. Comparing the coefficient before constant term we know:

$$s'v_i = v_{i/2}$$

Let $T = \langle 2 \rangle \subset (\mathbb{Z}/n\mathbb{Z})^\times$ be the subgroup generated by 2, then: for each $j \in (\mathbb{Z}/n\mathbb{Z})^\times/T$, we set:

$$N_j = \bigoplus_{t \in T} \mathbb{C} \cdot v_{\tilde{jt}}$$

for $\tilde{j}$ some representative j. This is well-defined (indepentent of the choice of the representative of j), and

$$N = \bigoplus_{j \in (\mathbb{Z}/n\mathbb{Z})^\times/T} N_j$$

Note that by design N_j is stable under the action of s' hence is a submodule. We claim that each N_j is simple. Indeed if $K \subset N_j$ is a nonzero submodule, we can find $0 \neq w \in K$ a t-eigenvector, so that $w \in \mathbb{C} \cdot v_{\tilde{j}}$ for some $\tilde{j}$ a lift of j. Then we can use s' to generate whole N_j from W.

In summary, we proved that M'/M_n is semisimple iff n is odd. In that case, its decomposition of simple modules is:

$$M'/M_n = \mathbb{C} \cdot v_0 \oplus \bigoplus_{j \in (\mathbb{Z}/n\mathbb{Z})^\times/T} N_j$$

for

$$N_j = \bigoplus_{t \in T} \mathbb{C} \cdot v_{\tilde{jt}}, \quad v_i = 1 + \zeta^{-i}z + \zeta^{-2i}z^2 + \cdots + \zeta^{-(n-1)i}z^{n-1}$$

□

C Homework 3

Exercise C.1. *For each of the following functors determine the existence of a left adjoint, of a right adjoint and describe the existing adjoint functors.*

1. *Consider the quiver $Q = (\bullet \to \bullet)$. Let $\mathcal{B}$ be the category of representations of Q over a field k, and $\mathcal{A}$ be the full subcategory consiting of such representations that the map between the two vector spaces is injective, and $F : \mathcal{A} \to \mathcal{B}$ be the embedding.*

2. *Let $\mathcal{A}$ be the category of vector spaces of a field k not of characteristic 2, $\mathcal{B}$ be the category of graded commutative k-algebra and let F send a vector space to its exterior algebra.*

Proof. 1. F has a left adjoint G given by:

$$G : \mathcal{B} \to \mathcal{A}, \quad (X \xrightarrow{f} Y) \mapsto (X \xrightarrow{(1_X, f)} X \oplus Y)$$

Note that each elements in

$$\mathrm{Hom}(G(X \to Y), N \hookrightarrow M)$$

corresponds to a commutative square:

$$\begin{array}{ccc} X & \xhookrightarrow{1_X \oplus f} & X \oplus Y \\ \downarrow & & \downarrow \\ N & \hookrightarrow & M \end{array}$$

Note to specify such a square it suffices to specify a map $X \to N$ and a compatible map $Y \to M$ (for the right vertical arrow, the component $X \to M$ is already determined once we specify $X \to N$), which is nothing but the data of an element in:

$$\mathrm{Hom}(X \to Y, F(N \hookrightarrow M))$$

But F doesn't have a right adjoint since it will not preserve cokernel. Let's consider:

$$\begin{array}{ccc} 0 & \longrightarrow & k \\ \downarrow & & \downarrow{\scriptstyle 1_k} \\ k & \xrightarrow[1_k]{} & k \end{array}$$

It is evident that

$$\mathrm{coker}_{\mathcal{B}}((0 \to k) \to (k \to k)) = (k \to 0)$$

But note that for any of the following diagram in $\mathcal{A}$:

$$\begin{array}{ccc} 0 & \longrightarrow & k \\ \downarrow & & \downarrow{\scriptstyle 1_k} \\ k & \xrightarrow[1_k]{} & k \\ \downarrow & & \downarrow \\ X & \hookrightarrow & Y \end{array}$$

with the columns compose to 0, we must have the map from the second row to the third row is the zero map. Therefore:

$$\mathrm{coker}_{\mathcal{A}}((0 \to k) \to (k \to k)) = 0$$

This shows F doesn't have right adjoint.

2. The fundamental relation is that if V is a vector space and $R = R_0 \oplus R_1$ a graded k-algebra, then:

$$\mathrm{Hom}_{\mathcal{B}}(\Lambda^\bullet V, R) \simeq \mathrm{Hom}_{\mathcal{A}}(V, R_1)$$

This shows that $\Lambda^\bullet$ has a right adjoint $R \mapsto R_1$. F cannot have a left adjoint since it will not preserve kernel. Indeed, suppose I have:

$$0 \to k \to k^3 \to k^2 \to 0$$

exact. Apply F I have:

$$\Lambda^\bullet k^3 \to \Lambda^\bullet k^2 \to 0$$

since F is right exact. But then the kernel ahs dimension $8 - 4 = 4$, which cannot be $\Lambda^\bullet k$. Therefore F has no left adjoint.

□

Exercise C.2. *Let Γ be a finite group acting on a ring R by automorphism. Then the smash product $\Gamma\#R$ is the abelian group $\bigoplus_{\gamma\in\Gamma} R$ with multiplication:*

$$(r_{\gamma_1})(r'_{\gamma_2}) = (r\gamma_1(r'))_{\gamma_1\gamma_2}$$

Suppose R is simple and $|G|$ is invertible in R. Suppose one of the following two condition holds:

1. *No nontrivial element $\gamma \in \Gamma$ act as an inner automorphism of R.*
2. *The element $\gamma \in \Gamma$ acts by an invertible element $r_\gamma \in R$ where the elements $r_\gamma \in R$ are linearly independent over the center of R.*

Prove that $\Gamma\#R$ and R^Γ are Morita equivalent.

Proof. Let $e = \frac{1}{n}\sum_{g\in\Gamma} g$ and $A = \Gamma\#R$, we claim that $eAe \simeq R^\Gamma$. Indeed, given an element $r\gamma \in A$, we compute:

$$e(r\gamma)e = \frac{1}{n^2}\sum_{g,h\in\Gamma} gr\gamma h = \frac{1}{n^2}\sum_{g,h\in\Gamma} g(r)g\gamma h$$

Now Fix g, and let h run through Γ then $g\gamma h$ runs through Γ. Hence:

$$e(r\gamma)e = (\frac{1}{n}\sum_{g\in\Gamma} g(r))e$$

Therefore $eAe = R^\Gamma e$. Moreover the ring structure is given by:

$$(re)(r'e) = re(r')e = rr'e$$

since $r' \in R^\Gamma$. It follows that $eAe \simeq R^\Gamma$. Therefore the result will follow if we can prove $AeA = A$. Note that we can decompose A into irreducible R-bimodules:

$$A = \bigoplus_{\gamma \in \Gamma} R \cdot \gamma$$

Moreover, we claim that AeA is R-bi-submodule. Indeed it suffices to show it is closed uner addition. To see this, note AeA is generated by elements of form:

$$r\gamma er'\gamma' = rer'\gamma' = \frac{1}{n} \sum_{g \in \Gamma} rg(r')g\gamma'$$

Hence:

$$AeA = \sum_{r' \in R} R \cdot (g(r'))_{g \in \Gamma}$$

is an abelian group.

1. In the first case, we claim that $R \cdot \gamma \not\simeq R \cdot \gamma'$ for $\gamma \neq \gamma'$. Suppose not, then we have a R-bimodule isomorphism:

   $$R \cdot \gamma \to R \cdot \gamma'$$
   $$t\gamma \mapsto ts\gamma'$$

   for some $s \in R^\times$, using the left module structure. But this map must also be right R-equivariant. So we need:

   $$\gamma r \mapsto s\gamma' r = s\gamma'(r)\gamma'$$

   On the other hand

   $$\gamma r = \gamma(r)\gamma \mapsto \gamma(r)s\gamma'$$

   It follows that:

   $$s\gamma'(r) = \gamma(r)s \implies \gamma'(r) = s^{-1}\gamma(r)s$$

   This implies that $\gamma^{-1}\gamma'$ is inner, a contradiction. Therefore we must have each irreducible piece not isomorphic. This implies that any sub-bimodule of A must take form:

   $$\bigoplus_{\gamma \in J} R \cdot \gamma$$

   for some subset $J \subset \Gamma$. Note that AeA, as a submodule contains $e = \frac{1}{n} \sum_{g \in \Gamma} g$, which has each component nonzero. This forces that $AeA = A$ as desired.

2. In the second case, we have isomorphism:

   $$R \xrightarrow{\sim} R \cdot \gamma$$

   $$t\gamma \mapsto t \cdot r_\gamma^{-1}\gamma$$

by the same computation as in the first case, where $\gamma(s) = r_\gamma s r_\gamma^{-1}$. Therefore, we have an isomorphism of R-bimodules:

$$A = \bigoplus_{\gamma\in\Gamma} R\cdot\gamma \to \bigoplus_{\gamma\in\Gamma} R$$

$$\gamma \mapsto (r_\gamma)_\gamma$$

Now recall that

$$AeA = \sum_{r'\in R} R\cdot(\sum_{g\in\Gamma} g(r')g)$$

which transformed to

$$\sum_{r'\in R} R\cdot(r_g r')_{g\in\Gamma}$$

to a submodule of the target. In particular, $AeA \subset A$ is a sub $R\oplus R^{\mathrm{op}}$-module containing $(r_g)_{g\in\Gamma}$. We claim that the cyclic $R\oplus R^{\mathrm{op}}$-module containing $(r_g)_{g\in\Gamma}$ is all of A. Indeed observe that:

$$\mathrm{End}_{R\oplus R^{\mathrm{op}}}(R)$$

is a quotient of $Z(R)$, therefore we're in the setting of density theorem by assumption of linearly independence of r_g. The claim then follows immediately.

□

Exercise C.3. *Let $R = \Gamma\#\mathbb{C}[x]$ where $\Gamma = \mathbb{Z}/2\mathbb{Z}$, with nontrivial element of Γ acting by $x\mapsto -x$.*

1. *Identify R with the subring in $\mathrm{Mat}_2(\mathbb{C}[x])$ consisting of matrices (P_{ij}) such $P_{ij}(-x) = (-1)^{i+j}P_{ij}(x)$.*
2. *Compute the center and cocenter of R and check that the latter is not a free module over the former. Conclude that R is not Morita equivalent to a commutative ring.*

Proof. 1. Consider:

$$\varphi : R \to \mathrm{Mat}_2(\mathbb{C}[x])$$

$$f \mapsto \begin{pmatrix} \frac{f+\overline{f}}{2} & \frac{f-\overline{f}}{2} \\ \frac{f-\overline{f}}{2} & \frac{f+\overline{f}}{2} \end{pmatrix}$$

$$f\gamma \mapsto \begin{pmatrix} \frac{f+\overline{f}}{2} & -\frac{f-\overline{f}}{2} \\ \frac{f-\overline{f}}{2} & -\frac{f+\overline{f}}{2} \end{pmatrix}$$

where $\overline{f}(x) = f(-x)$, and γ is the nontrivial element of Γ. This is clearly an isomorphism of $\mathbb{C}$-vector spaces. We must check the compatibility of multiplications. Indeed:

$$\varphi(f)\varphi(g) = \begin{pmatrix} \frac{f+\overline{f}}{2} & \frac{f-\overline{f}}{2} \\ \frac{f-\overline{f}}{2} & \frac{f+\overline{f}}{2} \end{pmatrix}\begin{pmatrix} \frac{g+\overline{g}}{2} & \frac{g-\overline{g}}{2} \\ \frac{g-\overline{g}}{2} & \frac{g+\overline{g}}{2} \end{pmatrix} = \begin{pmatrix} \frac{fg+\overline{f}\overline{g}}{2} & \frac{fg-\overline{f}\overline{g}}{2} \\ \frac{fg-\overline{f}\overline{g}}{2} & \frac{fg+\overline{f}\overline{g}}{2} \end{pmatrix} = \varphi(fg)$$

The additional relation to check is that:

$$\varphi(\gamma)\varphi(f) = \varphi(\gamma f) = \varphi(\gamma(f)\gamma) = \varphi(\overline{f}\gamma) = \varphi(\overline{f})\varphi(\gamma)$$

which is equivalent to:

$$\begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \begin{pmatrix} \frac{f+\overline{f}}{2} & \frac{f-\overline{f}}{2} \\ \frac{f-\overline{f}}{2} & \frac{f+\overline{f}}{2} \end{pmatrix} = \begin{pmatrix} \frac{\overline{f}+f}{2} & \frac{\overline{f}-f}{2} \\ \frac{\overline{f}-f}{2} & \frac{\overline{f}+f}{2} \end{pmatrix} \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$$

which is true.

2. If $f + g\gamma \in Z(R)$, then for all $h \in \mathbb{C}[x]$

$$hf + hg\gamma = fh + g\gamma h = fh + g\gamma(h)\gamma \implies hg = g\gamma(h), \forall h$$

This forces $g = 0$. Hence:

$$Z(R) = \mathbb{C}[x]^{\Gamma} = \{f \in \mathbb{C}[x] \mid f = \overline{f}\}$$

which corresponds to scalar matrices in $\mathrm{Mat}_2(\mathbb{C}[x])$. Now as a module over $Z(R)$, R has a basis:

$$E = \begin{pmatrix} 0 & x \\ 0 & 0 \end{pmatrix}, F = \begin{pmatrix} 0 & 0 \\ x & 0 \end{pmatrix}, H = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}, T = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}$$

From the standard $\mathfrak{sl}_2$-relation:

$$[H, E] = 2E, \quad [H, F = 2F], \quad [E, F] = x^2 H$$

It follows that the cocenter of R equals

$$Z \cdot \langle H, T \rangle / (x^2 H) \simeq Z \oplus Z/(x^2)$$

which is not free over $Z = Z(R)$. Now for a commutative ring S, the center and cocenter both equals S, in particular the latter is free over the formal. But Morita equivalence preserves center and cocenter, the last statement follows.

□

Exercise C.4. *Let R be a ring, and F the forgetful functor from the category of R-modules to abelian groups. Describe* $\mathrm{End}(F)$. *More generally, let $A \to R$ be a ring homomorphism, and let $F : R\text{-mod} \to A\text{-mod}$ be the pullback functor. Describe* $\mathrm{End}(F)$.

Proof. We just prove the "more generally" part. We claim that $\mathrm{End}(F) = \mathfrak{z}(A)$ the centralizer of A. Indeed, given any $\eta \in \mathrm{End}(F)$, we have:

$$\eta_R : F(R) \to F(R)$$

namely $\eta_R(1) \in \mathrm{End}_A(R)$. We claim that η_R determines η. Indeed, η consists of the data $\eta_M : F(M) \to F(M)$ for each R-module M. Now for each $m \in M$, let's consider the diagram:

$$\begin{array}{ccccc} 1 & & F(R) & \xrightarrow{\eta_R} & F(R) & & \eta_R(1) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ m & & F(M) & \xrightarrow{\eta_M} & F(M) & & \eta_R(1) \cdot m \end{array}$$

So: $\eta_M(m) = \eta_R(1) \cdot m$ is uniquely determined. Conversely, given $\eta_R \in \mathrm{End}_A(R)$, we consider the formula:

$$\eta_M : m \mapsto \eta_R(1) \cdot m$$

For this to determine an A-linear morphism for each M, we must have $\eta_R(1)$ commutes with all elements in A. The result follows. □

Exercise C.5. *Let k be a field and A a k-algebra. For a finite dimensional module M and $a \in A$, we have:*

$$\mathrm{Tr}(a, M) \in k$$

This defines a pairing:

$$\tau : C(A) \times K \to k$$

where $C(A) = A/[A, A]$ is the cocenter of A, and $K = K(A\text{-mod}_{\mathrm{fd}}) \otimes_{\mathbb{Z}} k$.

1. *Assume k is algebraically closed. Show that τ is nondegenerate in the second variable, i.e. a nonzero element in K defines a nonzero linear functional on $C(A)$.*
2. *Let G be a group (not necessarily finite), with $A = kG$. Identify $C(A)$ with the space of finitely supported k-valued functions on the set of conjugacy classes. Deduce the linear independence of characters of group representations when k is algebraically closed.*

Proof. 1. It suffices to show that the linear functional are linearly independent for finite dimensional irreducible modules. Assume:

$$\sum_{i=1}^{N} c_i \mathrm{Tr}(-, M_i) = 0$$

for a finite collection of irreducible modules M_i. Since k is algebraically closed we know $\mathrm{End}_A(M_i) = k$ for each i. As a consequence of density theorem, we have a surjection:

$$A \to \bigoplus \mathrm{End}_k(M_i)$$

In particular, we can find a such that its image in $\mathrm{End}_k(M_i)$ is

$$\begin{pmatrix} 1 & & & \\ & 0 & & \\ & & \cdots & \\ & & & 0 \end{pmatrix}$$

for $i = k$ and zero otherwise. Plug in this a we know $c_k = 0$. The result therefore follows.

2. We have a natural map:

$$A/[A,A] \to \mathrm{Cl}(G)$$

$$\mathbf{1}_g \mapsto \mathbf{1}_{g^G}$$

where g^G is the conjugacy class corresponding to group element g, with both sides endowed with convolution product structure. To see this is well-defined, note that:

$$(gh)^G = (hg)^G$$

for any $h, g \in G$, and therefore $\mathbf{1}_{gh} - \mathbf{1}_{hg} \in [A,A]$ is sent to 0. This is obviously surjective, with the inverse map given by:

$$\mathbf{1}_{g^G} \to \mathbf{1}_g$$

for some representative $g \in g^G$. This proves the identification claimed in the exercise. Now that the trace pairing is nothing but the definition of the character of a representation, the result then follows from the first part.

□

Exercise C.6. *We say A satisifes the density of characters property if the pairing τ from the previous problem is nondegenrate in the first variable, i.e. for every $a \in C(A)$, $a \neq 0$ there exists M such that $\mathrm{Tr}(a, M) \neq 0$. In each case below determine whether the group algebra kG satisifies density of characters.*

1. *G is finite p-group, and k is a field of characteristic p.*
2. *$G = \mathbb{Z}$, $k = \mathbb{C}$.*
3. *$G = (\mathbb{Z}/2) \ltimes \mathbb{Z}$, $k = \mathbb{C}$, where the nontrivial element in $\mathbb{Z}/2$ acting by $n \mapsto -n$.*
4. *$G = (\mathbb{Z}/2) \ltimes \mathbb{Z}^2$, $k = \mathbb{C}$, where the nontrivial element in $\mathbb{Z}/2$ acting by $(a,b) \mapsto (-a,-b)$.*

Proof. 1. In this case the only irreducible is the trivial representation. Now consider $a = g - 1$ for some nontrivial g, then $\mathrm{Tr}(a, \mathsf{triv}) = 0$ and therefore $\mathrm{Tr}(a, M) = 0$ for any finite dimensional M. But $a \notin [A,A]$, so that kG doesn't satisfy the density of characters.

2. Note that $k[G] = k[v, v^{-1}]$. Note that for each complex number $\lambda \in \mathbb{C}^*$, we have a $k[v,v^{-1}]$-module $\mathbb{C}_\lambda$ by letting v act as λ. Therefore for every $a = \sum_{n\in\mathbb{Z}} c_n v^n \in k[v,v^{-1}]$, a acts as

$$\sum_{n\in\mathbb{Z}} c_n \lambda^n$$

We can always pick λ so that this expression is nonzero. The densiity property holds in this case.

3. Let $A = \mathbb{C}[G]$, then A has a basis $(v^i, \gamma v^i)_{i\in\mathbb{Z}}$, where we still identify $\mathbb{C}[\mathbb{Z}] \simeq \mathbb{C}[v, v^{-1}]$ and γ is the nontrivial element in $\mathbb{Z}/2\mathbb{Z}$. Let M be an irreducible A-module, we can find $m \in M$ such that $\mathbb{C} \cdot m$ is preserved by the $\mathbb{C}[v, v^{-1}]$-action. Suppose $\mathbb{C} \cdot m \simeq \mathbb{C}_\lambda$, so that $v \cdot m = \lambda m$. Then:

$$M = \text{Span}(m, \gamma m)$$

Note that:

$$\lambda\gamma m = \gamma v m = \gamma(v)\gamma m = v^{-1}\gamma m$$

Therefore $v(\gamma m) = \lambda^{-1}\gamma m$. This shows $\lambda \neq \pm 1$, we have a 2-dimensional irreducible module:

$$M_\lambda = \mathbb{C} \cdot \langle m_\lambda, \gamma m_\lambda \rangle$$

For $\lambda = \pm 1$, we have a A-invariant subspace: $\mathbb{C} \cdot \langle m + \gamma m \rangle$, therefore we have 1-dimensional irreducibles:

$$M_1 = \mathbb{C} \cdot m, \quad vm = m, \gamma m = m$$

the trivial representation, and:

$$M_{-1} = \mathbb{C} \cdot m, \quad vm = -m, \gamma m = m$$

We thus classify all irreducible representations of A. Consider any

$$a = \sum_{i\in\mathbb{Z}} c_i v^i + \sum_{j\in\mathbb{Z}} d_j v^j \gamma$$

with the property that $\text{Tr}(a, M_\lambda) = 0$ for all λ. We compute that:

$$\text{Tr}(a, M_\lambda) = \sum_{i\in\mathbb{Z}} c_i(\lambda^i + \lambda^{-i}) = 0$$

for all $\lambda \neq 1, -1$. This forces $c_i = -c_{-i}$. But note that:

$$[v^i\gamma, \gamma] = (v^i - v^{-i})$$

So that we can adjust a so that each $c_i = 0$, and it represents the same element in $C(A)$. So without loss:

$$a = \sum_{j\in\mathbb{Z}} d_j v^j \gamma$$

Now:

$$\text{Tr}(a, M_1) = \sum d_j = 0$$

$$\text{Tr}(a, M_2) = \sum (-1)^j d_j = 0$$

Therefore, we can write:

$$a = \sum_{j\in\mathbb{Z}} \alpha_j (v^j - v^{j-2})$$

But then:

$$[v, v^{i-1}\gamma] = (v^i - v^{i-2})\gamma$$

This implies $a \in [A, A]$. Therefore, A satisifies the density property.

4. The algebra $A = \mathbb{C}[G]$ satisfies the density property. The proof is essentially the same as the previous case. We identify $\mathbb{C}[\mathbb{Z}^2] \simeq \mathbb{C}[v, v^{-1}, w, w^{-1}]$, so that A has a basis:

$$(v^i w^j, v^i w^j \gamma)_{i,j}$$

We can similarly classify irreducible A-modules: they're parametrized by $(\lambda, \mu) \in (\mathbb{C}^*)^2$. For $(\lambda, \mu) \neq (\pm 1, \pm 1)$, we have:

$$M_{\lambda,\mu} = \mathbb{C} \cdot \langle m, \gamma m \rangle$$

$$v^i w^j m = \lambda^i \mu^j m, \quad v^i w^j \gamma m = \lambda^{-i} \mu^{-j} \gamma m$$

For $(\lambda, \mu) = (\varepsilon_1, \varepsilon_2)$ for $\varepsilon_i \in \{0, 1\}$, we have:

$$M_{\varepsilon_1,\varepsilon_2} = \mathbb{C} \cdot \langle m \rangle$$

$$\gamma m = m, v^i w^j m = \varepsilon_1^i \varepsilon_2^j m$$

Now assume:

$$a = \sum_{i,j} c_{i,j} v^i w^j + \sum_{i,j} d_{i,j} v^i w^j \gamma$$

such that $\mathrm{Tr}(a, M) = 0$ for all M. Test on $M_{\lambda,\mu}$ we see that:

$$\sum_{i,j} c_{i,j} (\lambda^i \mu^j + \lambda^{-i} \mu^{-j}) = 0$$

This forces $c_{i,j} = -c_{-i,-j}$. But now we have:

$$[v^i \gamma, w^{-j} \gamma] = v^i w^j - v^{-i} w^{-j}$$

So that we can adjust by an element in $[A, A]$ and assume $c_{i,j} = 0$ for all i, j. Now test on $M_{\varepsilon_1,\varepsilon_2}$ we see that:

$$\sum_{i,j} d_{i,j} \varepsilon_1^i \varepsilon_2^j = 0$$

So that we can rewrite:

$$a = \sum_{i,j} d_{i,j} v^i w^j \gamma = \sum_{i,j} \alpha_{i,j} (v^i w^j - v^{i-2} w^{j-2}) \gamma$$

But then:

$$[vw, v^{i-1} w^{j-1} \gamma] = v^i w^j - v^{i-2} w^{j-2} \gamma$$

Proving that $a \in [A, A]$. We're done.

□

D Homweork 4

Exercise D.1. *Show that a self-injective Artinian ring has finite homological dimension if and only if it is semisimple.*

Proof. Let R be a self-injective Artinian ring. A basic observation is that every finitely projective module is injective, since it is a direct summand of a finitely generated free module. Now assume R is semisimple, then clearly R has homological dimension 0. Conversely, assume R has finite homolgoical dimension. Let L be a simple R-module. It will have a minimal resolution:

$$0 \to P_n \to P_{n-1} \to \cdots P_0 \to L \to 0$$

And each P_i will be finitely generated, hence injective. Let's $K_i = \ker(P_i \to P_{i-1}) = \text{im } (P_{i+1} \to P_i)$. Then we can break into short exact sequence:

$$0 \to P_n \to P_{n-1} \to K_{n-2} \to 0$$

and

$$0 \to K_i \to P_i \to K_{i-1} \to 0$$

Now P_n is injective, therefore:

$$P_{n-1} = P_n \oplus K_{n-2}$$

Hence K_{n-2} is projective and injective since it's a direct summand of a projective and injective module P_{n-1}. By induction we can show that each K_i are all projective and injective. In particular, $L = K_{-1}$ is projective and injective. Thus we've shown that every simple module is projective, this implies that R is semisimple. □

Exercise D.2. *Let Q be quiver with two vertices and two edges with opposite orientation connecting them. Set $A(Q)$ be the path algebra over $\mathbb{C}$, and $A = A(Q)/(ab)$ where a, b are the elements corresponding to the two edges. Prove the following structural results for A-modules.*

1. *A is finite dimensional and the Cartan matrix of A has the form*

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

2. *Both indecomposable projective modules have a simple socle (call it L_1).*

3. *Let P_1 be the projective cover of L_1. THen $B \coloneqq \text{End}(P_1) \simeq \mathbb{C}[t]/t^2$, and P_1 satisfies the double centralizer property:*

$$A \to \text{End}_B(P_1)$$

   is an isomorphism.

4. *Check that A has finite homological dimension, and compute $\text{Ext}^*(L_i, L_j)$ for $i, j = 1, 2$.*

Proof. Let v, w be the two vertices, and $a : v \to w, b : w \to v$.

1. It is clear that A has a basis:
$$\{e_v, e_w, a, b, ba\}$$
so is finite dimensional. Now that an A-modules is the same data as the following:
$$(V, W, \varphi, \psi)$$
such that V, W are $\mathbb{C}$-vector spaces, $\varphi : V \to W, \psi : W \to V$ such that: $\varphi \circ \psi = 0$. It is easy to see the irreducible objects are:
$$L_1 = \mathbb{C} \underset{0}{\overset{0}{\rightleftarrows}} 0 \ , L_2 = 0 \underset{0}{\overset{0}{\rightleftarrows}} \mathbb{C}$$
Note each L_i is one-dimensional. Now A is an Artinian ring so that:
$$A = P_1 \oplus P_2$$
for projective covers P_i of L_i. But as a path algebra, we always have idempotent e_v, e_w with $e_v + e_w = 1$. Hence by Krull-Schmidt, Ae_v, Ae_w are just P_1, P_2 in some order. Now:
$$Ae_v = \mathbb{C}\{e_v, a, ba\}$$
Note $e_v \cdot e_v = e_v, e_v \cdot a = 0, e_v \cdot ba = ba$, so the two vertices are $\mathbb{C}^2 = \mathbb{C}\langle e_v, ba\rangle$ and $\mathbb{C} = \mathbb{C}\langle a\rangle$, and the corresponding module is:
$$\mathbb{C}^2 \underset{0\oplus 1}{\overset{1\oplus 0}{\rightleftarrows}} \mathbb{C}$$
It has a quotient L_1. Therefore:
$$P_1 = \mathbb{C}^2 \underset{0\oplus 1}{\overset{1\oplus 0}{\rightleftarrows}} \mathbb{C}$$
Similarly:
$$P_2 = Ae_w = \mathbb{C}\{e_w, b\} = \mathbb{C} \underset{1}{\overset{0}{\rightleftarrows}} \mathbb{C}$$
Therefore the Cartan matrix equals:
$$C = \begin{pmatrix} [P_1 : L_1] & [P_1 : L_2] \\ [P_2 : L_1] & [P_2 : L_2] \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$
as required.

2. It is clear that the L_1 we defined in the previous part is the socle for both P_1 and P_2.

3. Note a morphism $P_1 \to P_2$ is given by the data $(A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \lambda) \in \mathrm{Mat}_2(\mathbb{C}) \times \mathbb{C}$ such that the following two diagrams commute:

$$\begin{array}{ccc} \mathbb{C}^2 & \xrightarrow{1\oplus 0} & \mathbb{C} \\ {\scriptstyle A}\downarrow & & \downarrow{\scriptstyle \lambda} \\ \mathbb{C} & \xrightarrow[1\oplus 0]{} & \mathbb{C} \end{array} \qquad \begin{array}{ccc} \mathbb{C}^2 & \xleftarrow{0\oplus 1} & \mathbb{C} \\ {\scriptstyle A}\downarrow & & \downarrow{\scriptstyle \lambda} \\ \mathbb{C} & \xleftarrow[0\oplus 1]{} & \mathbb{C} \end{array}$$

The first diagram tells us that: $a = \lambda, b = 0$. With the first diagram, the second diagram tells us $d = a$. So that:

$$\mathrm{End}(P_1) \simeq \{A = \begin{pmatrix} a & \\ c & a \end{pmatrix}\} \subset \mathrm{Mat}_2(\mathbb{C})$$

This is clearly isomorphic to $B = \mathbb{C}[t]/t^2$. Now let $\mathcal{A} = A$-mod, $\mathcal{B}$ be the Serre subcategory generated by L_2, i.e. $\mathcal{B} = P_1^{\perp}$. From Morita theory we know:

$$\mathcal{A}/\mathcal{B} \simeq \mathrm{End}_A(P_1)^{\mathrm{op}}\text{-mod} \simeq B^{\mathrm{op}}\text{-mod}$$

Therefore it suffices for us to show that the natural map:

$$\mathrm{End}_{\mathcal{A}}(A) \to \mathrm{End}_{\mathcal{A}/\mathcal{B}}(A)$$

is an isomorphism. Indeed, if so then:

$$A \simeq \mathrm{End}_{\mathcal{A}}(A)^{\mathrm{op}} \simeq \mathrm{End}_{\mathcal{A}/\mathcal{B}}(A)^{\mathrm{op}} \simeq \mathrm{End}_{B^{\mathrm{op}}}(P_1^{\vee})^{\mathrm{op}} \simeq \mathrm{End}_B(P_1)$$

where the last map:

$$\mathrm{End}_B(P_1) \to \mathrm{End}_{B^{\mathrm{op}}}(P_1^{\vee})^{\mathrm{op}}$$

is given by $\varphi \mapsto \varphi^{\vee}$. Now: $\mathrm{End}_A(\mathcal{A}) \to \mathrm{End}_{\mathcal{A}/\mathcal{B}}(A)$ is given by:

$$(\varphi : A \to A) \mapsto (f \mapsto \varphi \circ f \mid f \in \mathrm{Hom}_A(P_1, A))$$

Note that $P_1/L_1 \simeq L_2$, so that if $\varphi \circ f = 0$ for all f, then we have:

$$\varphi|_{P_1} = 0, \varphi|_{P_2} = 0$$

Together this implies $\varphi = 0$. So this map is injective. The surjectivity will follow from the following assertions:

(a) $\mathrm{Hom}_{\mathcal{A}}(P_1, M) \simeq \mathrm{Hom}_{\mathcal{A}/\mathcal{B}}(P_1, M)$. This is a general property proven in class.

(b) $\mathrm{Hom}_{\mathcal{A}}(M, P_1) \simeq \mathrm{Hom}_{\mathcal{A}/\mathcal{B}}(M, P_1)$. We need to check: $P_1 \in \mathcal{B}^{\perp}$, and $\mathrm{Ext}^1(B, P_1)$ for all $B \in \mathcal{B}$. But elements in $\mathcal{B}$ are just direct sums of L_2. So we only need to check:

$$\mathrm{Hom}(L_2, P_1) = 0, \quad \mathrm{Ext}^1(L_2, P_1) = 0$$

The first statement is true since P_1 has socle L_1. For the second statement, the minimal resolution of L_2 is:

$$0 \to P_2 \to P_1 \xrightarrow{f} P_2 \to 0$$

where $f : P_1 \twoheadrightarrow L_1 \hookrightarrow P_2$. Apply $\mathrm{Hom}(-, P_1)$, we get:

$$0 \to \mathrm{Hom}(P_2, P_1) \to \mathrm{Hom}(P_1, P_1) \to \mathrm{Hom}(P_2, P_1) \to 0$$

Note $\mathrm{Hom}(P_2, P_1) \simeq \mathbb{C}$, $\mathrm{Hom}(P_1, P_1) \simeq \mathbb{C}^2$ since we've already computed the Cartan matrix. We claim that the above sequence is exact. In fact, we only need to show each arrow is nonzero and the exactness follows from the dimension argument. The first map is injective because we already know $\mathrm{Hom}(L_2, P_1) = 0$. The second map is nonzero because the image of $1 \in \mathrm{Hom}(P_1, P_1)$ is nonzero.

(c) $\mathrm{End}_{\mathcal{A}/\mathcal{B}}(P_2) = \mathbb{C}$. This is because the image of P_2 in $\mathcal{A}/\mathcal{B}$ is:

$$\mathrm{Hom}_{\mathcal{A}}(P_1, P_2) \simeq \mathbb{C}$$

With these three relations, we can compute:

$$\mathrm{End}_{\mathcal{A}/\mathcal{B}}(A) = \mathrm{End}_{\mathcal{A}/\mathcal{B}}(P_1 \oplus P_2) = \begin{pmatrix} \mathrm{End}_{\mathcal{A}}(P_1) & \mathrm{Hom}_{\mathcal{A}}(P_2, P_1) \\ \mathrm{Hom}_{\mathcal{A}}(P_1, P_2) & \mathbb{C} \end{pmatrix}$$

which receives a surjection from:

$$\mathrm{End}_{\mathcal{A}}(A) = \begin{pmatrix} \mathrm{End}_{\mathcal{A}}(P_1) & \mathrm{Hom}_{\mathcal{A}}(P_2, P_1) \\ \mathrm{Hom}_{\mathcal{A}}(P_1, P_2) & \mathrm{End}_{\mathcal{A}}(P_2) \end{pmatrix}$$

4. We already know L_2 has minimal resolution:

$$0 \to P_2 \to P_1 \xrightarrow{f} P_2 \to 0$$

and L_1 has minimal resolution:

$$0 \to P_2 \to P_1 \to 0$$

Moreover, we know A is finite dimensional so any finitely generated module has finite length, which implies that A has finite homological dimension. Let's compute the Ext groups.

(a) $\mathrm{Ext}^*(L_1, L_1)$. We consider the complex:

$$0 \to \mathrm{Hom}(P_1, L_1) \to \mathrm{Hom}(P_2, L_1) \to 0$$

$$\iff 0 \to \mathbb{C} \to 0 \to 0$$

$$\mathrm{Ext}^i(L_1, L_1) = \begin{cases} \mathbb{C}, & \text{if } i = 0; \\ 0, & \text{otherwise.} \end{cases}$$

(b) $\mathrm{Ext}^*(L_1, L_2)$. We consider the complex:

$$0 \to \mathrm{Hom}(P_1, L_2) \to \mathrm{Hom}(P_2, L_2) \to 0 \iff 0 \to 0 \to \mathbb{C} \to 0$$

$$\mathrm{Ext}^i(L_1, L_2) = \begin{cases} \mathbb{C}, & \text{if } i = 1; \\ 0, & \text{otherwise.} \end{cases}$$

(c) $\mathrm{Ext}^*(L_2, L_1)$. We consider the complex:

$$0 \to \mathrm{Hom}(P_2, L_1) \to \mathrm{Hom}(P_1, L_1) \to \mathrm{Hom}(P_2, L_1) \to 0$$

$$\iff 0 \to 0 \to \mathbb{C} \to 0 \to 0$$

$$\mathrm{Ext}^i(L_2, L_1) = \begin{cases} \mathbb{C}, & \text{if } i = 1; \\ 0, & \text{otherwise.} \end{cases}$$

(d) $\mathrm{Ext}^*(L_2, L_2)$. We consider the complex:

$$0 \to \mathrm{Hom}(P_2, L_2) \to \mathrm{Hom}(P_1, L_2) \to \mathrm{Hom}(P_2, L_2) \to 0$$

$$\iff 0 \to \mathbb{C} \to 0 \to \mathbb{C} \to 0$$

$$\mathrm{Ext}^i(L_2, L_2) = \begin{cases} \mathbb{C}, & \text{if } i = 0, 2; \\ 0, & \text{otherwise.} \end{cases}$$

As a sanity check, the resulting matrix of pairing we get under the basis $[L_1], [L_2]$ is

$$E = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$$

which is indeed the inverse of the Cartan matrix.

□

Exercise D.3. *Let $R \subset \mathrm{Mat}_2(\mathbb{Q})$ consist of matrices $A = (a_{ij})$ such that $a_{11} \in \mathbb{Z}$ and $a_{21} = 0$. Show that the homological dimension of R equals two but the homological dimension of R^{op} equals one.*

Proof. Our ring R has distinguished idempotents $e_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $e_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. So we have projective objects $Re_1 = \begin{pmatrix} \mathbb{Z} & 0 \\ & 0 \end{pmatrix}$, $Re_2 = \begin{pmatrix} 0 & \mathbb{Q} \\ 0 & \mathbb{Q} \end{pmatrix}$ for left R-modules. Let's now see that $\mathrm{hdim}(R) \leq 2$. Let M be any finitely generated left R-module. Note that $e_1 M$ is actually a submodule of M, so we have a short exact sequence:

$$0 \to e_1 M \to M \to M/e_1 M \to 0$$

Note that for Re_2 is a two-sided ideal and that Re_2 acts trivially on $e_1 M$, so that the R-module structure on $e_1 M$ factors through the $R/Re_2 = \mathbb{Z}$-module structure. Now $\mathrm{hdim}(\mathbb{Z}) = 1$, so that we can find projective resolution of $\mathbb{Z}$-modules:

$$0 \to V_1 \to V_0 \to e_1 M \to 0$$

Then

$$0 \to Re_1 \otimes_{\mathbb{Z}} V_1 \to Re_1 \otimes_{\mathbb{Z}} V_0 \to e_1 M \to 0$$

is a projective resolution of R-modules. Therefore $\mathrm{pdim}(e_1 M) \leq 1$. Now let $V = M/e_1 M$, and $u = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in R$. We have a short exact sequence of R-modules

$$0 \to u \cdot V \to V \to V/u \cdot V \to 0$$

Indeed, for $r = \begin{pmatrix} n & a \\ & b \end{pmatrix}$, we have:

$$rum = num$$

Therefore $u \cdot V$ is again just a $\mathbb{Z}$-module, and as in the previous case we have $\mathrm{pdim}(u \cdot V) \leq 1$. On the other hand, $V/u \cdot V$ is essentially a $\mathbb{Q}$-vector space taking form:

$$\begin{pmatrix} 0 \\ \mathbb{Q} \end{pmatrix}^N, r \cdot \begin{pmatrix} 0 \\ q \end{pmatrix} = \begin{pmatrix} 0 \\ bq \end{pmatrix}$$

Now we have the following projective resolution of R-modules:

$$0 \to \begin{pmatrix} \ker \\ 0 \end{pmatrix} \xrightarrow{i} \begin{pmatrix} \mathbb{Z}^{\oplus \mathbb{Q}} \\ 0 \end{pmatrix} = (Re_1)^{\oplus \mathbb{Q}} \to Re_2 \xrightarrow{p} \begin{pmatrix} \mathbb{Q} \\ \mathbb{Q} \end{pmatrix} / \begin{pmatrix} \mathbb{Q} \\ 0 \end{pmatrix} \to 0$$

where $\mathrm{coker}\ (i) = \ker(p) = \begin{pmatrix} \mathbb{Q} \\ 0 \end{pmatrix}$. And this sequence includes a resolution for $\begin{pmatrix} \mathbb{Q} \\ 0 \end{pmatrix}$. Therefore:

$$\mathrm{pdim}(u \cdot V) \leq 1, \quad \mathrm{pdim}(V/u \cdot V) \leq 2 \implies \mathrm{pdim}(V) \leq 2 \implies \mathrm{pdim}(M) \leq 2$$

Moreover, the above construction tells me if $M = \begin{pmatrix} \mathbb{Q} \\ \mathbb{Q} \end{pmatrix} / \begin{pmatrix} \mathbb{Q} \\ 0 \end{pmatrix}$, then:

$$\mathrm{Ext}^2(M, Re_1 \simeq \mathbb{Z}) \neq 0$$

This justifies $\mathrm{hdim}(R) = 2$.

Let's now compute the homological dimension of R^{op}. Note now we have projective objects:

$$e_1 R = \begin{pmatrix} \mathbb{Z} & \mathbb{Q} \end{pmatrix}, \quad e_2 R = \begin{pmatrix} 0 & 0 \\ & \mathbb{Q} \end{pmatrix}$$

Let M be a finitely generated right R-module, we again have short exact sequence:

$$0 \to Me_2 \to M \to Me_2 \setminus M \to 0$$

As before Me_2 is direct sums of $e_2 R$, so that Me_2 is projective. We still take $V = Me_2 \setminus M$, $u = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then $V \cdot u$ is again a submodule, and fitting into the exact sequence:

$$0 \to V \cdot u \to V \to V \cdot u \setminus V \to 0$$

But now $V \cdot u$ is again just a $\mathbb{Q}$-vector space, isomorphic to direct sums of $e_2 R$. In particular $V \cdot u$ is projective. Moreover, $V \cdot u \setminus V$ is essentially a $\mathbb{Z}$-module: the R-action is given by:

$$v \begin{pmatrix} n & a \\ & b \end{pmatrix} = nv$$

Now we only to find resolution for $\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}$:

$$0 \to (0, \mathbb{Q}) \simeq e_2 R \to (\mathbb{Z}, \mathbb{Q}) \simeq e_1 R \to \mathbb{Z} \to 0$$

$$0 \to (\mathbb{Z}, \mathbb{Q}) \xrightarrow{n} (\mathbb{Z}, \mathbb{Q}) \simeq \mathbb{Z}/n\mathbb{Z} \to 0$$

Hence we know: $\operatorname{pdim}(V \cdot u \setminus V) \leq 1 \implies \operatorname{pdim}(M) \leq 1 \implies \operatorname{hdim}(R^{\mathrm{op}}) \leq 1$. On the other hand, the right R-module $\mathbb{Z}$ is not projective. This confirms the homological dimension is indeed 1. □

Exercise D.4. *Let A be a finite dimensional algebra over $k = \mathbb{R}$. Let $C_{ij} = \dim_k(P_i, P_j)$ be the Cartan matrix, where P_i, P_j run over the set of isomorphism classes of indecomposable projective modules. Assume in addition that A has finite homological dimension, prove that:*

$$\det(C) = \pm 2^n$$

for some n.

Proof. Let L_i be the corresponding irreducible modules. Recall we have a pairing:

$$K_0(A) \times K_0(A) \to \mathbb{Z}$$

$$([M], [N]) \mapsto \sum_{i \geq 0} (-1)^i \dim_{\mathbb{R}} \operatorname{Ext}^i(M, N)$$

As a consequence of Krull-Schmidt, we know $[P_i]$ and $[L_i]$ both freely generate the lattice $K = K_0(A)$. Therefore, we can write:

$$[P_i] = \sum_j \alpha_{ij} [L_j]$$

where the matrix (α_{ij}) must be invertible. Now let's compute:

$$C_{ij} = \dim_k(P_i, P_j) = \langle [P_i], [P_j] \rangle = \sum_j \alpha_{ij} \langle [P_i], [L_j] \rangle$$

But we know

$$\langle [P_i], [L_j] \rangle = \dim_{\mathbb{R}} \operatorname{Hom}(P_i, L_j) = \dim_{\mathbb{R}} \operatorname{Hom}(L_i, L_j) = \begin{cases} \dim_{\mathbb{R}} \operatorname{Aut}(L_i), & \text{if } i = j; \\ 0, & \text{otherwise.} \end{cases}$$

by Schur lemma. But $\operatorname{Aut}(L_i)$ is a divison ring finite over $\mathbb{R}$, so must be either $\mathbb{R}, \mathbb{C}, \mathbb{H}$. In any case:

$$\dim_{\mathbb{R}} \operatorname{Aut}(L_i) = 2^{n_i}$$

for $n_i \in \{0, 1, 2\}$. So:

$$C_{ij} = 2^{n_i} \alpha_{ij}$$

$$\implies \det(C) = 2^{n_1 + \cdots + n_r} \det(\alpha_{ij}) = \pm 2^{n_1 + \cdots + n_r}$$

since $(\alpha_{ij}) \in \mathrm{GL}_n(\mathbb{Z})$. □

E Homework 5

Exercise E.1. *Let $D \supset K$ be skew fields. Let d_l be the dimension of D as a left K-module, and d_r be the dimension of D as a right K-module. Show that if D is finite over its center then $d_l = d_r$ (the statement is false without finiteness condition).*

[Hint: if K contains the center of D then the question is easy. Reduce to this case by considering the subfield E in D generated by the center of K and the center of D and setting $D' = K \otimes_F E \subset D$, where $F = E \cap K$ is the center of K]

Proof. Follow the hint. If $K \supset Z(D)$, then:

$$d_l = d_r = \dim_{Z(D)} D / \dim_{Z(D)} K$$

is clear. Now for general D let's prove by induction on $n = \dim_{Z(D)} D$. The statement is clear for $n = 1$. Assume $\dim_{Z(D)} D = n$ and the statement is true for all $\leq n-1$. We set: $F = Z(K)$ be a field, $F(Z(D)) = E$ is a field extension of F, $D' = K \otimes_F E = K(Z(D))$. We look at:

$$Z(D) \subset E \subset D' \subset D, \quad K \subset D'$$

It follows that:

$$d_l = \dim_K^l D' \cdot \dim_{D'}^l D = \dim_K^l D' \cdot \dim_E(D) / \dim_E(D')$$

and similarly for d_r. It follows that it suffices to show:

$$\dim_K^l D' = \dim_K^r D'$$

But observe that:

$$Z(D) \subset Z(D') \subset D' \subset D$$

We're therefore done by induction unless $D' = D$. So the remaining case is:

$$D = D' = K \otimes_F E$$

But then a F-basis for E will become both left and right K-basis for D, and we're therefore done. □

Exercise E.2. *Let F be a field of characteristic different from 2. For $a, b \in F^\times$, let $A_{a,b}$ be the four dimensional algebra over k with basis $1, i, j, k$ such that:*

$$ij = k = -ji, i^2 = a, j^2 = b$$

It was propved in lecture that $A_{a,b}$ is a CSA over F, which splits iff $b = \mathrm{Nm}_{E/F}(z)$ for some $z \in E = F(\sqrt{a})$. The Brauer class of $A_{a,b}$ is denoted as $\left(\frac{a,b}{F}\right)$. Check that:

$$\left(\frac{a,b}{F}\right)^2 = 1 = \left(\frac{a,1-a}{F}\right)$$

and

$$\left(\frac{a,bc}{F}\right) = \left(\frac{a,b}{F}\right)\left(\frac{a,c}{F}\right)$$

As a consequence we have a homomorphism:

$$K_2(F)/K_2(F)^2 \to \text{Br }(F)[2]$$

where $K_2(F)$ is the Milnor K_2 group of F: generated by symbols $\{a,b\}$ for $a,b \in F^\times$ subject to relation:

$$\{a,bc\} = \{a,b\}\{a,c\}, \{a,b\} = \{b,a\}^{-1}, \{a,1-a\} = 1$$

In fact Merkuriev(1981) proved this is an isomorphism.

Proof. First observe that:

$$1 - a = \text{Nm}(1+\sqrt{a}) \implies 1 = \left(\frac{a, 1-a}{F}\right)$$

Moreover, we have an isomorphism:

$$A_{a,b} \simeq A_{b,a}^{\text{op}}$$
$$i \mapsto j, j \mapsto i, k \mapsto k$$

But it's also clear that $A_{a,b} \simeq A_{b,a}$ by sending: $i \mapsto j, j \mapsto i, k \mapsto -k$. It follows that:

$$A_{a,b} \simeq A_{a,b}^{\text{op}} \implies \left(\frac{a,b}{F}\right) = \left(\frac{a,b}{F}\right)^{-1}$$

Hence we checked the first claim. For the second part, we claim there is an isomorphism:

$$A_{a^2,b} \otimes_F A_{a,bc} \to A_{a,b} \otimes_F A_{a,c}$$

Indeed one has:

$$A_{a^2,b} \to A_{a,b} \otimes_F A_{a,c}$$
$$1 \mapsto 1, \quad i \mapsto i \otimes i, \quad j \mapsto 1 \otimes j, \quad k \mapsto i \otimes k$$
$$A_{a,bc} \to A_{a,b} \otimes A_{a,c}$$
$$1 \mapsto 1, \quad i \mapsto i \otimes 1, \quad j \mapsto j \otimes j, \quad k \mapsto k \otimes j$$

Clearly both are algebra homomorphisms, and their image commutes. Therefore we obtain an algebra homorphism:

$$A_{a^2,b} \otimes_F A_{a,bc} \to A_{a,b} \otimes_F A_{a,c}$$

It is also straightforward to check that it is surjective. By comparing F-dimension on both sides (both 16) we see its an isomorphism. The second claim follows. □

Exercise E.3. *Recall that a module is called uniform if it is nonzero and any two nonzero submodules have a nonzero intersection. Let R be the ring of continuous $\mathbb{C}$-valued functions on $[0,1]$ with pointwise operations. Show that R has no uniform ideals.*

Proof. Let $I \subset R$, we need to show I is not uniform. Choose $0 \neq f \in I$. Suppose $(a, a+\varepsilon) \subset$ supp (f). Let b_1, b_2 be bump functions with:

$$(a + \frac{2\varepsilon}{9}, a + \frac{\varepsilon}{3}) \subset \text{supp } (b_1) \subset (a + \frac{\varepsilon}{9}, a + \frac{4\varepsilon}{9})$$

$$(a + \frac{2\varepsilon}{3}, a + \frac{7\varepsilon}{9}) \subset \text{supp } (b_2) \subset (a + \frac{5\varepsilon}{9}, a + \frac{8\varepsilon}{9})$$

Set $f_1 = fb_1, f_2 = fb_2 \in I$. So that for $g \in (f_1) \cap (f_2)$, we have:

$$V(g) \supset V(f_1) \cup V_f(_2) = [0, 1]$$

Hence $g = 0$. It follows I is not uniform. □

Exercise E.4. *Let D be a skew field which is not algebraic over its center k. Show that $R = D \otimes_k k(t)$ is a simple Noetherian ring which is not isomorphic to a matrix ring over a skew field.*

[Hint: Define a R-action on D. Show that the sum of finitely many copies of that module cannot contain a free submodule.]

Proof. Since D is a central simple algebra (not necessarily finite-dimensional) over k, and $k(t)$ is a simple k-algebra, by a lemma in lecture we see R is simple. Now R is a localization of a noetherian ring $D[t]$ (at a a multiplicative subset of the center), hence is again noetherian. It also follows that R has no nonzero zero divisor.

Now let's see it's not a matrix ring. Pick $x \in D$ be a transcendental element. We can therefore define an R-action on D by letting D acting by left multiplication, and t acting by right multiplication by x. We now show that $D^{\oplus n}$ cannot contain a copy a free submodule for any n. Indeed, if so, then there exists:

$$v = (\lambda_1, \cdots, \lambda_n) \in D^{\oplus n}$$

that freely generates a copy of R. Note if $\lambda_1 \neq 0$, we can set:

$$0 \neq c_1 = x\lambda_1^{-1} \otimes 1 - \lambda_1^{-1} \otimes t$$

and $c_1 v = (0, \lambda_2', \cdots)$. So that we can select c_i's such that their product kills v. But R has no nonzero divisor so these product is nonzero. This is a contradiction. Therefore $D^{\oplus n}$ cannot contain a copy of a free submodule. But if R is a matrix algebra, then any R-module M will have the property that $M^{\oplus n}$ contains a copy of free rank 1 module, for n large enough. This implies that R is not a matrix algebra. □

Exercise E.5. *Let $k = \mathbb{F}_p(t_1, t_2)$ be the field of rational functions in two variables over $\mathbb{F}_p$. Set $D = k\langle x, y\rangle/(xy - yx = 1, x^p = t_1, y^p = t_2)$.*

1. *Show that D is a skew field of dimension p^2 over its center k. Give an example of a splitting field of D.*

2. *Let us generalize the definition of D as follows. For $P \in \mathbb{F}_p[t_1]$ let $D(P)$ be given by $x^p = P(t_1), y^p = t_2, xy - yx = 1$. Check that we get a group homomorphism:*

$$P \mapsto [D(P)]$$

3. *Show the kernel of this homomorphism is $\mathbb{F}_p[t_1^p]$. Conclude that p-torsion in the Brauer group of k is infinite.*

Proof. 1. It is clear that D has a basis:

$$x^i y^j, \quad 0 \leq i, j \leq p-1$$

So that D has dimension p^2. Now $R = \mathbb{F}_p\langle x, y\rangle/(xy - yx = 1)$ has no nonzero divisor. To see this, their is a natural degree filtration on this, and we have $\mathrm{gr}R \simeq \mathbb{F}_p[x, y]$ has no zero divisor. It follows R has no zero divisor. Now observe that $\mathbb{F}_p[x^p, y^p] \subset Z(R)$, so we can localize at $S = \mathbb{F}_p[x^p, y^p] \setminus \{0\}$. Now we claim there is an isomorphism:

$$R[S^{-1}] \to D$$

$$x \mapsto x, y \mapsto y$$

Indeed this is well defined because the image of S^{-1} lies in $k^\times$, and this is clearly an isomorphism by presenting a basis on both sides. Therefore D is a domain. But any finite dimensional domain is a division algebra, so D is a division algebra. Moreover, D has center k. Suppose not, let $f \in Z(D)$ be the central element with smallest lowest degree, except for the constant term. We may also assume f has no constant term. Let f_0 be the lowest degree term, then we must have $f_0 \in Z(D)$. Therefore, we may assume f is homogenize:

$$f = \sum_{i+j=k} a_{ij} x^i y^j$$

But now:

$$[x, f] = \sum_{i+j=k, j\neq 0} a_{ij} x^i y^{j-1} = 0$$

this implies $a_{ij} = 0$ for $j \neq 0$. Similarly using commutation with y we see $a_{ij} = 0$ for $i \neq 0$. It follows that $f = 0$. This checks $Z(D) = k$ and hence D is indeed a CSA. Now a choice of splitting field can be: $F = k[x]/(x^p - t_1) \simeq \mathbb{F}_p(t_1^{\frac{1}{p}}, t_2)$. This is because F/k has degree p so is maximal commutative subfield, hence splits.

2. First if $P = 0$, then $D(P)$ has a zero divisior x, and it follows that $D(P) \simeq \mathrm{Mat}_p(k) \implies [D(0)] = 0$. Now to check this is a homomorphism, we must check:

$$[D(P)] \otimes_k [D(Q)] = [D(P+Q)]$$

In fact, as in the second problem we in fact has an isomorphism of algebras:

$$D_0 \otimes D(P+Q) \xrightarrow{\sim} D(P) \otimes D(Q)$$

where $D_0 = k\langle x, y\rangle/(xy - yx = 1, x^p = t_1, y^p = 0) \simeq \mathrm{Mat}_p(k)$. The isomorphism is given by:

$$x \otimes 1 \mapsto x \otimes 1 + 1 \otimes x$$

$$y \otimes 1 \mapsto 1 \otimes y$$

$$1 \otimes x \mapsto x \otimes 1$$
$$t \otimes y \mapsto y \otimes 1 - 1 \otimes y$$

It is straightforward to check this is well-defined and surjective, between algebras of same dimension over k. So we get an isomorphism.

3. It is clear that $\mathbb{F}_p[t_1^p]$ lies in the kernel. Indeed, this is because if I have $f = g^p \in \mathbb{F}_p[t_1^p]$, for some $g \in \mathbb{F}_p[t_1]$, then $D(f) \simeq D(0)$ by change of variables $x' = x - g$. I don't know how to prove other direction. But suppose we know this, then the imag of the homomorphism is contained in Br $(k)[p]$, and is isomorphic to:

$$\mathbb{F}_p[t]/\mathbb{F}_p[t^p]$$

which is clearly infinite.

□

Exercise E.6. *Let A be an algebra over a field k. Recall that an n-th order deformation of A is an associative algebra $\widetilde{A}$ over $k[t]/(t^{n+1})$, free as a module over $k[t]/(t^{n+1})$, together with an isomoprhism of k-algebras:*

$$f : \widetilde{A}/t\widetilde{A} \to A$$

Two deformations $(\widetilde{A}, f), (\widetilde{A}', f')$ are equivalent if there exists an algebra isomorphism $g : \widetilde{A} \to \widetilde{A}'$ such that $f' \circ g = f$.

1. *Construct a bijection between first order defromation of A and $\mathrm{HH}^2(A)$.*
2. *Given an n-th order deformation $\widetilde{A}_n$, there is an obstruciton to extending it to the $n+1$-th order which lies in $\mathrm{HH}^3(A)$. If it vanishes, then the set of extensions is a torsor over $\mathrm{HH}^2(A)$.*
3. *Suppose A is commutative and $\widetilde{A}_\sigma$ is a first order deformation. Set $\{a, b\} = \sigma(a, b) - \sigma(b, a)$. We have:*
$$\{ab, c\} = b\{a, c\} + a\{b, c\}$$
If $\widetilde{A}$ extends to a second order deformation then $\{-, -\}$ satisfies the Jacobi identity. This is saying, $(A, \{-, -\})$ is a **Poisson algebra**.

Proof. 1. A Hochschild cocycle is a map $\sigma : A \otimes A \to A$ satisfying:

$$(d\sigma)(a, b, c) = a\sigma(b, c) - \sigma(ab, c) + \sigma(a, bc) - \sigma(a, b)c = 0$$

A coboundary takes form
$$d\iota : A \otimes A \to A$$
$$d\iota(a, b) = a\iota(b) - \iota(ab) + \iota(a)b$$

for some $\iota \in \mathrm{Hom}_k(A, A)$. Let's see what is the data needed to define a first order deformation. Indeed we must have:

$$\widetilde{A} = A \oplus A \cdot t$$

as k vector spaces. Note if $\mu : A \otimes A \to \widetilde{A}$ is the multiplication map restricted to A, this extends uniquely:

$$(a+bt)\cdot(c+dt) = \mu(a,c) + (\mu(bc)+\mu(ad))t = \mu(a,c) + (bc+ad)t$$

Therefore it suffices to specify μ, but to define a deformation we must have $\mu(a,b) = ab + \sigma(a,b)t$ for some $\sigma : A \otimes A \to A$. Let's check associativity of $\widetilde{A}$ is equivalent to cocycle condion. That is:

$$((a+bt)\cdot(c+dt))\cdot(e+ft) = (a+bt)\cdot((c+dt)\cdot(e+ft))$$

$$\iff (ac + (\sigma(a,c)+bc+ad)t)\cdot(e+ft) = (a+bt)\cdot(ce + (\sigma(c,e)+cf+de)t)$$

$$\begin{aligned}\iff & ace + (\sigma(ac,e)+\sigma(a,c)e + bce + ade + acf)t \\ & = ace + (\sigma(a,ce) + bce + a\sigma(c,e) + acf + ade)t\end{aligned}$$

$$\iff \sigma(ac,e) + \sigma(a,c)e = \sigma(a,ce) + a\sigma(c,e)$$

this is precisley the cocycle conditon. Now assume σ gives rise to the trivial deformations. So that there exists an algebra isomorphism $g : \widetilde{A} \to \widetilde{A}_0$ extending identity map on A. In other words,

$$g(a+bt) = a + \lambda(a,b)t$$

for some λ. Let's now translate g being a homomorphism in terms of $\lambda : A \otimes A \to A$. That is:

$$g((a+bt)\cdot_\sigma (c+dt)) = g(a+bt)\cdot_{\sigma_0} g(c+dt)$$

$$\iff g(ac + (\sigma(a,c)+bc+ad)t) = (a+\lambda(a,b)t)\cdot_{\sigma_0}(c+\lambda(c,d)t)$$

$$\iff ac + \lambda(ac, \sigma(a,c)+bc+ad)t = ac + (\lambda(a,b)c + a\lambda(c,d))t$$

$$\iff \lambda(ac,\sigma(a,c)+bc+ad) = \lambda(a,b)c + a\lambda(c,d)$$

Moreover, observe that $k[t]/t^2$-linearity implies we must have:

$$\lambda(a,b) = \lambda(a,0) + b$$

So if we set $f(a) = \lambda(a,0)$, we get:

$$f(ac) + \sigma(a,c) + bc + ad = f(a)c + bc + af(c) + ad$$

$$\iff \sigma(a,b) = af(c) - f(ac) + f(a)c$$

This is precisely saying σ is a 2-coboundary.

2. Again let's write:
$$\widetilde{A}_n = A \oplus At \cdot A \cdot t^2 \oplus \cdots \oplus At^n$$
To define an order n-deformation, it suffices to specify:
$$a \cdot b = ab + \sigma_1(a,b)t + \cdots + \sigma_n(a,b)t^n$$

Suppose I want to extend $\widetilde{A}_n$ to $\widetilde{A}_{n+1}$, via:

$$a \cdot b = ab + \sigma_1(a,b)t + \cdots + \sigma_n(a,b)t^n + \sigma_{n+1}(a,b)t^{n+1}$$

We need this multiplication to be associative. In other words, for $a, b, c \in A$:

$$\begin{aligned}
&(a \cdot b) \cdot c = a \cdot (b \cdot c) \\
\iff &(ab + \sigma_1(a,b)t + \cdots \sigma_{n+1}(a,b)t^{n+1})c = a(bc + \sigma_1(b,c)t + \cdots \sigma_{n+1}(b,c)t^{n+1}) \\
\iff &\sigma_{n+1}(ab,c) + \sigma_n(\sigma_1(a,b),c) + \sigma_{n-1}(\sigma_2(a,b),c) + \cdots + \sigma_{n+1}(a,b)c \\
&= \sigma_{n+1}(a,bc) + \sigma_n(a,\sigma_1(b,c)) + \cdots + a\sigma_{n+1}(b,c)
\end{aligned}$$

The last equivalence is because we only need to check the equality of the coefficient before t^{n+1} (the equality for lower degrees follows from the fact we are extending an n-th order deformation). Hence we can write:

$$(d\sigma_{n+1})(a,b,c) = f(\sigma_1, \cdots, \sigma_n)(a,b,c)$$

where:

$$f(\sigma_1, \cdots, \sigma_n) = \sum_{i=1}^{n} (\sigma_{n+1-i}(\sigma_i(a,b),c) - \sigma_{n+1-i}(a,\sigma_i(b,c)))$$

We need to checkthat f is a 3-cocycle. The fundamental observation is the following:

Lemma. *Let $F \in C^2(A)$, and set:*

$$\Phi(F)(a,b,c) = F(F(a,b),c) - F(a,F(b,c))$$

Then:

$$\begin{aligned}
d\Phi(F)(a,b,c,d) = &F(dF(a,b,c),d) + F(a,dF(b,c,d)) - \\
&(dF(F(a,b),c,d) - dF(a,F(b,c),d) + dF(a,b,F(c,d)))
\end{aligned}$$

This can be seen by brute-force computation:

$$\begin{aligned}
&(d\Phi(F))(a,b,c,d) \\
&= aF(F(b,c),d) - F(F(ab,c),d) + F(F(a,bc),d) - F(F(a,b),cd) + F(F(a,b),c)d \\
&- aF(b,F(c,d)) + F(ab,F(c,d)) - F(a,F(bc,d)) + F(a,F(b,cd)) - F(a,F(b,c))d
\end{aligned}$$

Let's pair first term and last term:

$$aF(F(b,c),d) - F(a,F(b,c))d = F(aF(b,c),d) - F(a,F(b,c)d) + dF(a,F(b,c),d)$$

Let's pair the second term and the third term:

$$F(F(a,bc),d) - F(F(ab,c),d) = F(F(a,b)c,d) - F(aF(b,c),d) + F(dF(a,b,c),d)$$

Let's pair the 4th term and the 5th term:

$$F(F(a,b),c)d - F(F(a,b),cd) = F(a,b)F(c,d) - F(F(a,b)c,d) - dF(F(a,b),c,d)$$

Let's pair the 6th and 7th term:

$$F(ab,F(c,d)) - aF(b,F(c,d)) = F(a,bF(c,d)) - F(a,b)F(c,d) - dF(a,b,F(c,d))$$

Let's pair the 8th and 9th term:

$$F(a,F(b,cd)) - F(a,F(bc,d)) = F(a,F(b,c)d) - F(a,bF(c,d)) + F(a,dF(b,c,d))$$

Collect all terms we proved the lemma.

Now let $\Phi(F,G)(a,b,c) = F(G(a,b),c) + G(F(a,b),c) - F(a,G(b,c)) - G(a,F(b,c))$, then it is clear that:

$$\Phi(F,G) = \Phi(F+G) - \Phi(F) - \Phi(G)$$

And therefore:

$$\begin{aligned}
&d\Phi(F,G)(a,b,c,d)\\
&= F(dG(a,b,c),d) + G(dF(a,b,c),d) + F(a,dG(b,c,d)) + G(a,dF(b,c,d))\\
&- (dF(G(a,b),c,d) + dG(F(a,b),c,d)) + (dF(a,G(b,c),d) + dG(a,F(b,c),d))\\
&- (dF(a,b,G(c,d)) + dG(a,b,F(c,d)))
\end{aligned}$$

Now note that:

$$2f(\sigma_1,\cdots,\sigma_n) = \sum_{i,j\geq 1, i+j=n+1} \Phi(\sigma_i,\sigma_j)$$

We thus compute that:

$$2df = \sum_{i+j=n+1} d\Phi(\sigma_i,\sigma_j)$$

But one can compute:

$$\begin{aligned}
&\sum_{i+j=n+1} d\Phi(\sigma_i,\sigma_j)(a,b,c,d)\\
&= 2\sum_{i+j+k=n+1} \sigma_i(\sigma_j(\sigma_k(a,b),c),d) - \sigma_i(\sigma_j(a,\sigma_k(b,c)),d)\\
&- \sigma_i(\sigma_j(\sigma_k(a,b),c),d) + \sigma_i(\sigma_k(a,b),\sigma_j(c,d))\\
&+ \sigma_i(a,\sigma_j(\sigma_k(b,c),d)) - \sigma_i(a,\sigma_j(b,\sigma_k(c,d)))\\
&+ \sigma_i(\sigma_j(a,\sigma_k(b,c)),d) - \sigma_i(a,\sigma_j(\sigma_k(b,c),d))\\
&- \sigma_i(\sigma_j(a,b),\sigma_k(,c,d)) + \sigma_i(a,\sigma_j(b,\sigma_k(c,d)))\\
&= 0
\end{aligned}$$

This proves $df = 0$, i.e. f is a cocycle.

3. The Leibniz identity is equivalent to σ being a cocycle. As we computed previously, $\widetilde{A}$ extens to a second order deformation iff there exists $\tau \in C^2(A)$ such that

$$d\tau(a,b,c) = \sigma(\sigma(a,b),c) - \sigma(a,\sigma(b,c))$$

$$\iff a\tau(b,c) - \tau(ab,c) + \tau(a,bc) - \tau(a,b)c = \sigma(\sigma(a,b),c) - \sigma(a,\sigma(b,c))$$

Let's now verify Jacobi identity:

$$\begin{aligned}
J &= \{\{a,b\},c\} + \{\{b,c\},a\} + \{\{c,a\},b\} \\
&= \{\sigma(a,b) - \sigma(b,a),c\} + \{\sigma(b,c) - \sigma(c,b),a\} + \{\sigma(c,a) - \sigma(a,c),b\} \\
&= \sigma(\sigma(a,b),c) - \sigma(c,\sigma(a,b)) - \sigma(\sigma(b,a),c) + \sigma(c,\sigma(b,a)) \\
&+ \sigma(\sigma(b,c),a) - \sigma(a,\sigma(b,c)) - \sigma(\sigma(c,b),a) + \sigma(a,\sigma(c,b)) \\
&+ \sigma(\sigma(c,a),b) - \sigma(b,\sigma(c,a)) - \sigma(\sigma(a,c),b) + \sigma(b,\sigma(a,c)) \\
&= d\tau(a,b,c) - d\tau(b,a,c) + d\tau(b,c,a) - d\tau(c,b,a) + d\tau(c,a,b) - d\tau(a,c,b)
\end{aligned}$$

Now observe that:

$$\begin{aligned}
&d\tau(a,b,c) + d\tau(b,c,a) + d\tau(c,a,b) \\
&= a\tau(b,c) - \tau(ab,c) + \tau(a,bc) - \tau(a,b)c \\
&+ b\tau(c,a) - \ \tau(bc,a) + \tau(b,ca) + \tau(b,c)a \\
&+ c\tau(a,b) - \tau(ca,b) + \tau(c,ab) - \tau(c,a)b \\
&= \tau(a,bc) - \tau(bc,a) + \tau(b,ca) - \tau(ca,b) + \tau(c,ab) - \tau(ab,c)
\end{aligned}$$

using A is commutative. Now similarly, $d\tau(b,a,c) + d\tau(c,b,c) + d\tau(a,c,b)$ must equal the same expression. It follows that $J = 0$.

□

F Homework 6

Exercise F.1. *An element x in a ring R is said to be* ad *locally nilpotent if* $\mathrm{ad}(x) : a \mapsto xa - ax$ *is locally nilpotent. Show S consisting of* ad *locally nilpotent elements satisfies Ore's condition.*

Proof. Let's first check S satisifes O1. Let $a \in R, s \in S$ we must show $aS \cap sR \neq \emptyset$. But this is easy, say $\mathrm{ad}(s)^n(a) = 0$. We consider $x = as^n \in aS$. We only need to show $x \in sR$. Indeed:

$$as^n = (sa + \mathrm{ad}(s)(a))s^{n-1} = sas^{n-1} + (s\mathrm{ad}(s)(a) + \mathrm{ad}(s)^2(a))s^{n-2} = \cdots$$

Eventually we can write:

$$as^n = sas^{n-1} + s \cdot \mathrm{ad}(s)(a) \cdot s^{n-2} + s \cdot \mathrm{ad}(s)(a) \cdot s^{n-3} + \cdots + \mathrm{ad}(s)^n(a) \in sR$$

since the last term vanishes.

For O2, we need to check if $sa \neq 0$ with $s \in S, a \in R$, we can find $t \in S$ such that $at = 0$. But observe the fact that for all $y \in R$, if $sy = 0$ then:

$$s \cdot \mathrm{ad}(s)(y) = s^2y - sys = 0$$

By induction $s \cdot \mathrm{ad}(s)^n(y) = 0$ for all n. But now we can just take $t = s^n$, then $sa = 0 \implies as^n = 0$ by the computation for O1. □

Exercise F.2. *Let K be a skew field and $\varphi : K \to K$ a homomorphism. Set $A = K\langle x \rangle / (xa = \varphi(a)x)$. Show that the set of powers of x is a left Ore set, and is a right Ore set iff φ is surjective. In the latter case prove that A is a right Ore domain.*

Proof. Let S consist of powers of x. Let's check S is a left Ore set. For O1, note that for all $a \in R, n \in \mathbb{Z}_{\geq 0}$:

$$x^n a = \varphi^{\circ n}(a)x^n \in Rx^n \cap Sa$$

For O2, suppose $ax^n = 0$, we must find other m such that $x^m a = 0$. But if $a = 0$ this is clear. Assume $a \neq 0$, then a has an iverse since K is a skew field. Therefore $x^n = 0 \implies x^n a = 0$. This checks O2.

Now assume φ is surjective (hence an automorphism). Then checking S is a right Ore set is similar:

$$ax^n = x^n(\varphi^{-1})^{\circ n}(a) \in x^n R \cap aS$$

Checking O2 is similar by using K is a skew field. Let's finally check A is a right Ore domain in this case. First of all the reltion guarantees that every element in A can be uniquely write as:

$$a_0 + a_1x + a_2x^2 \cdots + a_nx^n$$

for $a_i \in K$. Now suppose $f = \sum_{i=0}^n a_i x^i, g = \sum_{i=0}^m b_i x^i$ with $a_n, b_m \neq 0$. Then the leading coeffient of fg equals $a_n\varphi^{\circ n}(b_m) \neq 0$. So that $fg \neq 0$. This check A is a domain. To check A is an Ore domain we prove that A is right noetherian. In fact, we will prove A is right principal ideal domain. Indeed left $0 \neq I \subset A$ be any left ideal, and pick $f \in I$ has minimal degree. We claim that $I = (f)$. Indeed, if $0 \neq g \in I$ then $\deg g \geq \deg f$. Use the commuting relation we can do "division with remainder" and obtain an element of degree $< \deg f$, unless g is a right multiple of f. □

Exercise F.3. *Let us say that an ideal $I \subset R$ is right localizable if the set of elements regular modulo I is a right Ore set.*

Let k be a field, and $R \subset \mathrm{Mat}_2(k[x])$ be given by $\left\{ \begin{pmatrix} f(0) & g(x) \\ 0 & f(x) \end{pmatrix} \mid f, g \in k[x] \right\}$. Show that R is right noetherian, the ideal of strictly upper triangular matrices is prime, has square zero and is not localizable.

Proof. Let's first classify right ideals of R. Let I be a right ideal of R.

1. There exists $r \in I$ such that $r_{11} \neq 0$. Let's choose

$$r = \begin{pmatrix} f(0) & g(x) \\ & f(x) \end{pmatrix} \in I$$

with $f(0) \neq 0$ and $\deg f$ minimal. Now consider:

$$r \cdot \begin{pmatrix} 1 & -f(0)^{-1}g(x) \\ & 1 \end{pmatrix} = \begin{pmatrix} f(0) & 0 \\ & f(x) \end{pmatrix}$$

Replace r by $\begin{pmatrix} f(0) & \\ & f(x) \end{pmatrix}$. We claim for every $r' \in I$ with $r'_{11} \neq 0$, it must be a multiple of f. Indeed, we can use the same method to multiply a unit and make r' diagonal:

$$r' = \begin{pmatrix} f'(0) & \\ & f'(x) \end{pmatrix}$$

Then Bezout's theorem implies that we will get:

$$r'' = \begin{pmatrix} f''(0) & \\ & f''(x) \end{pmatrix}, \quad f'' = \gcd(f, f')$$

The minimality of $\deg f$ tells me $f'' = f \implies f \mid f'$. Now consider the element:

$$r' = \begin{pmatrix} 0 & g'(x) \\ & f'(x) \end{pmatrix}, \quad f'(0) = 0$$

Choose $a, b \in k[x]$ such that $af + bf' = \gcd(f, f')$, then:

$$r \cdot \begin{pmatrix} a(0) & \\ & a(x) \end{pmatrix} + r' \cdot \begin{pmatrix} b(0) & \\ & b(x) \end{pmatrix} = \begin{pmatrix} \gcd(f, f')(0) & * \\ & \gcd(f, f') \end{pmatrix}$$

Now $f(0) \neq 0 \implies \gcd(f, f')(0) \neq 0 \implies f \mid f'$. Therefore we can substract some multiples of r from r' to get:

$$\begin{pmatrix} 0 & g'(x) \\ 0 & 0 \end{pmatrix}$$

We rename the new element to be r'. Now choose element r' to be such that $\deg g'$ is minimal. Then it is evident that I is generated by r, r'.

2. Otherwise for all $r \in I$ we have $r_{11} = 0$. Let's take:

$$r = \begin{pmatrix} 0 & g(x) \\ & f(x) \end{pmatrix} \in I$$

with $\deg f$ minimal. Again by some division of remainder we know every other $r' = \begin{pmatrix} 0 & g'(x) \\ & f'(x) \end{pmatrix}$ will have $f \mid f'$. So after substraction we can transform any other element in the form:

$$\begin{pmatrix} 0 & g'(x) \\ & 0 \end{pmatrix}$$

again. Now let r' be the one such that $\deg g'$ is minimal. Then again: $I = (r, r')$.

From the above classification we see that any right ideal of R is generated by 2 elements, and hence is noetherian. Let P be the 2-sided ideal consisting of strictly upper triangular matrices. Then:

$$R/P \xrightarrow{\sim} k[x]$$

$$r \mapsto r_{22}$$

is an isomorphism. So P is a prime ring. It is also evident that $P^2 = 0$. Now the set of elements regular modulo P is:

$$S = \{r \in R \mid r_{22} \neq 0\}$$

We must show this is not a right Ore set. So we need to pick $a \in R, s \in S$ such that:

$$aS \cap sR = \emptyset$$

Indeed, we can pick

$$s = \begin{pmatrix} 0 & 0 \\ & x \end{pmatrix}, a = \begin{pmatrix} 0 & 1 \\ & 0 \end{pmatrix}$$

then:

$$aS = \{\begin{pmatrix} 0 & f(x) \\ & 0 \end{pmatrix} \mid f \neq 0\}, \quad sR = \{\begin{pmatrix} 0 & 0 \\ & xf(x) \end{pmatrix}\}$$

It is clear that $aS \cap sR = \emptyset$. □

Exercise F.4. *The goal of this exercise is to show that GK dimension does not behave well under short exact sequences. Show that the following provide an example of PI algebra R, and R-module M with a submodule N such that:*

$$\mathrm{GKdim}(N) = \mathrm{GKdim}(M/N) = 1, \quad \mathrm{GKdim}(M) = 2$$

Set $R = \mathbb{C}\langle x, y\rangle/(yx)$, let M have two generators α, β subject to relations:

$$x^{n+1}y^n\alpha = 0$$

$$xy^n\beta = \begin{cases} xy^m\alpha, & \text{if } n = m^2; \\ 0, & \text{otherwise.} \end{cases}$$

Let $N = R\beta$. Check that R satisifies the identity $[a, b]^2 = 0$, thus it is PI.

Proof. Let's first show R is PI. Indeed, R has a $\mathbb{C}$ basis consisting of monomials $x^i y^j$ for $i, j \geq 0$. From this presentation it is clear that:

$$[x^a y^b, x^c y^d] = \begin{cases} x^{a+c} y^d, & \text{if } b = 0, a \neq 0, d \neq 0; \\ x^a y^{b+d}, & \text{if } c = 0, d \neq 0, a \neq 0; \\ x^{a+c} y^b, & \text{if } b \neq 0, c \neq 0, d = 0; \\ x^c y^{b+d}, & \text{if } b \neq 0, c \neq 0, a = 0; \\ 0, & \text{otherwise.} \end{cases}$$

Therefore $[R, R] = C \cdot \{x^i y^j \mid i, j \geq 1\}$ so that $[R, R]^2 = 0$ and hence R is PI.

To compute GK dimension, we use generating set $W = \mathbb{C} \cdot \{\alpha, \beta\}$ and $V = \mathbb{C} \cdot \{x, y\}$. Then M_n consists of monomials:

$$x^i y^j \alpha \text{ for } \quad i \leq j, i + j \leq n$$

Therefore $\dim(M_n) \geq \sum_{i+j \leq n, i \leq j} 1 = O(n^2)$. This implies $\mathrm{GKdim}(M) \geq 2$. But $\mathrm{GKdim}(M) \leq \mathrm{GKdim}(R) = 2$, therefore $\mathrm{GKdim}(M) = 2$.

Now N consists of elements of form:

$$y^n \beta, x^i y^m \alpha$$

for $n \geq 0, m \geq 0, i \geq 1$. Therefore M/N has a basis:

$$\{y^m \beta \mid m \geq 0\}$$

On which x acts by 0 and y acts naturally. So this is just the free rank 1 module of $R/(x) \simeq \mathbb{C}[y]$. Therefore $\mathrm{GKdim}(M/N) = 1$.

Finally let's compute the GK dimension for N. We take $W = \mathbb{C} \cdot \{\beta\}$ and $V = C \cdot \{x, y\}$ as before. Then:

$$N_n = \mathrm{Span}(\{y^r \beta \mid 0 \leq r \leq n\} \cup \{x^i y^s \alpha \mid i + s^2 \leq n, i \geq 1, i \leq s\})$$

It is also clear that the above set also forms a basis for N_n. Note the first piece consists of $n + 1$ elements, and the set piece consists of elements of number smaller than or equal to:

$$\sqrt{n} \cdot \sqrt{n} = n$$

It follows that:

$$n + 1 \leq \dim(N_n) \leq 2n + 1 \implies \mathrm{GKdim}(N) = 1$$

as required. □

Exercise F.5. *Let G be the group of transformations of the line generated by $x \mapsto x + 1$ and $x \mapsto 2x$. Show that the group algebra of G has exponential growth.*

Proof. Let $\alpha(x) = x + 1, \beta(x) = 2x$. Then our algebra A is isomorphic to:

$$k\langle \alpha, \beta, \alpha^{-1}, \beta^{-1} \rangle / (\alpha^2 \beta - \beta \alpha)$$

Let V be generated by $\alpha, \beta, \alpha^{-1}, \beta^{-1}$, $d : TV \to A$, and let $d_V(n) = \dim(A_n) = \dim(d(T^{\leq n}V))$. Note that:

$$\alpha^m \beta^{-i_1} \alpha^{\pm 1} \beta^{-i_2} \alpha^{\pm 1} \cdots \beta^{-i_k} \alpha^{\pm 1} \beta^{i_1 + \cdots + i_k} \in A_n$$

for $m + (i_1 + 1) + \cdots + (i_k + 1) + (i_1 + \cdots i_k) \leq n$,; $i_1, \cdots, i_k \geq 1$; $m \geq 0$. Moreover, they represents different elements in the group G since their value at 0 are distinct (the above expression is equivalent to give a binary expansion of a number). Let's try to give a lower bound of the number of monomials of the above form. A naive bound is given by setting all $i_j = 1$ and $m = 0$, the choice will be each power of α to take $+1$ or -1. This gives lower bound: $2^{[\frac{n}{3}]}$. In particular:

$$d_V(n) \geq 2^{[\frac{n}{3}]}$$

Hence A has exponential growth. □